

62. Flash Memory (FLASH)

This MCU incorporates code flash memory, data flash memory, and option-setting memory. The code flash memory stores instructions and operands, and the data flash memory stores only data. For option-setting memory, refer to [section 7, Option-Setting Memory \(OFSM\)](#).

62.1 Overview

Table 62.1 lists the specifications of the flash memory, and Figure 62.1 is a block diagram of the flash memory related modules.

The I/O pins used in boot mode, see [Table 62.17](#).

The FCU (flash control unit) controls programming and erasure of the flash memory. The FACI (flash application command interface) controls the FCU according to the specified FACI commands.

Regarding the configuration of the code flash memory, see [Figure 62.2](#) to [Figure 62.4](#), and for the configuration of the data flash memory, see [Figure 62.5](#).

Table 62.1 Specifications of Flash Memory (1/2)

Item	Code Flash Memory	Data Flash Memory
Memory capacity	User area: 4 Mbytes max.	Data area: 32 Kbytes
ROM cache	- Capacity: 8 Kbytes - Mapping method: direct mapping - Line size: 16 bytes	Not available
Read cycle	- While ROM cache operation is enabled: - When the cache is hit, one cycle; - When the cache is missed, one to two cycles if ICLK ≤ 120 MHz, two to three cycles if ICLK > 120 MHz.*1 - When ROM cache operation is disabled: - One cycle if ICLK ≤ 120 MHz - Two cycles if ICLK > 120 MHz*1	Reading proceeds in every cycle of FCLK*2
Value after erase	FFh	Undefined
Programming/erasing method	- Programming and erasing the code flash memory and data flash memory, and programming the option-setting memory are handled by the FACI commands specified in the FACI command issuing area (007E 0000h) (self-programming). - Programming/erasure through transfer by a serial-programmer via a serial interface (serial programming)	
Security function	Protects against illicit tampering with or reading out of data in flash memory	
Protection	Protects against erroneous rewriting of the flash memory	
Dual bank function	The dual-bank structure makes a safe update possible in cases where programming is suspended. - Linear mode: the code flash memory is used as one area - Dual mode: the code flash memory is divided into two areas	Not available
Trusted memory (TM) function	Protects against illicit reading of the code flash memory - Linear mode: blocks 8 and 9 - Dual mode: blocks 8, 9, 78, and 79	Not available
Background operations (BGOs)*3	- The code flash memory can be read while the code flash memory is being programmed or erased.*4 - The data flash memory can be read while the code flash memory is being programmed or erased. - The code flash memory can be read while the data flash memory is being programmed or erased.	

Table 62.1 Specifications of Flash Memory (2/2)

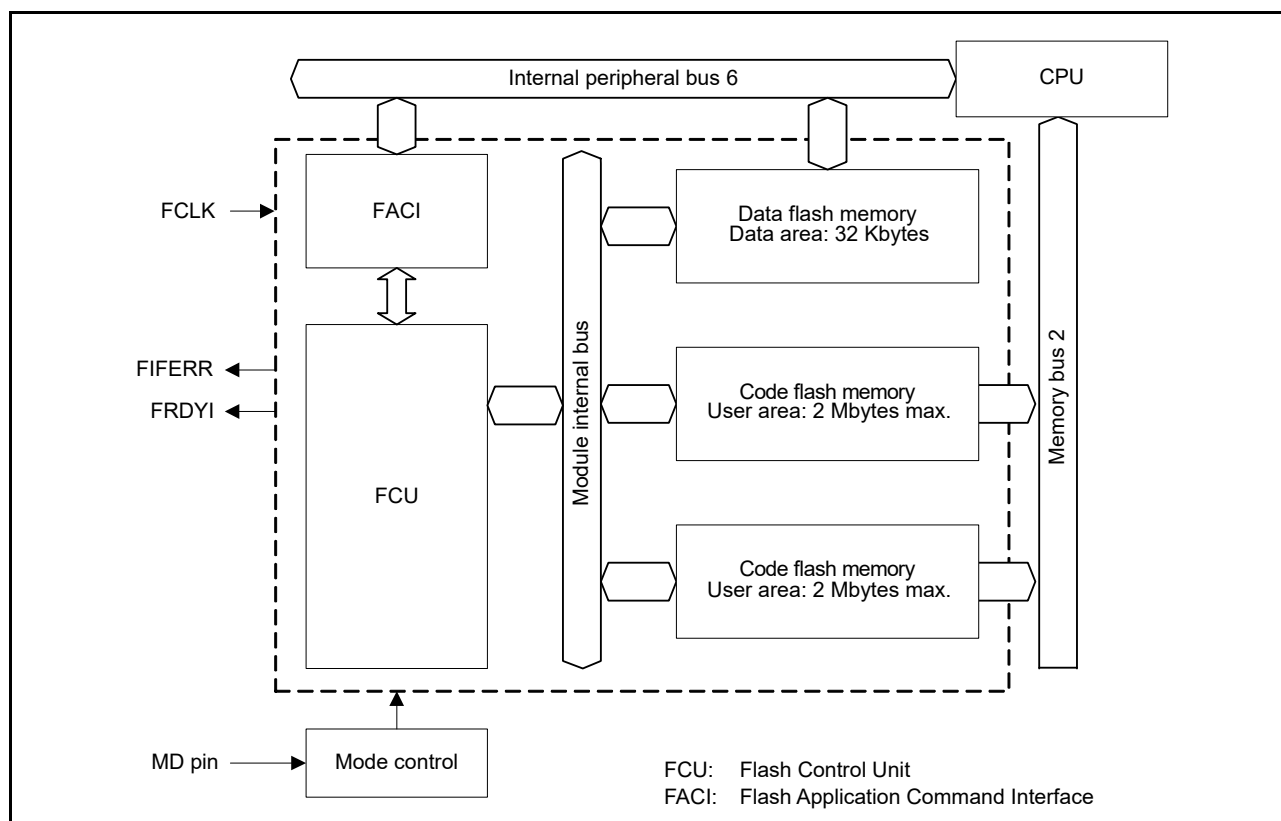
Item	Code Flash Memory	Data Flash Memory
Units of programming and erasure	- Units of programming for the user area: 128 bytes - Units of erasure for the user area: Block units	- Unit of programming for the data area: 4 bytes - Unit of erasure for the data area: 64/128/256 bytes
Other functions	Interrupts can be accepted during self-programming. In the initial settings of this MCU, an expansion area of the option-setting memory can be set.	
On-board programming (Serial programming/Self-programming)	Program/erase in boot mode (SCI interface) - The asynchronous serial interface (SCI1) is used. - The transfer rate is adjusted automatically. Program/erase in boot mode (USB interface) - USB0 is used. - Dedicated hardware is not required, so direct connection to a PC is possible. Program/erase in boot mode (FINE interface) - FINE is used Program/erase in single-chip mode or on-chip ROM enabled extended mode - Programming or erasure by a routine for writing to code flash memory or data flash memory within the user program is possible.	
Off-board programming	Programming and erasure of the code flash memory and option-setting memory by using a parallel programmer is possible.	Programming or erasure of the data flash memory by a parallel programmer is not possible.
Unique ID	A 16-byte ID code provided for each MCU	

Note 1. If the frequency of ICLK is above 120 MHz, the setting of the MEMWAIT register must be changed. For details, see section 9.2.2, Memory Wait Cycle Setting Register (MEMWAIT).

Note 2. The required number of cycles for access must be set according to the FCLK frequency. For details, refer to section 62.4.22, Data Flash Memory Access Frequency Setting Register (EEPFCCLK).

Note 3. The BGO function cannot be used to rewrite the option-setting memory.

Note 4. Limitations apply to the combinations of the address ranges for programming/erasure process and reading process: see Table 62.25.

**Figure 62.1 Block Diagram of Flash Memory Related Modules**

62.2 Hardware Interface Area

Using the hardware interface with the flash memory requires accessing to the area containing registers of the hardware, that for the issuing of FACI commands. Table 62.2 summarizes information on all of these areas.

Table 62.2 Information on the Hardware Interface Area

Area	Address	Capacity
Area containing the various registers of the hardware	See section 62.4, Register Descriptions.	See section 62.4, Register Descriptions.
FACI command-issuing area	007E 0000h	4 bytes
Option-setting memory (configuration setting area)	FE7F 5D00h to FE7F 5D7Fh	128 bytes

62.3 Structure of Memory

Figure 62.2 shows the memory map of code flash memory in linear mode. Figure 62.3 and Figure 62.4 show the memory map of code flash memory in dual mode. The code flash memory can be used as 2 bank areas by using the dual bank function. This dual-bank structure allows a safe update of a program while a user program is running.

The user area of the code flash memory in this MCU is divided into 8- and 32-Kbyte blocks, which serve as the units of erasure.

When the TM function is enabled in linear mode, blocks 8 and 9 are the TM target areas. The user area is available as areas for storing the user program.

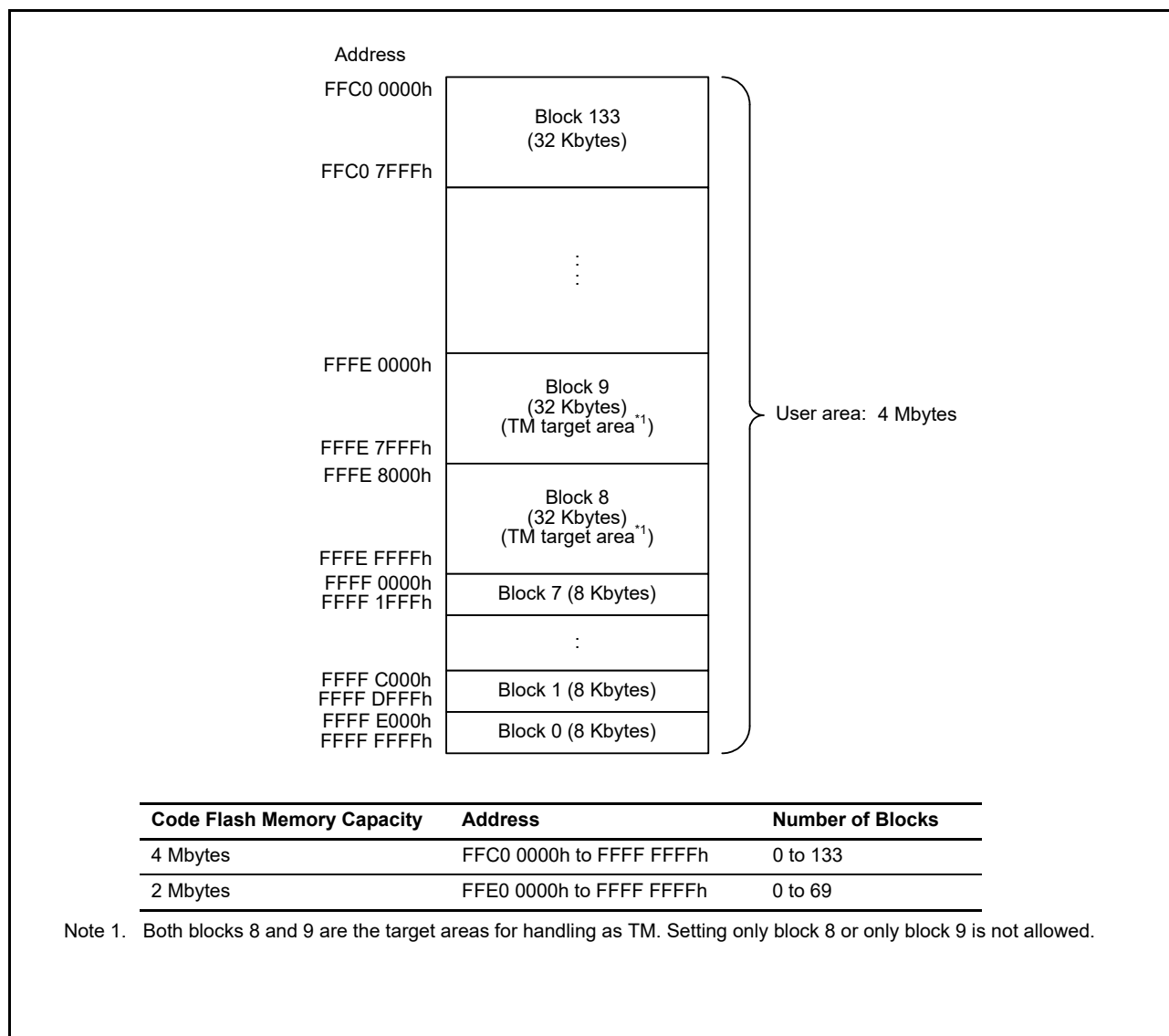


Figure 62.2 Map of the Code Flash Memory in Linear Mode

When the TM function is enabled in dual mode, blocks 8 and 9, and blocks 78 and 79 are the TM target areas. The user area can be used as storage area of the user program.

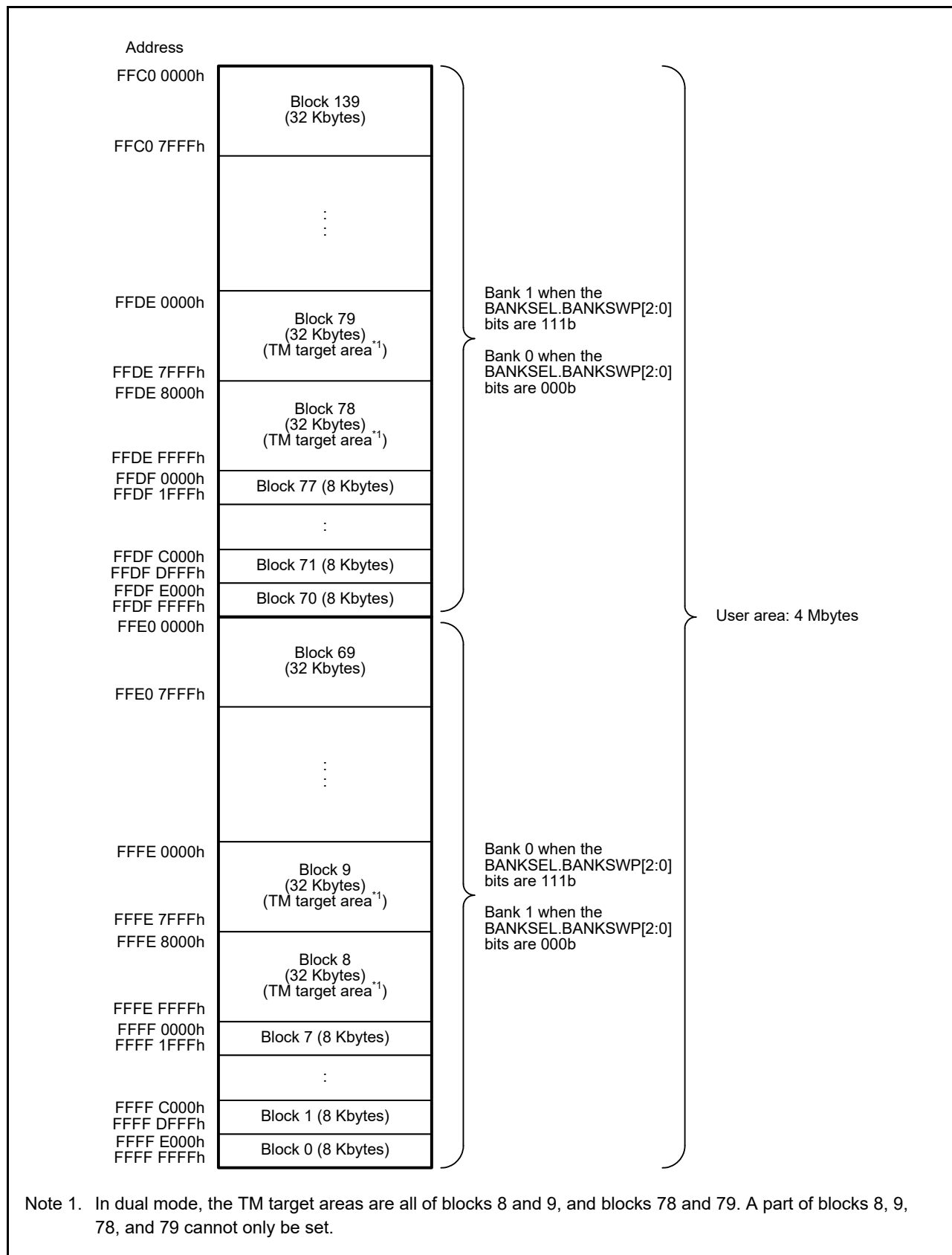


Figure 62.3 Map of the Code Flash Memory in Dual Mode (Products with 4 Mbytes of code flash memory)

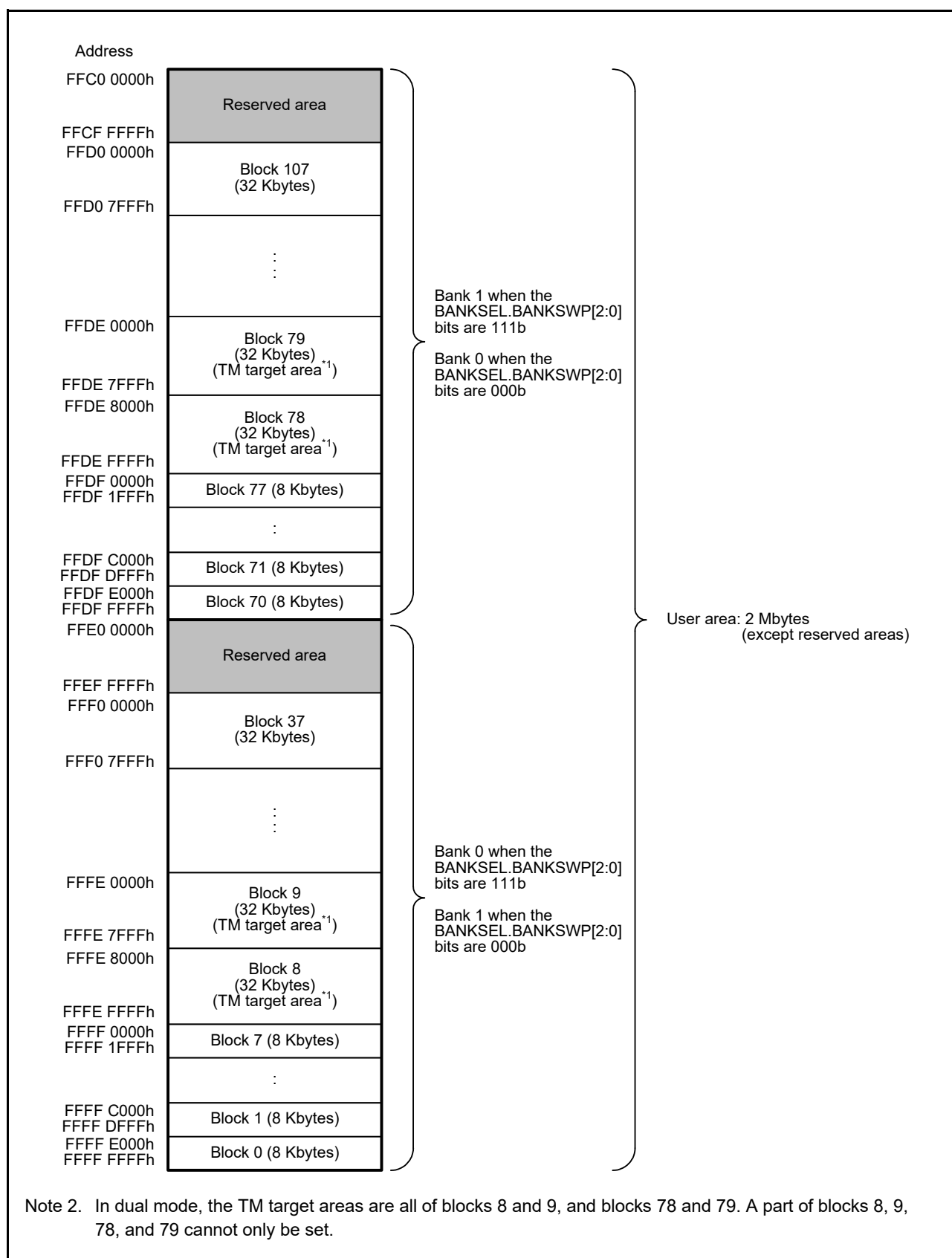


Figure 62.4 Map of the Code Flash Memory in Dual Mode (Products with 2 Mbytes of code flash memory)

The data area of the data flash memory in this MCU is divided into 64-byte blocks, with each being a unit for erasure. Figure 62.5 shows the mapping of the data flash memory.

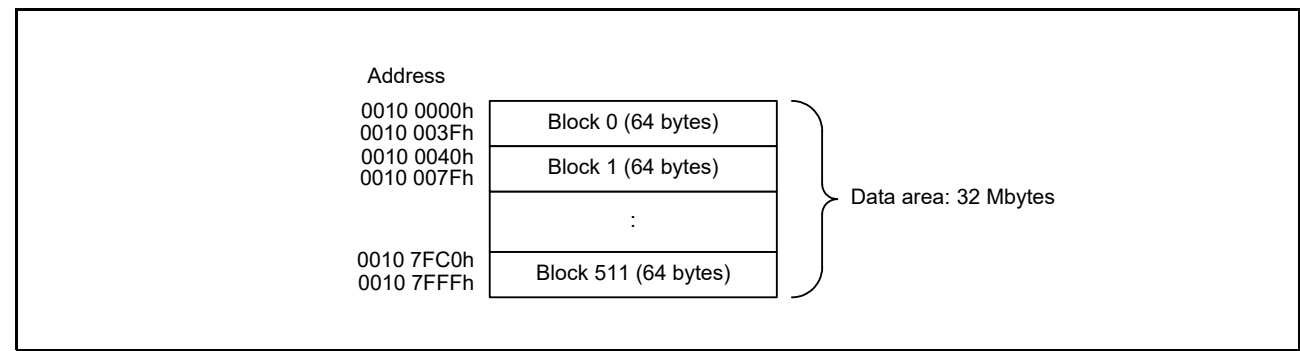


Figure 62.5 Map of the Data Flash Memory

62.4 Register Descriptions

62.4.1 ROM Cache Enable Register (ROMCE)

Address(es): FLASH.ROMCE 0008 1000h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ROMC EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ROMCEN	ROM Cache Enable	0: ROM cache is disabled. 1: ROM cache is enabled.	R/W
b15 to b1	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R

ROMCEN Bit (ROM Cache Enable)

When setting the ROMCEN bit to 1, the ROM cache starts operation. While ROM cache is enabled, the data is served from the cache when a cache hit occurs. Using ROM caching helps to reduce power consumption by the MCU.

Set a non-cacheable area while the ROMCEN bit is 0.

62.4.2 ROM Cache Invalidate Register (ROMCIV)

Address(es): FLASH.ROMCIV 0008 1004h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ROMCI V
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

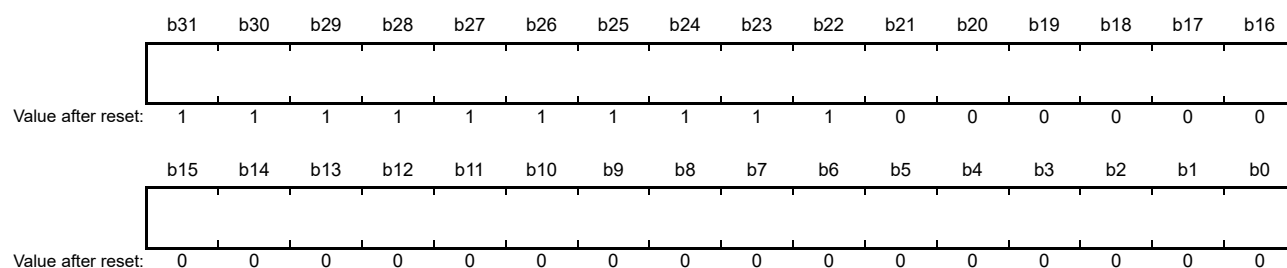
Bit	Symbol	Bit Name	Description	R/W
b0	ROMCIV	ROM Cache Invalidate	When reading 0: Invalidation is not started/Invalidation is completed. 1: Invalidation is in progress. When writing Writing 1 invalidates the cache line. Writing 0 has no effect.	R/W
b15 to b1	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R

ROMCIV Bit (ROM Cache Invalidate)

When writing 1 to the ROMCIV bit, the ROM cache invalidation is performed.

62.4.3 Non-Cacheable Area n Address Register (NCRGn) (n = 0, 1)

Address(es): FLASH.NCRG0 0008 1040h, FLASH.NCRG1 0008 1048h



The NCRGn register specifies the address where an area in which caching is disabled starts (non-cacheable area).

The higher-order 10 bits (b31 to b22), which are fixed to 1 and the lower-order 4 bits (b3 to b0), which are fixed to 0, are reserved. When writing, write 1 and 0 to the respective sets of bits.

Set the NCRGn register while the ROMCE.ROMCEN bit is 0.

62.4.4 Non-Cacheable Area n Setting Register (NCRCn) (n = 0, 1)

Address(es): FLASH.NCRC0 0008 1044h, FLASH.NCRC1 0008 104Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	NCSZ[16:12]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	NCSZ[11:0]												NC3E	NC2E	NC1E	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b1	NC1E	IF Non-Cacheable Area Specification Enable	This bit selects whether an area of memory is non-cacheable by the IF cache. 0: Disabled 1: Enabled	R/W
b2	NC2E	OA Non-Cacheable Area Specification Enable	This bit selects whether an area of memory is non-cacheable by the OA cache. 0: Disabled 1: Enabled	R/W
b3	NC3E	DM Non-Cacheable Area Specification Enable	This bit selects whether an area of memory is non-cacheable by the DM cache. 0: Disabled 1: Enabled	R/W
b20 to b4	NCSZ[16:0]	Non-Cacheable Area Size Specification	These bits specify the size of the given non-cacheable area. b20 b4 0 0000 0000 0000 0000: 16 bytes 0 0000 0000 0000 0001: 32 bytes 0 0000 0000 0000 0011: 64 bytes 0 0000 0000 0000 0111: 128 bytes 0 0000 0000 0000 1111: 256 bytes 0 0000 0000 0001 1111: 512 bytes 0 0000 0000 0011 1111: 1 Kbyte 0 0000 0000 0111 1111: 2 Kbytes 0 0000 0000 1111 1111: 4 Kbytes 0 0000 0001 1111 1111: 8 Kbytes 0 0000 0011 1111 1111: 16 Kbytes 0 0000 0111 1111 1111: 32 Kbytes 0 0000 1111 1111 1111: 64 Kbytes 0 0001 1111 1111 1111: 128 Kbytes 0 0011 1111 1111 1111: 256 Kbytes 0 0111 1111 1111 1111: 512 Kbytes 0 1111 1111 1111 1111: 1 Mbyte 1 1111 1111 1111 1111: 2 Mbytes Settings other than above are prohibited.	R/W
b31 to b21	—	Reserved	These bits are read as 0. The write value should be 0.	R

The NCRCn register specifies enabling or disabling of non-cacheable areas n (n = 0, 1) and the size of the area. Write to the register while the ROMCE.ROMCEN bit is 0.

NC1E Bit (IF Non-Cacheable Area Specification Enable)

The NC1E bit enables or disables the settings of the non-cacheable area in the cache (IF cache) used for fast instruction fetching (IF) by the CPU.

NC2E Bit (OA Non-Cacheable Area Specification Enable)

The NC2E bit enables or disables the settings of the non-cacheable area in the cache (OA cache) used for fast operand access (OA) by the CPU.

NC3E Bit (DM Non-Cacheable Area Specification Enable)

The NC3E bit enables or disables the settings of the non-cacheable area in the cache (DM cache) used for fast access to data by bus masters other than the CPU.

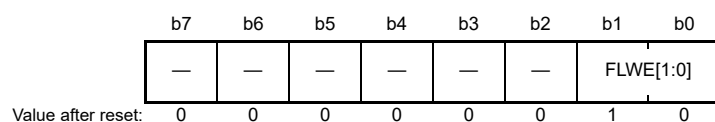
NCSZ[16:0] Bit (Non-Cacheable Area Size Specification)

The NCSZ[16:0] bits specify the size of the non-cacheable area. They are used to mask address bits 4 to 20.

When an address generated by the bus master is compared to the value of the NCRGn register, bits corresponding to NCSZ[16:0] bits with the value 1 are ignored.

62.4.5 Flash P/E Protect Register (FWEPROR)

Address(es): FLASH.FWEPROR 0008 C296h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	FLWE[1:0]	Flash Program/Erase Enable	b1 b0 0 0: Program, block erase, and blank check are disabled. 0 1: Program, block erase, and blank check are enabled. 1 0: Program, block erase, and blank check are disabled. 1 1: Program, block erase, and blank check are disabled.	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register enables or disables the following three FACL commands for the flash memory: program, block erase, and blank check.

This register is initialized not only by a reset but also by a transition to deep software standby mode or a transition to software standby mode.

62.4.6 Flash Access Status Register (FASTAT)

Address(es): FLASH.FASTAT 007F E010h

b7	b6	b5	b4	b3	b2	b1	b0
CFAE	—	—	CMDLK	DFAE	—	—	—
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b3	DFAE	Data Flash Memory Access Violation Flag	0: No data flash memory access violation has occurred. 1: A data flash memory access violation has occurred.	R/W*1
b4	CMDLK	Command Lock Flag	0: The flash sequencer is not in the command-locked state. 1: The flash sequencer is in the command-locked state.	R
b6, b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	CFAE	Code Flash Memory Access Violation Flag	0: No code flash memory access violation has occurred. 1: A code flash memory access violation has occurred.	R/W*1

Note 1. Only 0 can be written to clear the flag after 1 is read.

This register indicates whether a code flash memory or data flash memory access violation has occurred. If either of the CFAE, and DFAE flags is 1, the CMDLK flag is set to 1 and the flash sequencer enters the command-locked state (see section 62.10.2, Error Protection). To release it from the command-locked state, a status clear command or forced stop command must be issued by the FACI.

DFAE Flag (Data Flash Memory Access Violation Flag)

This flag indicates whether a data flash memory access violation has occurred. If this flag is set to 1, the FSTATR.ILGLERR flag is set to 1, placing the flash sequencer in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When 0 is written after reading of 1
- When a status clear command or forced stop command is issued

CMDLK Flag (Command Lock Flag)

This flag indicates that the flash sequencer is in the command-locked state.

[Setting Condition]

- When the flash sequencer detects any of errors listed in Table 62.16, Error Protection Type and transitions to the command-locked state

[Clearing Condition]

- When the flash sequencer starting to process a status clear or forced stop command

CFAE Flag (Code Flash Memory Access Violation Flag)

This flag indicates whether a code flash memory access violation has occurred. If this flag is set to 1, the FSTATR.ILGLERR flag is set to 1, placing the flash sequencer in the command-locked state.

[Setting Conditions]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When 0 is written after reading of 1

- When a status clear command or forced stop command is issued

62.4.7 Flash Access Error Interrupt Enable Register (FAEINT)

Address(es): FLASH.FAEINT 007F E014h

	b7	b6	b5	b4	b3	b2	b1	b0
	CFAEIE	—	—	CMDLKIE	DFAEIE	—	—	—
Value after reset:	1	0	0	1	1	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b3	DFAEIE	Data Flash Memory Access Violation Interrupt Enable	0: Generation of an FIFERR interrupt request is disabled when FASTAT.DFAE is set to 1. 1: Generation of an FIFERR interrupt request is enabled when FASTAT.DFAE is set to 1.	R/W
b4	CMDLKIE	Command Lock Interrupt Enable	0: Generation of an FIFERR interrupt request is disabled when FASTAT.CMDLK is set to 1. 1: Generation of an FIFERR interrupt request is enabled when FASTAT.CMDLK is set to 1.	R/W
b6, b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	CFAEIE	Code Flash Memory Access Violation Interrupt Enable	0: Generation of an FIFERR interrupt request is disabled when FASTAT.CFAE is set to 1. 1: Generation of an FIFERR interrupt request is enabled when FASTAT.CFAE is set to 1.	R/W

This register enables or disables generation of a flash access error (FIFERR) interrupt request.

DFAEIE Bit (Data Flash Memory Access Violation Interrupt Enable)

This bit enables or disables generation of an FIFERR interrupt request when a data flash memory access violation occurs and the FASTAT.DFAE flag is set to 1.

CMDLKIE Bit (Command Lock Interrupt Enable)

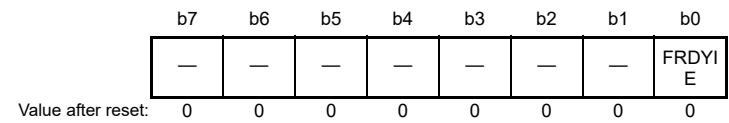
This bit enables or disables generation of an FIFERR interrupt request when the flash sequencer enters the command-locked state and the FASTAT.CMDLK flag is set to 1.

CFAEIE Bit (Code Flash Memory Access Violation Interrupt Enable)

This bit enables or disables generation of an FIFERR interrupt request when a code flash memory access violation occurs and the FASTAT.CFAE flag is set to 1.

62.4.8 Flash Ready Interrupt Enable Register (FRDYIE)

Address(es): FLASH.FRDIYE 007F E018h



Bit	Symbol	Bit Name	Description	R/W
b0	FRDYIE	Flash Ready Interrupt Enable	0: Generation of an FRDY interrupt request is disabled. 1: Generation of an FRDY interrupt request is enabled.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

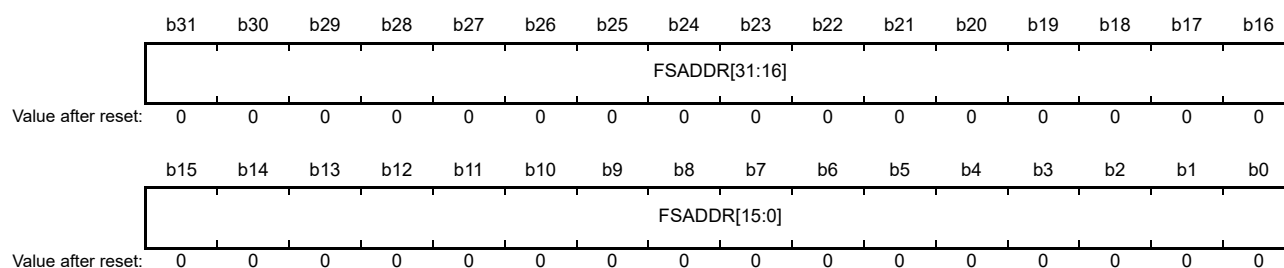
This register enables or disables generation of a flash ready (FRDY) interrupt request.

FRDYIE Bit (Flash Ready Interrupt Enable)

This bit is used to enable or disable generation of an FRDY interrupt request when the FASTAT.FRDY flag is changed from 0 to 1 upon completion of processing by the flash sequencer of program, block erase, or blank check command.

62.4.9 FACI Command Start Address Register (FSADDR)

Address(es): FLASH.FSADDR 007F E030h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	FSADDR[31:0]	Start Address for FACI Command Processing	Start Address for FACI Command Processing	R/W*1

Note 1. Writing to these bits is possible only when the FSTATR.FRDY flag is 1. Writing to these bits while the FSTATR.FRDY flag = 0 is ignored. The value of the lower 2 bits (b1 and b0) is fixed to 00b.

This register specifies the address where the target area for command processing starts when the FACI command for program, block erase, multi-block erase, blank check, or configuration set is issued.

The FSADDR register is initialized when the FSUINITR.SUINIT bit is set to 1. It is also initialized by a reset.

FSADDR[31:0] Bits (Start Address for FACI Command Processing)

These bits specify the start address for FACI command processing. Bits 31 to 24 are ignored in FACI command processing for the code flash memory. Bits 31 to 17 are ignored in FACI command processing for the data flash memory. Bits that do not reach the address boundaries are also ignored. Table 62.3 shows the address boundary for each command.

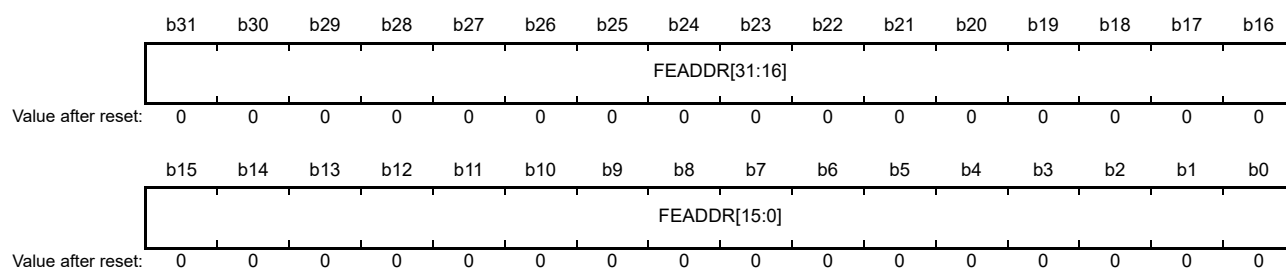
Table 62.3 Address Boundary for Each of the Commands

Command	Address Boundary
Program (code flash memory)	128-byte
Program (data flash memory)	4-byte
Block erase (code flash memory)	8-Kbyte or 32-Kbyte
Block erase (data flash memory)	64-byte
Multi-block erase (data flash memory)	64-, 128-, or 256-byte
Blank check (data flash memory)	4-byte
Configuration set	16-byte

See Table 62.15, Address Used by Configuration Set Command for the start address of the option-setting memory (configuration setting area).

62.4.10 FACI Command End Address Register (FEADDR)

Address(es): FLASH.FEADDR 007F E034h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	FEADDR[31:0]	End Address for FACI Command Processing	The end address for FACI command processing	R/W*1

Note 1. Writing to these bits is possible only when the FSTATR.FRDY flag is 1. Writing to these bits while the FSTATR.FRDY flag = 0 is ignored. The value of the lower 2 bits (b1 and b0) is fixed to 00b.

This register is used to specify the end address of the area targeted for the multi-block erase command and blank check command handling. When executing the multi-block erase command, the setting value of the register must be the setting of the FEADDR register or lower. If the setting value of the FSADDR register is larger than the FEADDR register setting value, the flash sequencer enters the command-locked state.

When the FBCCNT.BCDIR bit is 0, the value of the FSADDR register must be that of this register or less. When the FBCCNT.BCDIR bit is 1, the value of the FSADDR register must be at least that of this register. If the settings of the FBCCNT.BCDIR bit and the FSADDR and FEADDR registers are inconsistent with the above rules, the flash sequencer enters the command-locked state (see section 62.10.2, Error Protection).

The FEADDR register is initialized when the FSUINITR.SUINIT bit is set to 1. It is also initialized by a reset.

FEADDR[31:0] Bits (End Address for FACI Command Processing)

These bits are used to specify the end address for handling of the multi-block erase command and blank check command. In command processing, bits 31 to 17 and any bits that do not reach the address boundaries listed in Table 62.3, Address Boundary for Each of the Commands are ignored.

62.4.11 Flash Status Register (FSTATR)

Address(es): FLASH.FSTATR 007F E080h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	ILGCOMERR	FESETERR	SECERR	OTERR	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	FRDY	ILGLERR	ERSERR	PRGERR	SUSRDY	DBFULL	ERSSPD	PRGSPD	—	FLWEERR	—	—	—	—	—	—
Value after reset:	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b6	FLWEERR	Flash P/E Protect Error Flag	0: An error has not occurred. 1: An error has occurred.	R
b7	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b8	PRGSPD	Program Suspend Status Flag	0: The flash sequencer is in a state other than those corresponding to the value 1. 1: The flash sequencer is in the program suspend processing state or the program-suspended state.	R
b9	ERSSPD	Erase Suspend Status Flag	0: The flash sequencer is in a state other than those corresponding to the value 1. 1: The flash sequencer is in the erase suspend processing state or the erase-suspended state.	R
b10	DBFULL	Data Buffer Full Flag	0: The data buffer is empty. 1: The data buffer is full.	R
b11	SUSRDY	Suspend Ready Flag	0: The flash sequencer cannot receive P/E suspend commands. 1: The flash sequencer can receive P/E suspend commands.	R
b12	PRGERR	Program Error Flag	0: Program operation has been completed successfully. 1: An error has occurred during program operation.	R
b13	ERSERR	Erase Error Flag	0: Erase operation has been completed successfully. 1: An error has occurred during erase operation.	R
b14	ILGLERR	Illegal Error Flag	0: The flash sequencer has not detected an illegal FACI command or illegal flash memory access. 1: The flash sequencer has detected an illegal FACI command or illegal flash memory access.	R
b15	FRDY	Flash Ready Flag	0: Program, block erase, multi-block erase, P/E suspend, P/E resume, forced stop, blank check, or configuration set command processing is in progress. 1: None of the above is in progress.	R
b19 to b16	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b20	OTERR	Other Error Flag	0: An error has not occurred. 1: An error has occurred.	R
b21	SECERR	Security Error Flag	0: Write protection by the FAW.FSPR bit is not violated 1: Write protection by the FAW.FSPR bit is violated	R
b22	FESETERR	FENTRY Setting Error Flag	0: A setting error in the FENTRYR register has not been detected. 1: A setting error in the FENTRYR register has been detected.	R
b23	ILGCOMERR	Illegal Command Error Flag	0: The flash sequencer has not detected an illegal FACI command error. 1: The flash sequencer has detected an illegal FACI command error.	R
b31 to b24	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register indicates the state of the flash sequencer.

FLWEERR Flag (Flash P/E Protect Error Flag)

This flag indicates a violation of the flash memory program/erase protection setting in the FWEPROR register. When this flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a forced stop command

PRGSPD Flag (Program Suspend Status Flag)

This flag indicates that the flash sequencer is in the processing of program suspend or has transitioned to the program-suspended state.

[Setting Condition]

- When the flash sequencer starts processing in response to a program suspend command

[Clearing Conditions]

- When the flash sequencer has received a P/E resume command (after write access to the FACI command-issuing area is completed)
- When the flash sequencer starts processing of a forced stop command

ERSSPD Flag (Erase Suspend Status Flag)

This flag indicates that the flash sequencer is the processing of erase suspend or has transitioned to the erase-suspended state.

[Setting Condition]

- When the flash sequencer starts processing in response to an erase suspend command

[Clearing Conditions]

- Reception of the P/E resume command by the flash sequencer (after write access to the FACI command-issuing area is completed)
- When the flash sequencer starts processing of a forced stop command

DBFULL Flag (Data Buffer Full Flag)

This flag indicates the state of the data buffer when a program command or configuration set command is issued. The FACI incorporates a buffer for write data (data buffer). When data for writing to the flash memory are issued to the FACI command-issuing area while the data buffer is full, the FACI inserts a wait cycle in the peripheral bus 6.

[Setting Condition]

- When the data buffer becomes full while a program command or configuration set command is being issued

[Clearing Condition]

- When the data buffer becomes empty

SUSRDY Flag (Suspend Ready Flag)

This flag indicates whether the flash sequencer can receive a P/E suspend command.

[Setting Condition]

- When the flash sequencer is ready to accept the P/E suspend command after the program or erase sequence has started

[Clearing Conditions]

- When the flash sequencer has accepted the P/E suspend command or forced stop command (after write access to the FACI command-issuing area is completed)

- When the flash sequencer enters the command-locked state during programming or erasure
- When programming or erasure has been completed

PRGERR Flag (Program Error Flag)

This flag indicates the result of programming the flash memory. If this flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command

ERSERR Flag (Erase Error Flag)

This flag indicates the result of erasing the flash memory. If this flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command

ILGLERR Flag (Illegal Error Flag)

This flag indicates that the flash sequencer has detected an illegal FACI command or flash memory access. When this flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command.

FRDY Flag (Flash Ready Flag)

This flag indicates the command processing state of the flash sequencer.

[Setting Conditions]

- When the flash sequencer completes command processing
- When the flash sequencer receives a P/E suspend command and suspends programming or erasure of the flash memory
- When the flash sequencer has received a forced stop command and ended command processing

[Clearing Conditions]

- When the flash sequencer receives the FACI command of the setting of the program and configuration and after the first write access is made to the FACI command-issuing area
- When the flash sequencer receives any FACI command other than of the setting of the program and configuration and after the last write access is made to the FACI command issuing area

OTERR Flag (Other Error Flag)

This flag indicates that an FACI command has been issued when the condition of accepting commands is not satisfied. The OTERR flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command

SECERR Flag (Security Error Flag)

This flag indicates that write protection by the FAW.FSPR bit is violated. When the SECERR flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command

FESETERR Flag (FENTRY Setting Error Flag)

This flag indicates that a value of AA81h is written in the FENTRYR register or the value in the FENTRYR register differs when P/E is suspended and resumed. When the FESETERR flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command

ILGCOMERR Flag (Illegal Command Error Flag)

This flag indicates that the flash sequencer has detected an illegal FACI command. When the ILGCOMERR flag is 1, the flash sequencer is in the command-locked state.

[Setting Condition]

- See Table 62.16, Error Protection Type

[Clearing Condition]

- When the flash sequencer starts processing of a status clear or forced stop command

62.4.12 Flash P/E Mode Entry Register (FENTRYR)

Address(es): FLASH.FENTRYR 007F E084h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	KEY[7:0]								FENTR YD	—	—	—	—	—	—	FENTR YC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	FENTRYC	Code Flash Memory P/E Mode Entry	0: Code flash memory is in read mode. 1: Code flash memory is in P/E mode.	R/W*1, *2
b6 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	FENTRYD	Data Flash Memory P/E Mode Entry	0: Data flash memory is in read mode. 1: Data flash memory is in P/E mode.	R/W*1, *2
b15 to b8	KEY[7:0]	Key Code	Key code	R/W*3

Note 1. Writing to these bits is possible only when the FSTATR.FR DY flag is 1. Writing to these bits while the FSTATR.FR DY flag = 0 is ignored.

Note 2. Writing to this bit is possible only when this register is written in 16-bit units with the KEY[7:0] bits set to AAh.

Note 3. Written values are not retained by these bits. These bits are read as 0.

This register is used to specify code flash memory P/E mode and data flash memory P/E mode. To specify code flash memory P/E mode or data flash memory P/E mode so that the flash sequencer can receive FACI commands, set either the FENTRYD or FENTRYC bit to 1 to place the flash sequencer in P/E mode.

Note that writing AA81h in this register causes the FSTATR.ILGLERR and FSTATR.FESETERR flags to be set to 1, and the flash sequencer to enter the command-locked state.

The FENTRYR register is initialized when the FSUINTR.SUINIT bit is set to 1. It is also initialized by a reset.

FENTRYC Bit (Code Flash Memory P/E Mode Entry)

This bit specifies the P/E mode for code flash memory.

[Setting Condition]

- When AA01h is written to the FENTRYR register while the FSTATR.FR DY flag is 1 and the FENTRYR register is 0000h

[Clearing Conditions]

- When the FENTRYR register is written in 8-bit units while the FSTATR.FR DY flag is 1
- When the FENTRYR register is written in 16-bit units with the KEY[7:0] bits set to a value other than AAh while the FSTATR.FR DY flag is 1
- When AA00h is written to the FENTRYR register while the FSTATR.FR DY flag is 1
- When the FENTRYR register is written while the FSTATR.FR DY flag is 1 and the FENTRYR register is not 0000h
- When the flash sequencer set-up is initialized by writing 1 to the FSUINTR.SUINIT bit.

FENTRYD Bit (Data Flash Memory P/E Mode Entry)

This bit specifies the P/E mode for data flash memory.

[Setting Condition]

- When AA80h is written to the FENTRYR register while the FSTATR.FR DY flag is 1 and the FENTRYR register is 0000h

[Clearing Conditions]

- When the FENTRYR register is written in 8-bit units while the FSTATR.FR DY flag is 1
- When the FENTRYR register is written in 16-bit units with the KEY[7:0] bits set to a value other than AAh while

the FSTATR.FRDY flag is 1

- When AA00h is written to the FENTRYR register while the FSTATR.FRDY flag is 1
- When the FENTRYR register is written while the FSTATR.FRDY flag is 1 and the FENTRYR register is not 0000h
- When the flash sequencer set-up is initialized by writing 1 to the FSUINTR.SUINIT bit.

KEY[7:0] Bits (Key Code)

These bits control permission and prohibition of writing to the FENTRYD and FENTRYC bits.

62.4.13 Flash Sequencer Set-Up Initialization Register (FSUINITR)

Address(es): FLASH.FSUINITR 007F E08Ch

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	KEY[7:0]							—	—	—	—	—	—	—	—	SUINIT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SUINIT	Set-Up Initialization	0: The FEADDR, FCPSR, FSADDR, FENTRYR, and FBCCNT flash sequencer set-up registers keep their current values. 1: The FEADDR, FCPSR, FSADDR, FENTRYR, and FBCCNT flash sequencer set-up registers are initialized.	R/W*1, *2
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b8	KEY[7:0]	Key Code	Key code	R/W*3

Note 1. Writing to these bits is possible only when the FSTATR.FRDY flag is 1. Writing to these bits while the FSTATR.FRDY flag = 0 is ignored.

Note 2. Writing to these bits is possible only when this register is written in 16-bit units with the KEY[7:0] bits set to 2Dh.

Note 3. Written values are not retained by these bits. This bit is read as 0.

This register is used for initialization of the flash sequencer set-up.

SUINIT Bit (Set-Up Initialization)

This bit initializes the following flash sequencer set-up registers.

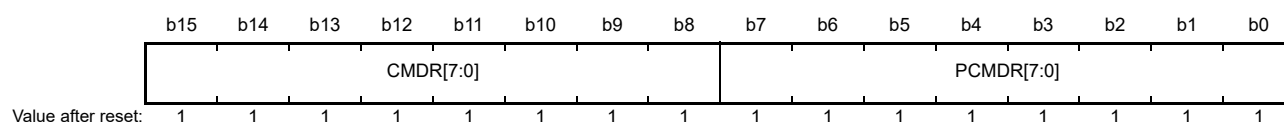
- FEADDR
- FCPSR
- FSADDR
- FENTRYR
- FBCCNT

KEY[7:0] Bits (Key Code)

These bits control permission and prohibition of writing to the SUINIT bit.

62.4.14 FACL Command Register (FCMDR)

Address(es): FLASH.FCMDR 007F E0A0h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	PCMDR[7:0]	Precommand Flag	The command immediately before the latest command is stored.	R
b15 to b8	CMDR[7:0]	Command Flag	The latest command is stored.	R

This register records the two most recent commands accepted by the FACL.

PCMDR[7:0] Flags (Precommand Flag)

These flags indicate the command received immediately before the last command received by the FACL.

CMDR[7:0] Flags (Command Flag)

These flags indicate the latest command received by the FACL.

Table 62.4 States of FCMDR after Receiving Commands

Command	CMDR	PCMDR
Program	E8h	Previous command
Block erase	D0h	20h
Multi-block erase	D0h	21h
P/E suspend	B0h	Previous command
P/E resume	D0h	Previous command
Status clear	50h	Previous command
Forced stop	B3h	Previous command
Blank check	D0h	71h
Configuration set	40h	Previous command

62.4.15 Data Flash Blank Check Control Register (FBCCNT)

Address(es): FLASH.FBCCNT 007F E0D0h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	BCDIR
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BCDIR	Blank Check Direction	0: Blank checking is executed from lower addresses to higher addresses (incremental mode). 1: Blank checking is executed from higher addresses to lower addresses (decremental mode).	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register specifies the addressing mode in processing of a blank check command. The register is initialized when the FSUINTR.SUINIT bit is set to 1. It is also initialized by a reset.

BCDIR Bit (Blank Check Direction)

This bit specifies the addressing mode for blank checking.

62.4.16 Data Flash Blank Check Status Register (FBCSTAT)

Address(es): FLASH.FBCSTAT 007F E0D4h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	BCST
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	BCST	Blank Check Status Flag	0: The target area is in the non-programmed state (i.e. is blank; the area has been erased but has not yet been re-programmed). 1: The target area has been programmed with 0s or 1s.	R
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register stores the results of checking in response to a blank check command.

BCST Flag (Blank Check Status Flag)

This flag indicates the results of checking in response to a blank check command.

[Setting condition]

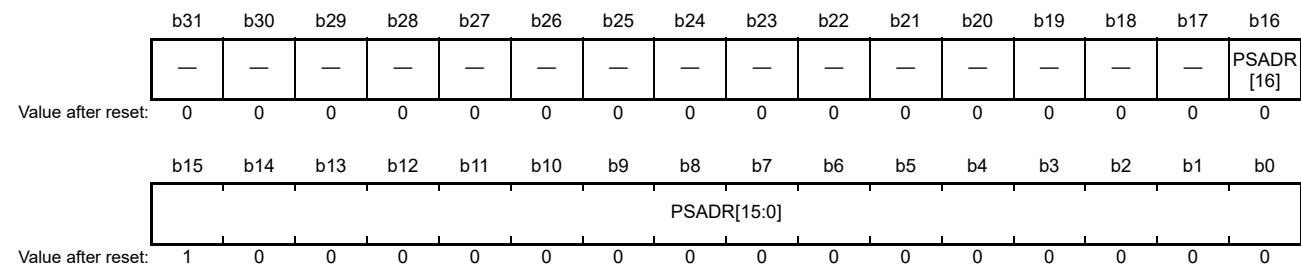
- When the blank checking command is issued and data of 0 or 1 is written to the target area

[Clearing condition]

- When the blank checking command is executed and the target area is blank

62.4.17 Data Flash Programming Start Address Register (FPSADDR)

Address(es): FLASH.FPSADDR 007F E0D8h



Bit	Symbol	Bit Name	Description	R/W
b16 to b0	PSADR[16:0]	Programmed Area Start Address	The starting address of the programmed area to be found firstly	R
b31 to b17	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register indicates the starting address of the programmed area to be found firstly in processing of a blank check command.

PSADR[16:0] Bits (Programmed Area Start Address)

These bits indicate the starting address of the programmed area to be found firstly in processing of a blank check command. The offset value is stored from the first address of the data flash memory. The value of the bits is valid only while the FBCSTAT.BCST bit is 1.

62.4.18 Flash Sequencer Processing Switching Register (FCPSR)

Address(es): FLASH.FCPSR 007F E0E0h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ESUSPMD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ESUSPMD	Erase Suspend Mode	0: Suspend priority mode 1: Erase priority mode	R/W
b15 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

This register is for selecting the erase suspend mode.

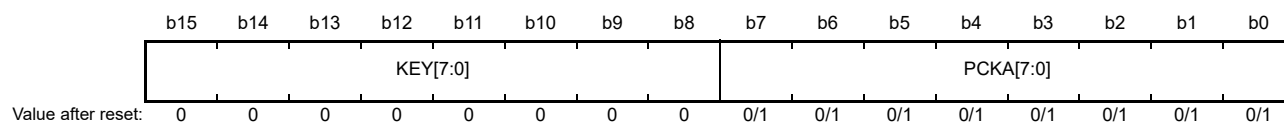
The register is initialized when the FSUINITR.SUINIT bit is set to 1. It is also initialized by a reset.

ESUSPMD Bit (Erase Suspend Mode)

This bit is for selecting the erase suspend mode when a P/E suspend command is issued while the flash sequencer is executing erase processing (see section 62.8.3.9, P/E Suspend Command). This bit should be set before issuing a block erase command.

62.4.19 Flash Sequencer Processing Clock Frequency Notification Register (FPCKAR)

Address(es): FLASH.FPCKAR 007F E0E4h



Bit	Symbol	Bit Name	Description	R/W
b7 to b0	PCKA[7:0]	Flash Sequencer Processing Clock Frequency Notification	These bits are used to set the frequency of the FlashIF clock (FCLK) and notify the flash sequencer of the frequency used	R/W*1, *2
b15 to b8	KEY[7:0]	Key Code	Key code	R/W*3

Note 1. Writing to these bits is possible only when the FSTATR.FR DY flag is 1. Writing to these bits while the FSTATR.FR DY flag = 0 is ignored.

Note 2. Writing to these bits is possible only when this register is written in 16-bit units with the KEY[7:0] bits set to 1Eh.

Note 3. Written values are not retained by these bits. These bits are read as 0.

This register specifies the frequency of the FlashIF clock (FCLK) generated in the clock generator and notifies the flash sequencer of the frequency used. The flash sequencer determines the FSCI command processing time based on the frequency notified by the FPCKAR register. The initial value is set to the maximum operating frequency of the FCLK.

PCKA[7:0] Bits (Flash Sequencer Processing Clock Frequency Notification)

These bits are used to specify the frequency of the FCLK generated in the clock generator and to notify the flash sequencer of the frequency used. Set the desired frequency in these bits before issuing an FSCI command. Specifically, convert the frequency represented in MHz into a binary number and set it in these bits.

Example: When frequency is 35.9 MHz (PCKA[7:0] = 24h)

Round up the first decimal place of 35.9 MHz to a whole number (= 36) and convert it into a binary number.

If the value set in these bits is smaller than the frequency of the FCLK, the rewriting characteristics of the flash memory cannot be guaranteed. Conversely, if the value set in these bits is greater than the frequency of the FCLK, the rewriting characteristics of the flash memory can be guaranteed although the FSCI command processing time such as time for rewriting will increase (the FSCI command processing time becomes the shortest when the frequency of the FCLK is the same as the value set in the bits).

KEY[7:0] Bits (Key Code)

These bits control permission and prohibition of writing to the PCKA[7:0] bits.

62.4.20 Flash Access Window Monitor Register (FAWMON)

Address(es): FLASH.FAWMON 007F E0DCh

b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
BTFLG	—	—	—												
FAWE[11:0]															
Value after reset: The value set by the user*1															
b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
FSPR	—	—	—												
FAWS[11:0]															
Value after reset: The value set by the user*1															

Bit	Symbol	Bit Name	Description	R/W
b11 to b0	FAWS[11:0]	Flash Access Window Start Address	Flash access window start address	R
b14 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15	FSPR	Access Window Protection Flag	0: With protection (P/E disabled) 1: Without protection (P/E enabled)	R
b27 to b16	FAWE[11:0]	Flash Access Window End Address	Flash access window end address	R
b30 to b28	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b31	BTFLG	Start-Up Area Select Flag*2	0: Startup area 0 is in the range from FFFF C000h to FFFF DFFFh, startup area 1 is in the range from FFFF E000h to FFFF FFFFh. 1: Startup area 1 is in the range from FFFF C000h to FFFF DFFFh, startup area 0 is in the range from FFFF E000h to FFFF FFFFh.	R

Note 1. The value of the blank product is FFFF FFFFh. This register is set to a specified value after programming of the flash memory with the user program.

Note 2. When the FSUACR.SAS[1:0] bits are changed to 1xb, the startup area is dependent on the setting of the FSUACR.SAS[1:0] bits regardless of the setting of the FAW.BTFLG bit.

This register indicates the values of the write protection flag and start-up area select flag for setting the flash access window start/end address, and the access window. When a reset or configuration set command is executed, the FAW transfers data from the option-setting memory to this register and the setting of the option-setting memory is enabled.

FAWS[11:0] Bits (Flash Access Window Start Address)

These bits are used to verify the access window start address setting value.

FSPR Flag (Access Window Protection Flag)

This flag indicates whether or not protection for a configuration set command for the access window setting or for writing to the FSUACR register is available.

FAWE[11:0] Bits (Flash Access Window End Address)

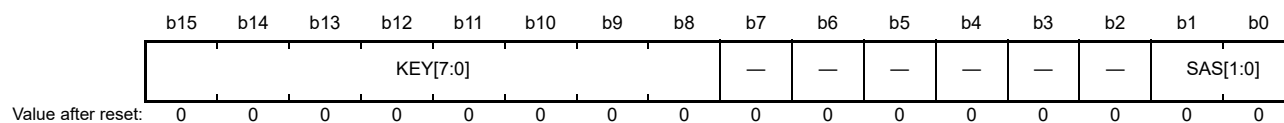
These bits are used to verify the access window end address setting value. The value of these bits indicates the first address of the next P/E-enabled block as configured in the access window.

BTFLG Flag (Start-Up Area Select Flag)

This flag indicates whether the start-up area is switched by using start-up program protection.

62.4.21 Start-Up Area Control Register (FSUACR)

Address(es): FLASH.FSUACR 007F E0E8h



Bit	Symbol	Bit Name	Description	R/W
b1, b0	SAS[1:0]	Start-Up Area Select	b1 b0 0 x: A start-up area is selected based on the setting of the FAW.BTFLG bit. 1 0: The area from FFFF E000h to FFFF FFFFh is selected as the address range for startup area 0, and the area from FFFF C000h to FFFF DFFFh is selected as the address range for startup area 1, regardless of the setting of the FAW.BTFLG bit. 1 1: The area from FFFF E000h to FFFF FFFFh is selected as the address range for startup area 1, and the area from FFFF C000h to FFFF DFFFh is selected as the address range for startup area 0, regardless of the setting of the FAW.BTFLG bit.	R/W*1, *2
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b15 to b8	KEY[7:0]	Key Code	Key codes	R/W*3

x: Don't care

Note 1. Writable only when the FAW.FSPR bit is 1. When the FAW.FSPR bit is 0, writing to these bits are ignored.

Note 2. Writing to these bits is possible only when this register is written in 16-bit units with the KEY[7:0] bits set to 66h.

Note 3. Written values are not retained by these bits. These bits are read as 0.

This register is used to switch startup areas 0 and 1 by startup program protection.

Do not use this register in dual mode (the MDE.BANKMD[2:0] bits are 000b). In dual mode, starting up proceeds from startup area 0.

SAS[1:0]Bits (Start-Up Area Select)

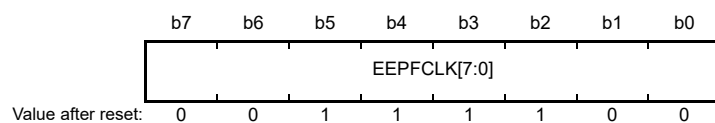
These bits are used to switch startup areas 0 and 1.

KEY[7:0] Bits (Key Code)

These bits enables or disable overwriting to the SAS[1:0] bits.

62.4.22 Data Flash Memory Access Frequency Setting Register (EEPFCLK)

Address(es): FLASH.EEPFCLK 007F C040h



This register optimizes the speed of reading the data flash memory.

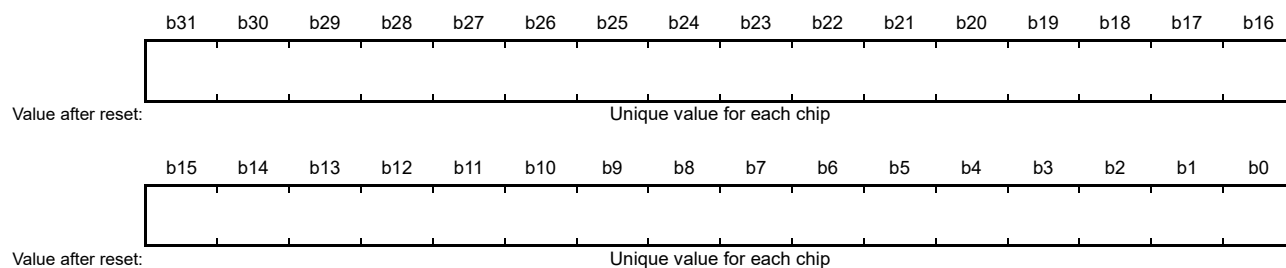
Set the frequency of the peripheral module clock (FCLK) of internal peripheral bus 6 which is the clock for access to the data flash memory, in MHz units. For example, 35.9 MHz should be rounded up and set the frequency to 36. Number of cycles required for access to the data flash memory are inserted according to the frequency.

When changing the frequency of the FCLK, follow the procedure below to modify the value of the data flash memory access frequency setting register (EEPFCLK) in either of the following ways according to whether operation is at a lower frequency before or after the change.

- When changing the speed from low to high: Modify EEPFCLK. After confirming the change by reading EEPFCLK, change the frequency.
- When changing the speed from high to low: Change the frequency. After the frequency is changed, modify EEPFCLK.

62.4.23 Unique ID Register n (UIDRn) (n = 0 to 3)

Address(es): FLASHCONST.UIDR0 FE7F 7D90h, FLASHCONST.UIDR1 FE7F 7D94h, FLASHCONST.UIDR2 FE7F 7D98h, FLASHCONST.UIDR3 FE7F 7D9Ch



Note: These registers are only readable when the SYSCR0.ROME bit is 1 (the on-chip ROM is enabled).

The UIDRn is a read-only register that stores a 16-byte ID code (unique ID) for identifying the individual MCU. The UIDRn register should be read in 32-bit units.

62.5 Operating Modes Associated with Flash Memory

Figure 62.6 is a diagram of the mode transitions associated with the flash memory. For the procedures for setting the modes, refer to section 3, Operating Modes.

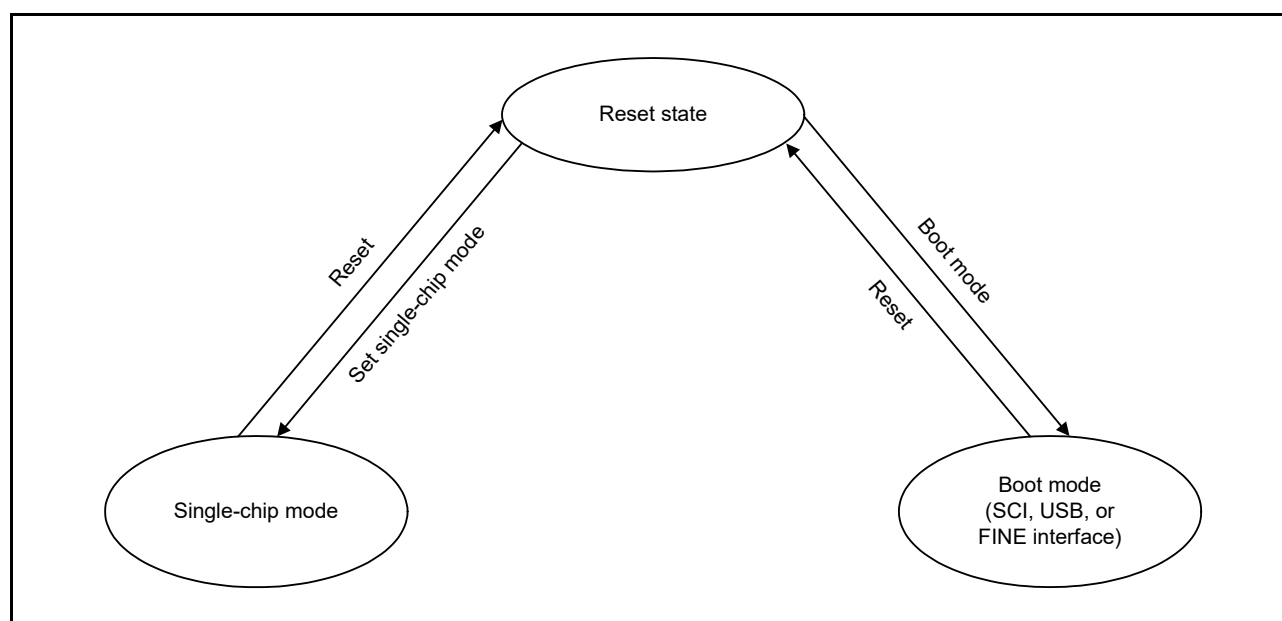


Figure 62.6 Mode Transitions Associated with Flash Memory

The flash memory area where programming and erasure are permitted and the boot program after a reset are different according to each mode. The differences between modes are listed in Table 62.5.

Table 62.5 Differences between Modes

Item	Single-chip Mode or On-chip ROM Enabled Extended Mode	Boot Mode (SCI, USB, or FINE Interface)
Programmable and erasable area	Code flash memory Data flash memory Option-setting memory (programming only)	Code flash memory Data flash memory Option-setting memory
Boot program at a reset	Program of the code flash memory	Boot program

62.6 Operating Modes of the Flash Sequencer

The flash sequencer has three operating modes as shown in Figure 62.7. Transitions between modes are initiated by changing the value of the FENTRYR register.

When the value of the FENTRYR register is 0000h, the flash sequencer is in read mode. In this mode, it does not receive FACI commands. The code flash memory and data flash memory are readable.

When the value of the FENTRYR register is 0001h, the flash sequencer is in code flash memory P/E mode where the code flash memory can be programmed or erased by FACI commands. In this mode, reading from the code flash memory is disabled under the conditions where BGO cannot be used. Under the conditions where BGO can be used, reading from the code flash memory is enabled.

When the value of the FENTRYR register is 0080h, the flash sequencer is in data flash memory P/E mode where the data flash memory can be programmed or erased by FACI commands. In this mode, the data flash memory is not readable. However, the code flash memory is readable.

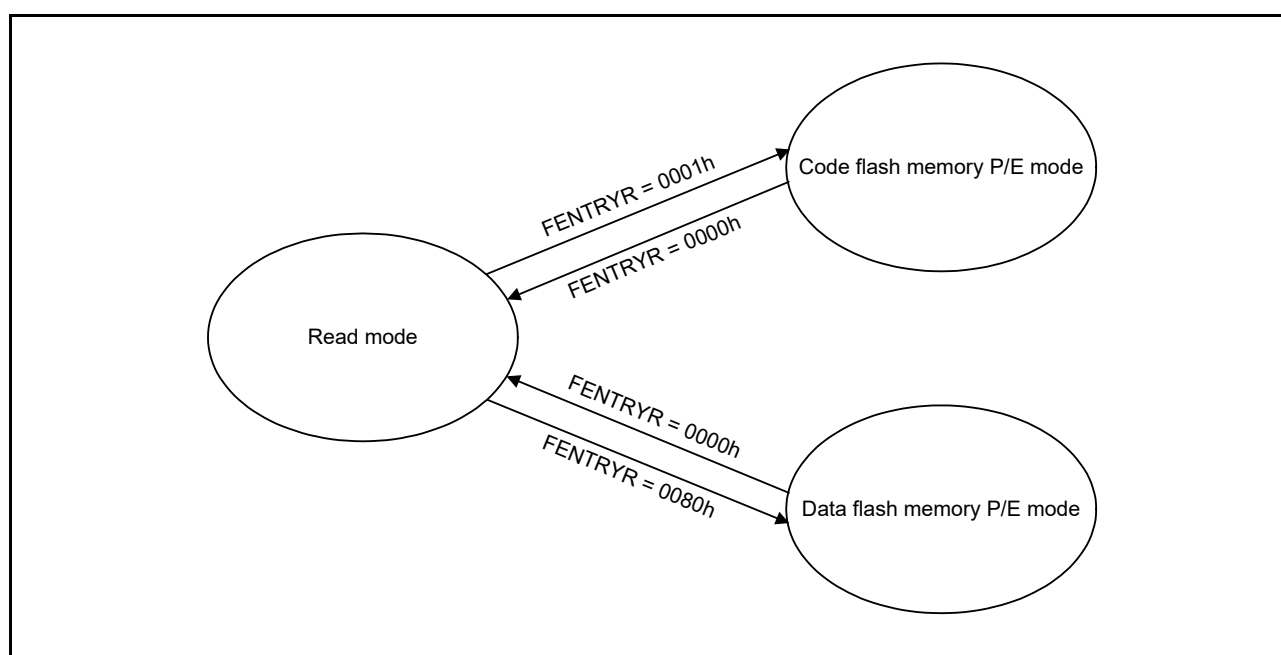


Figure 62.7 Modes of the Flash Sequencer

62.7 Overview of Functions

62.7.1 ROM Cache

This MCU has three types of cache as listed below.

- Instruction fetching cache (IF cache): 8 Kbytes
- Data cache (OA cache): 16 bytes
- Data cache (DM cache): 16 bytes

The IF cache and OA cache are for the CPU, and the DM cache is for bus masters other than the CPU, such as the DMAC and DTC.

After release from a reset, the caches are automatically invalidated following programming and erasure of the flash memory.

The area subject to caching is the 4-Mbyte address range from FFC0 0000h through FFFF FFFFh, in which up to two areas can be set as not within the scope of caching (non-cacheable areas).

62.7.1.1 Setting Non-Cacheable Area

Parts of the flash memory of this MCU that are within the cacheable areas can be set as not in the scope of caching by setting the NCRGn and NCRCn registers (n = 0, 1). Figure 62.8 shows the settings of each register and the non-cacheable areas.

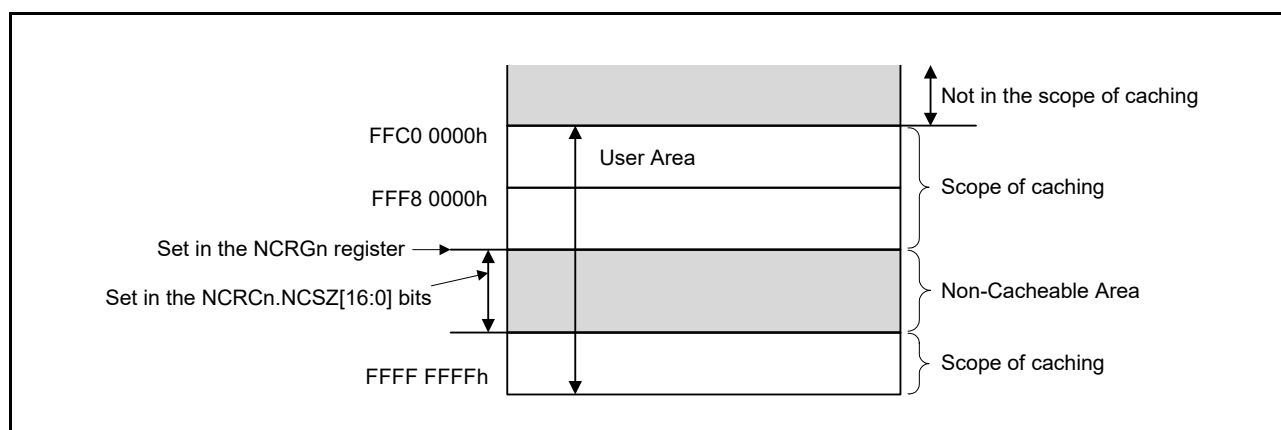


Figure 62.8 Specifying a Non-Cacheable Area (n = 0, 1)

The restrictions shown below in Table 62.6 apply to the settings of the NCRCn.NCSZ[16:0] bits and of the NCRGn register.

Table 62.6 Restrictions Applying to the Settings of the NCRCn.NCSZ[16:0] Bits and of the NCRGn Register

Settings of the NCRCn.NCSZ[16:0] Bits	Restrictions Applying to the Settings of the NCRGn Register
0 0000 0000 0000 0000: 16 bytes	No restriction
0 0000 0000 0000 0001: 32 bytes	Bit 4 in the NCRGn register is ignored.
0 0000 0000 0000 0011: 64 bytes	Bits 5 and 4 in the NCRGn register are ignored.
0 0000 0000 0000 0111: 128 bytes	Bits 6 to 4 in the NCRGn register are ignored.
0 0000 0000 0000 1111: 256 bytes	Bits 7 to 4 in the NCRGn register are ignored.
0 0000 0000 0001 1111: 512 bytes	Bits 8 to 4 in the NCRGn register are ignored.
0 0000 0000 0011 1111: 1 Kbyte	Bits 9 to 4 in the NCRGn register are ignored.
0 0000 0000 0111 1111: 2 Kbytes	Bits 10 to 4 in the NCRGn register are ignored.
0 0000 0000 1111 1111: 4 Kbytes	Bits 11 to 4 in the NCRGn register are ignored.
0 0000 0001 1111 1111: 8 Kbytes	Bits 12 to 4 in the NCRGn register are ignored.
0 0000 0011 1111 1111: 16 Kbytes	Bits 13 to 4 in the NCRGn register are ignored.
0 0000 0111 1111 1111: 32 Kbytes	Bits 14 to 4 in the NCRGn register are ignored.
0 0000 1111 1111 1111: 64 Kbytes	Bits 15 to 4 in the NCRGn register are ignored.
0 0001 1111 1111 1111: 128 Kbytes	Bits 16 to 4 in the NCRGn register are ignored.
0 0011 1111 1111 1111: 256 Kbytes	Bits 17 to 4 in the NCRGn register are ignored.
0 0111 1111 1111 1111: 512 Kbytes	Bits 18 to 4 in the NCRGn register are ignored.
0 1111 1111 1111 1111: 1 Mbyte	Bits 19 to 4 in the NCRGn register are ignored.
1 1111 1111 1111 1111: 2 Mbytes	Bits 20 to 4 in the NCRGn register are ignored.

62.7.2 Methods of Programming/Erase

By using a dedicated flash-memory programmer to program the flash memory of this MCU, the device can be rewritten regardless of whether this is before or after it is mounted on the target system.

Furthermore, security functions to prohibit rewriting or reading of the user program written to the flash memory are incorporated, and this can prevent falsification and illicit reading of the programs by third parties. TM target areas of the code flash memory can also be protected against reading by using the TM function.

Programming by the user program (self-programming) is available to suit applications where the application on the target system may require updating after manufacturing or shipment. Protection features for the safe rewriting of the flash memory are also incorporated. Furthermore, interrupt processing during self-programming is supported, so programming can proceed at the same time as processing for external communications, etc., and this is the case in various situations.

Table 62.7 lists the overview of the methods of programming and the corresponding operating modes.

Table 62.7 Methods of Programming

Method of Programming	Functional Overview	Operating Mode
Programming by a flash-memory programmer	A serial programmer is capable of on-board programming of the flash memory after the device is mounted on the target system. The TM function can be in enabled or disabled at this time.	Boot mode
	A parallel programmer and dedicated programming adapter board allow off-board programming of the flash memory, i.e. programming of the device before it is mounted on the target system.	
Self-programming	The execution of the user program that is written to code flash memory in advance by serial programming executing is also capable of programming the flash memory. The TM function can be in the enabled state at this time. The background operation capability makes it possible to fetch instructions or otherwise read data from code flash memory while the data flash memory is being programmed. Therefore, the data flash memory can be rewritten by executing a program for writing contained by the code flash memory. Furthermore, background operation can also be used for reading and writing to code flash memory alone, but only when the address range of code flash memory that is the target for programming and the address range of code flash memory that is the target for reading satisfy particular conditions (see Table 62.25). When this is the case, at the time of self-programming, a program for programming that is in the code flash memory can be executed to rewrite the code flash memory. In cases where background operation is not possible, instructions in the code flash memory cannot be fetched and data cannot be accessed while code flash memory is being re-written by self-programming. In such cases, a program for programming must be transferred to the internal RAM or external memory in advance and executed.	Single-chip mode On-chip ROM enabled extended mode

Table 62.8 lists the functions of the flash memory. Serial programmer commands realize each function of serial programming, while reading of the flash memory by an FACI command or the user program realizes each function of self-programming.

For security function settings, see section 7.2.1, Serial Programmer Command Control Register (SPCC), in section 7, Option-Setting Memory (OFSM).

Table 62.8 List of Basic Functions

Function	Functional Overview	Support Status	
		Serial Programming	Self-programming
Blank check	This is used to check a specified block to ensure that writing to it has not already proceeded. Results of reading from data flash memory to which nothing has been written after erasure are not guaranteed, so use blank checking to confirm that programming to memory has not proceeded after erasure.	Supported	Supported
Block erase	Erases the memory contents in the specified block.	Supported	Supported
Multi-block erase	Erases the contents of the specified block of the data flash memory.	Not supported	Supported
Program	Write to the specified address.	Supported	Supported
Verify/Checksum	Compares the data read from flash memory and the data transferred from a serial programmer.	Supported	Not supported (read by user program is possible)
Read	Read data programmed to flash memory.	Supported	Supported
Setting of control or ID codes	Sets the OSIS register.	Supported	Supported
Prohibition of serial programmer connection	Sets the permission/prohibition of serial programmer connection.	Supported	Supported with conditions (Only switching the configuration from enabled to disabled is possible)
Area protection and start-up program protection functions	Configures the area protection and start-up program protection functions.	Supported	Supported
Option function selection	Selects the option function, and modifies the initial settings of this MCU.	Supported	Supported
Configuration clear	Erases the option-setting memory and the TM target areas.	Supported	Not supported
Dual bank function	Switches different modes (linear or dual).	Supported	Supported
Setting the TM function	Sets the TM function.	Supported	Supported with conditions (Only switching the configuration from disabled to enabled is possible)

62.7.3 Security Functions

The flash memory supports various security functions.

The security function includes serial programmer ID code protection, prohibition of serial programmer connection, on-chip debugger ID code protection, and the ROM code protection.

In serial programming, serial programmer ID code protection and prohibition of serial programmer connection can be used. On-chip debugger ID code protection can be used when using the on-chip debugger. In off-board programming, ROM code protection can be used.

Table 62.9 lists the security functions supported by the flash memory.

Table 62.9 **Lists of Security Functions**

Function	Description
Serial programmer ID code protection	Connection of a serial programmer can be controlled by judging the control code or ID code.
Prohibition of serial programmer connection	Connection of a serial programmer for serial programming is prohibited. Since execution of the configuration clear command is also prohibited when the connection of a serial programmer is prohibited, changing a security setting from “disabled” to “enabled” is not possible.
On-chip debugger ID code protection	Connection to the on-chip debugger can be controlled by judging the ID code.
ROM code protection	This function is to prohibit reading, programming, and erasure of the flash memory when a parallel programmer is in use.

62.8 FACI Commands

62.8.1 List of FACI Commands

Table 62.10 List of FACI Commands

FACI Command	Description
Program	This is used to program the code or data flash memory. Units of programming are 128 bytes for the code flash memory and 4 bytes for the data flash memory.
Block erase	This is used to erase the code or data flash memory. The unit of erasure is one block. (code flash memory: 8 K or 32 Kbytes, data flash memory: 64 bytes)
Multi-block erase	This is used to erase the data flash memory. The unit of erasure is 64, 128, 256 bytes
P/E suspend	This suspends programming or erasure processing.
P/E resume	This resumes suspended programming or erasure processing.
Status clear	This initializes the ILGLERR, ERSERR, PRGERR, ILGCOMERR, FESETERR, SECERR, and OTERR flags in the FSTATR register and the CMDLK, CFAE, and DFAE flags in the FASTAT releases the flash sequencer from the command-locked state.
Forced stop	This forcibly stops processing of FACI commands and initializes the FASTAT and FSTATR registers.
Blank check	This is used to blank-check the data flash memory. Range of checking: 4 bytes to 64 Kbytes (specified in 4-byte units).
Configuration set	This is used to set the option-setting memory (configuration setting area). Units of setting: 16 bytes.

The FACI commands are issued by writing to the FACI command-issuing area (see Table 62.2). When write access as shown in Table 62.11 proceeds in the specified state, the flash sequencer executes the processing corresponding to the given command (see section 62.8.2, Relationship between the Flash Sequencer State and FACI Commands).

Table 62.11 FACI Command Formats

FACI Commands	Number of Write Access	Data to be Written to the FACI Command-Issuing Area			
		1st Access	2nd Access	3rd to (N+2)th Access	(N+3)th Access
Program (code flash memory) 128-byte programming, N = 64	67	E8h	40h (= N)	WD ₁ to WD ₆₄	D0h
Program (data flash memory) 4-byte programming, N = 2	5	E8h	02h (= N)	WD ₁ to WD ₂	D0h
Block erase (code flash memory)	2	20h	D0h	—	—
Block erase (data flash memory 64-byte)	2	20h	D0h	—	—
Multi-block erase (data flash memory 64-byte/128-byte/256-byte)	2	21h	D0h	—	—
P/E suspend	1	B0h	—	—	—
P/E resume	1	D0h	—	—	—
Status clear	1	50h	—	—	—
Forced stop	1	B3h	—	—	—
Blank check	2	71h	D0h	—	—
Configuration set N = 8	11	40h	08h (= N)	WD ₁ to WD ₈	D0h

Note: WD_N (N = 1, 2,...): Nth 16-bit data to be programmed.

The flash sequencer clears the FSTATR.FRDY flag to 0 at the start of processing of a command other than the status clear command and sets this bit to 1 upon completion of command processing.

When the setting of the FRDYIE.FRDYIE bit is 1 and when the FSTATR.FRDY flag is set to 1, a flash ready (FRDY) interrupt is generated.

62.8.2 Relationship between the Flash Sequencer State and FSCI Commands

Each FSCI command can be accepted in a specific mode or state of the flash sequencer. FSCI commands should be issued after the transition of the flash sequencer to the code flash memory P/E mode or data flash memory P/E mode and checking of the state of the flash sequencer. Use the FSTATR and FSTAT registers to check the state of the flash sequencer. The value of the FSTAT.CMDLK flag is the logical OR of values of the ILGLERR, ILGCOMERR, FESETERR, SECERR, OTERR, ERSERR, PRGERR, and FLWEERR flags in the FSTATR register. Therefore, the occurrence of errors can be checked by reading the value of the FSTAT.CMDLK flag.

Table 62.12 lists the available commands in each operating mode.

Table 62.12 Operating Mode and Available Commands

Operating Mode	FENTRYR Register Value	Available Commands
Read mode	0000h	None
Code flash memory P/E mode	0001h	Program Block erase P/E suspend P/E resume Status clear Forced stop Configuration set
Data flash memory P/E mode	0080h	Program Block erase Multi-block erase P/E suspend P/E resume Status clear Forced stop Blank check

Table 62.13 shows the state of the flash sequencer and acceptable FSCI commands. An appropriate mode is assumed to be set before the commands are executed.

Table 62.13 Acceptable FSCI Commands and the State of the Flash Sequencer

	Processing of Programming or Erasure	Processing of Configuration Setting	Processing to Suspend Programming or Erasure	Processing of Blank Checking	Programming Suspended	Erasure Suspended	Programming while Erasure is Suspended	Command-Locked State (FRDY = 1)	Command-Locked State (FRDY = 0)	Processing of Forced Stop Command	Other State
FRDY flag	0	0	0	0	1	1	0	1	0	0	1
SUSRDY flag	1	0	0	0	0	0	0	0	0	0	0
ERSSPD flag	0	0	0/1	0/1	0	1	1	0/1	0/1	0	0
PRGSPD flag	0	0	0/1	0/1	1	0	0	0/1	0/1	0	0
CMDLK flag	0	0	0	0	0	0	0	1	1	0	0
Program	x	x*4	x	x	x	✓*3	x	x	x	x	✓
Block erase	x	x*4	x	x	x	x	x	x	x	x	✓
Multi-block erase	x	x*4	x	x	x	x	x	x	x	x	✓
P/E suspend	✓	x*4	x	x	x	x	x	—	x	x	—
P/E resume	x	x*4	x	x	✓	✓	x	x	x	x	x
Status clear	x	x*4	x	x	✓	✓	x	✓	x	x	✓
Forced stop	✓	✓*4	✓	✓	✓	✓	✓	✓	✓	✓	✓
Blank check	x	x	x	x	✓*1	✓*1	x	x	x	x	✓*1
Configuration set	x	x*4	x	x	x	x	x	x	x	x	✓*2

✓: Acceptable

x: Not acceptable (the sequencer in the command-locked state)

—: Ignored

Note 1. Acceptable only in data flash memory P/E mode

Note 2. Acceptable only in code flash memory P/E mode

Note 3. Program command can be issued to a block other than the erase-suspended block.

Note 4. Do not issue a FSCI command when configuration setting is being processed and the FSTATR.DBFULL bit is 1.

62.8.3 Usage of FACL Commands

This section gives an overview of the usage of FACL commands.

62.8.3.1 Transition to Code Flash Memory P/E Mode

To use the FACL commands for the code flash memory, a transition to code flash memory P/E mode is required. To cause shift to code flash memory P/E mode, set the FENTRYR.FENTRYC bit to 1.

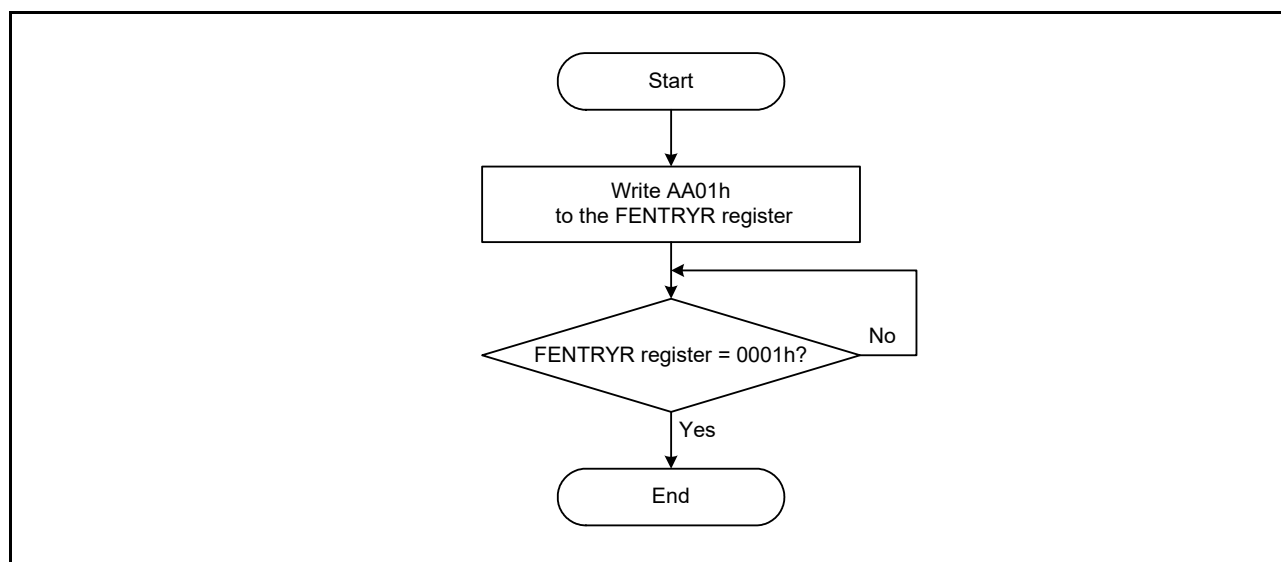


Figure 62.9 Procedure for Transition to Code Flash Memory P/E Mode

62.8.3.2 Transition to Data Flash Memory P/E Mode

To use the FACL commands for the data flash memory, a transition to data flash memory P/E mode is required. To shift to data flash memory P/E mode, set the FENTRYR.FENTRYRD bit to 1.

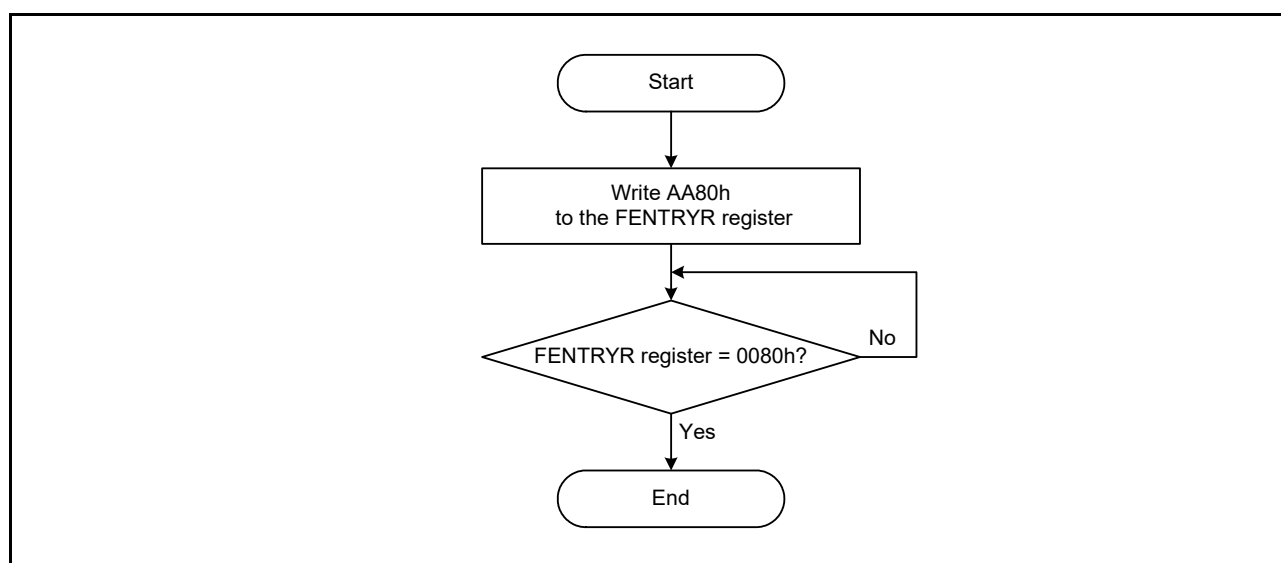


Figure 62.10 Procedure for Transition to Data Flash Memory P/E Mode

62.8.3.3 Transition to Read Mode

To read the flash memory without using the BGO function, a transition to read mode is required. To shift to read mode, set the FENTRYR register to 0000h. The transition to read mode should be made after processing by the flash sequencer is completed and while operation is in other than in the command-locked state. In addition, operation is started in read mode after release from the reset state.

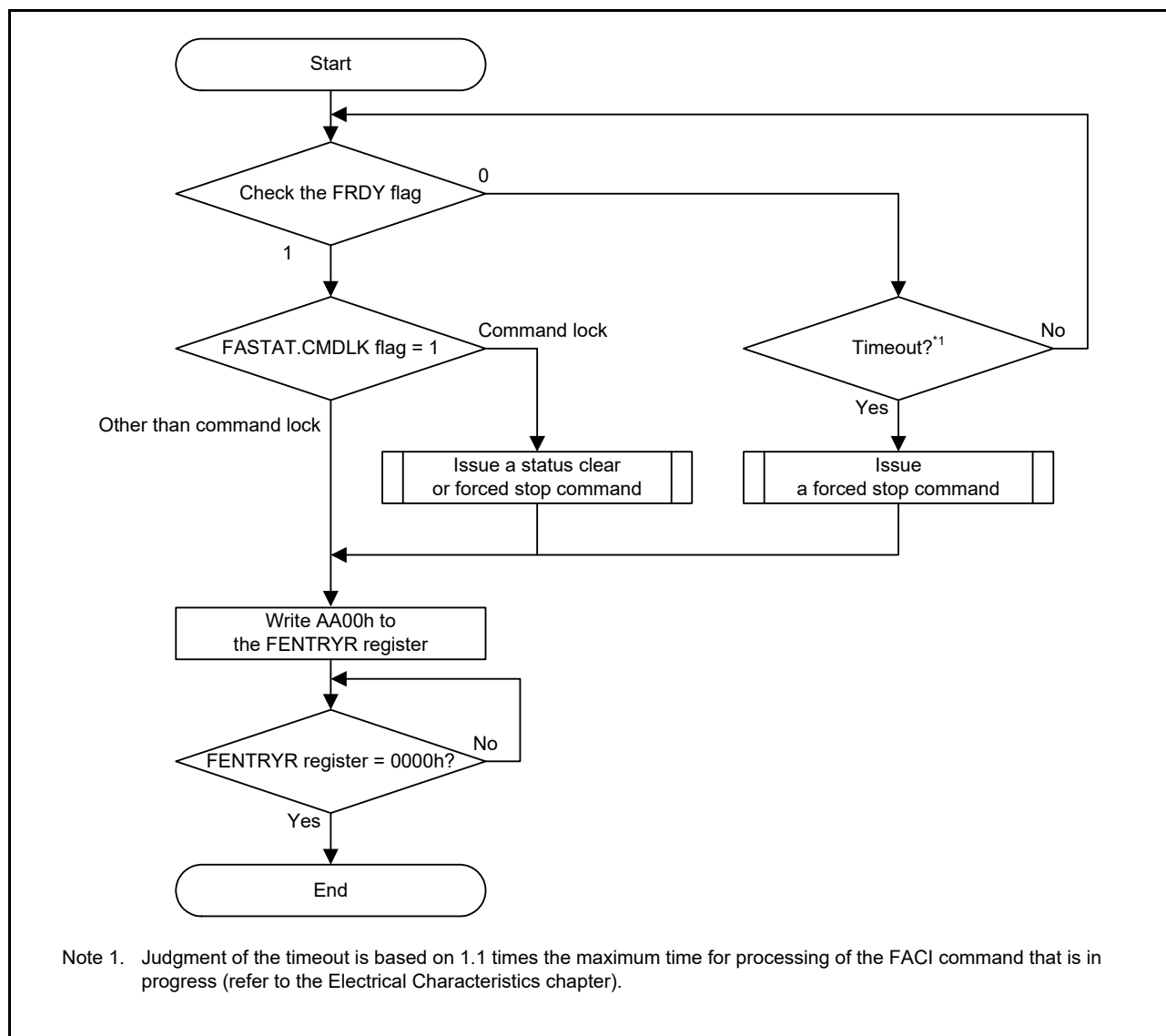


Figure 62.11 Procedure for Transition to Read Mode

62.8.3.4 Overview Flow when FACL Command is Used

Figure 62.12 shows an overview flow when the FACL command is used.

If BGO is enabled, the jump to the internal RAM or external area (other than code flash memory) is not required because an FACL command can be issued for the code or data flash memory by using the rewriting program in the code flash memory.

When the FCLK is changed, changing the FPCKAR register shortens time for processing the FACL command. For details, refer to section 62.4.19, Flash Sequencer Processing Clock Frequency Notification Register (FPCKAR).

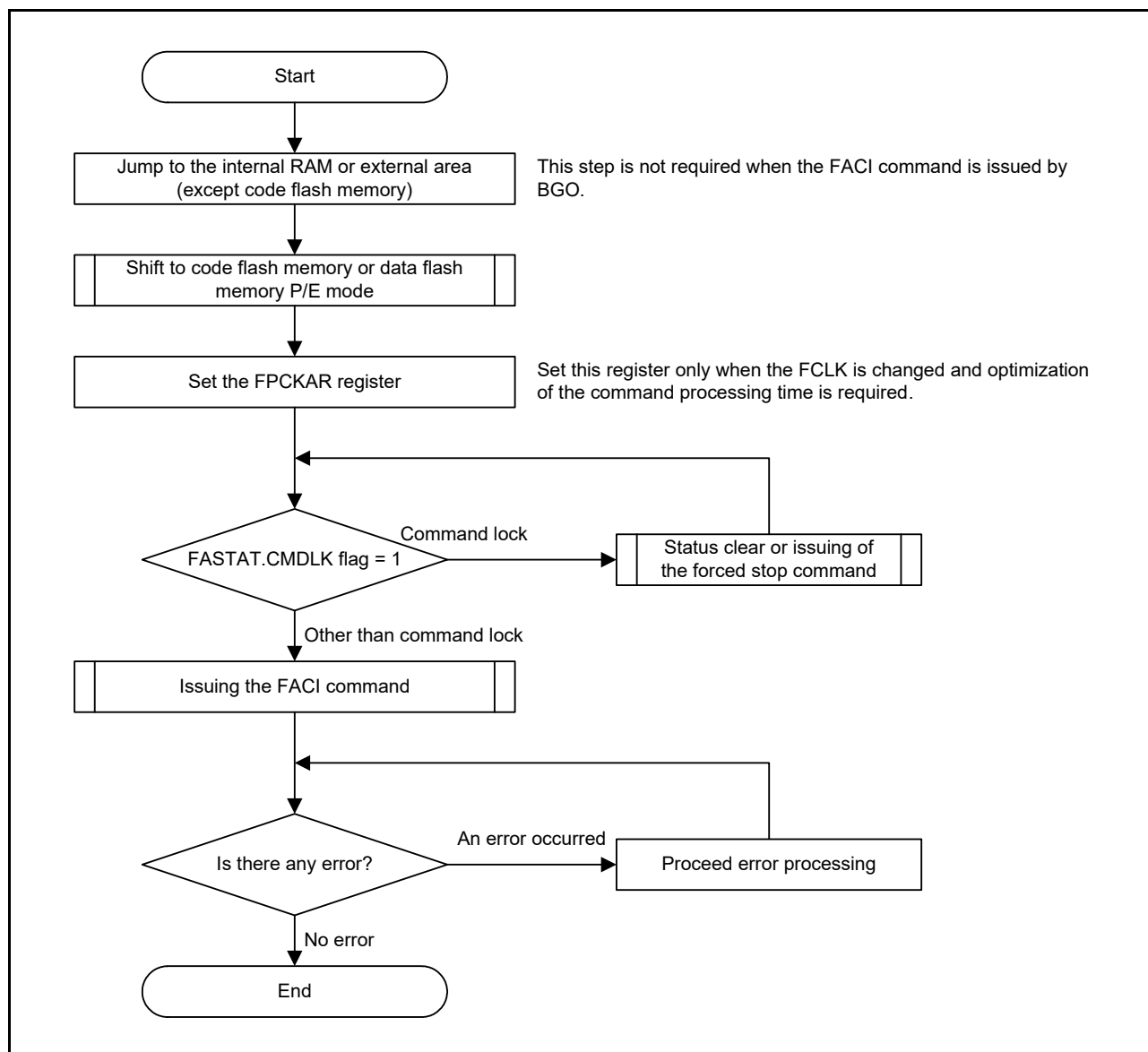


Figure 62.12 Overview Flow when FACL Command is Used

62.8.3.5 Recovery from the Command-Locked State

When the flash sequencer enters the command-locked state, FACL commands cannot be accepted. To release the sequencer from the command-locked state, use the status clear command, or forced stop command.

When the command-locked state is detected by checking for an error before issuing the P/E suspend command, the FSTATR.FRDY flag may hold 0 as the command processing has not been completed. When the processing is not completed even after time equal to the maximum program or erase time specified in the section 63, Electrical Characteristics times 1.1 has elapsed, this is considered a time-out and the flash sequencer should be stopped by the forced stop command.

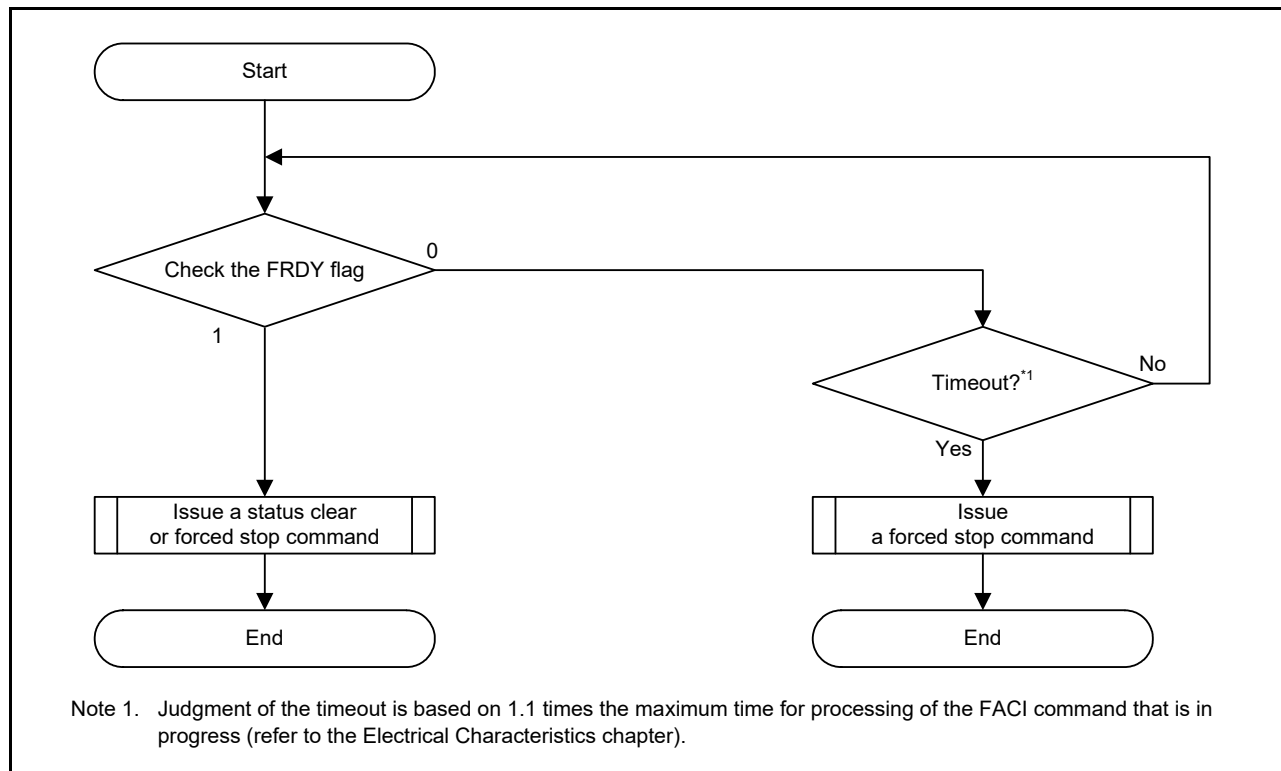


Figure 62.13 Recovery from the Command-Locked State

62.8.3.6 Program Command

A program command is used for programming the code or data flash memory.

Before issuing a program command, set the first address of the target block in the FSADDR register.

Writing the final data (WD₆₄ or WD₂) to the FSCI command-issuing area when the FSCI command is issued starts the program command processing. To prevent program malfunction, make sure that interrupts are not accepted between writing the final data and writing D0h. Completion of command processing can be checked by reading the FSTATR.FRDY flag. If the target area of program command processing contains the area not for writing, write FFFFh to the corresponding area.

Issuing a program command consecutively while the FSCI internal data buffer is full leads to a wait on the peripheral bus 6 and this may affect on the bus accesses of the other peripheral IP modules. To avoid the generation of such a wait, issue an FSCI command while the FSTATR.DBFULL flag is 0.

In addition, the data buffer never becomes full during programming of the data flash memory.

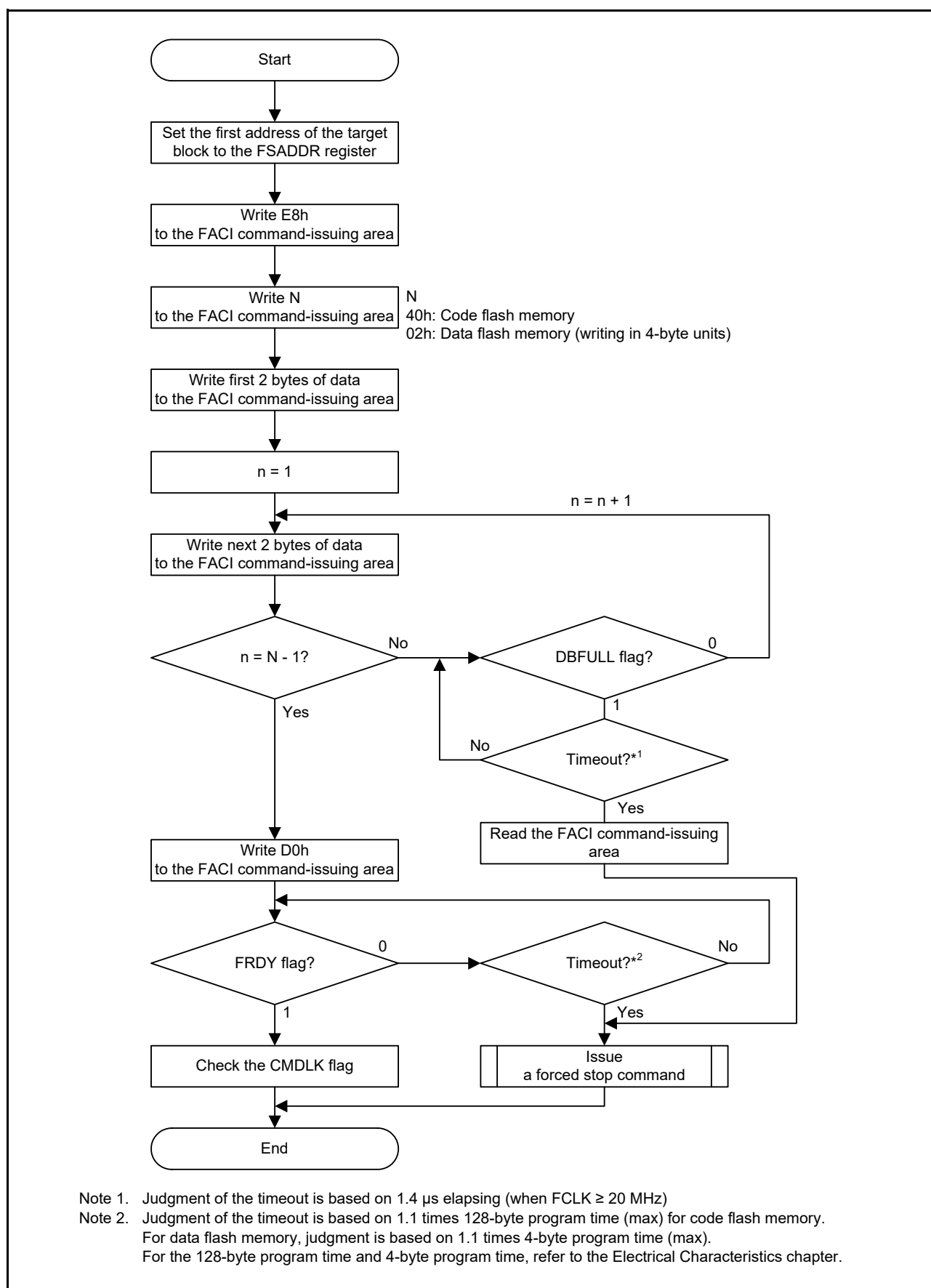


Figure 62.14 Usage of the Program Command

62.8.3.7 Block Erase Command

A block erase command is used to erase the code or data flash memory in single-block units.

Before issuing a block erase command, set the first address of the target block in the FSADDR register. Writing 20h and D0h to the FACL command-issuing area starts processing of a block erase command. Completion of command processing can be confirmed by reading the FSTAT.FRDY flag.

The FCPSR register must be set before issuing the block erase command. To change the erase suspend mode (suspend priority mode/erase priority mode) when the erase process is interrupted by the P/E suspend command, it is necessary to change the FCPSR register.

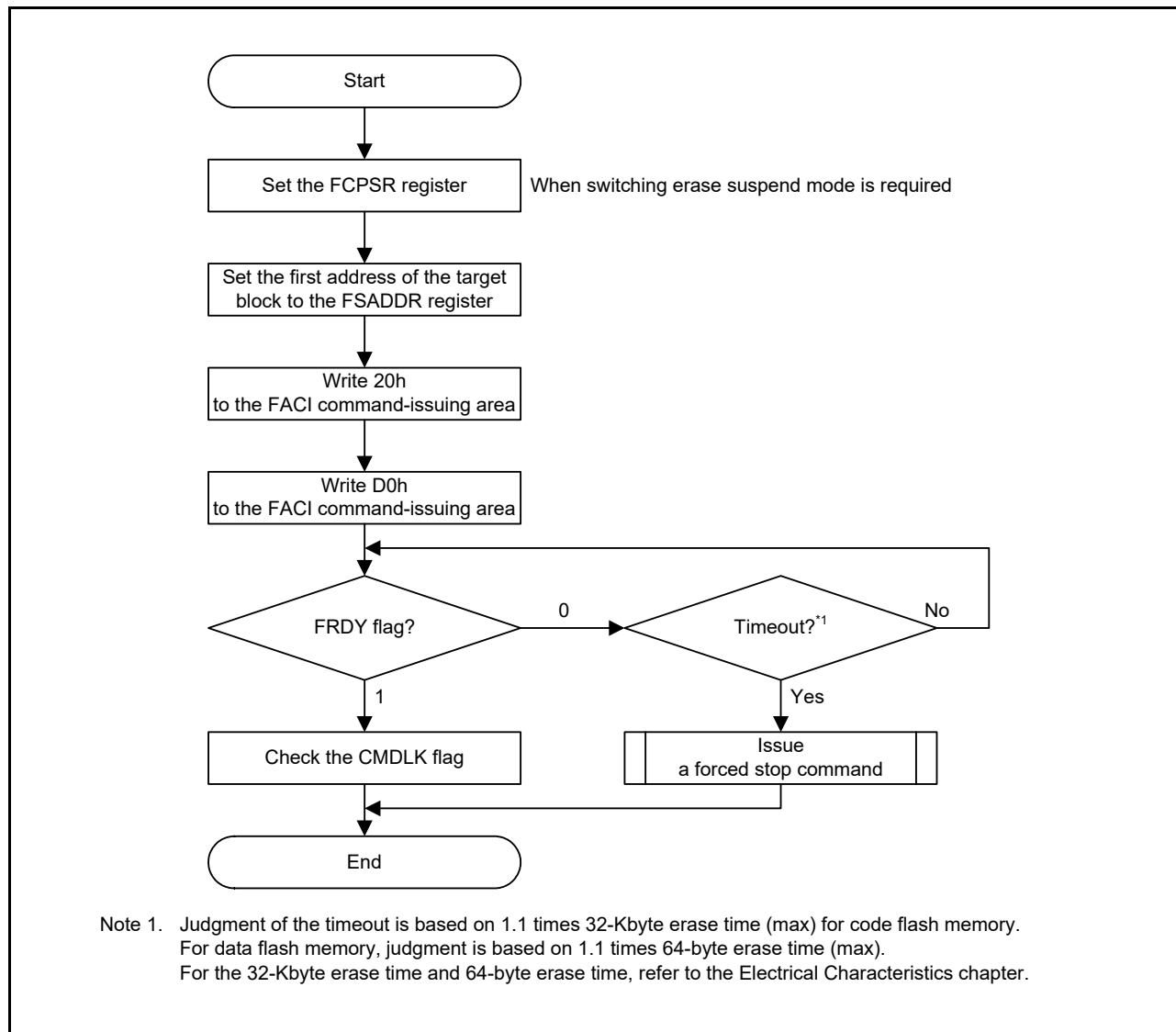


Figure 62.15 Usage of the Block Erase Command

62.8.3.8 Multi-Block Erase Command

The multi-block erase command can also be issued to erase data in the data flash memory. The unit for erasure is 64, 128, or 256 bytes.

Set the first address of the target area for erasure to the FSADDR register and the end address to the FEADDR register before issuing a multi-block erase command. When 21h and D0h are written to the FACL command issuing area, the process of the multi-block erase command proceeds. The end of command processing can be confirmed by the FSTATR.FRDY flag.

The FCPSR register must be set before issuing the multi-block erase command. To change the erase suspend mode (suspend priority mode/erase priority mode) when the erase process is interrupted by the P/E suspend command, it is necessary to change the FCPSR register.

Table 62.14 Size Setting for Erasure

Erasure Size	FSADDR	FEADDR
64 bytes	FSA0 to FSA5 = 0 (64-byte boundary)	FSADDR + 3Ch
128 bytes	FSA0 to FSA6 = 0 (128-byte boundary)	FSADDR + 7Ch
256 bytes	FSA0 to FSA7 = 0 (256-byte boundary)	FSADDR + FCh

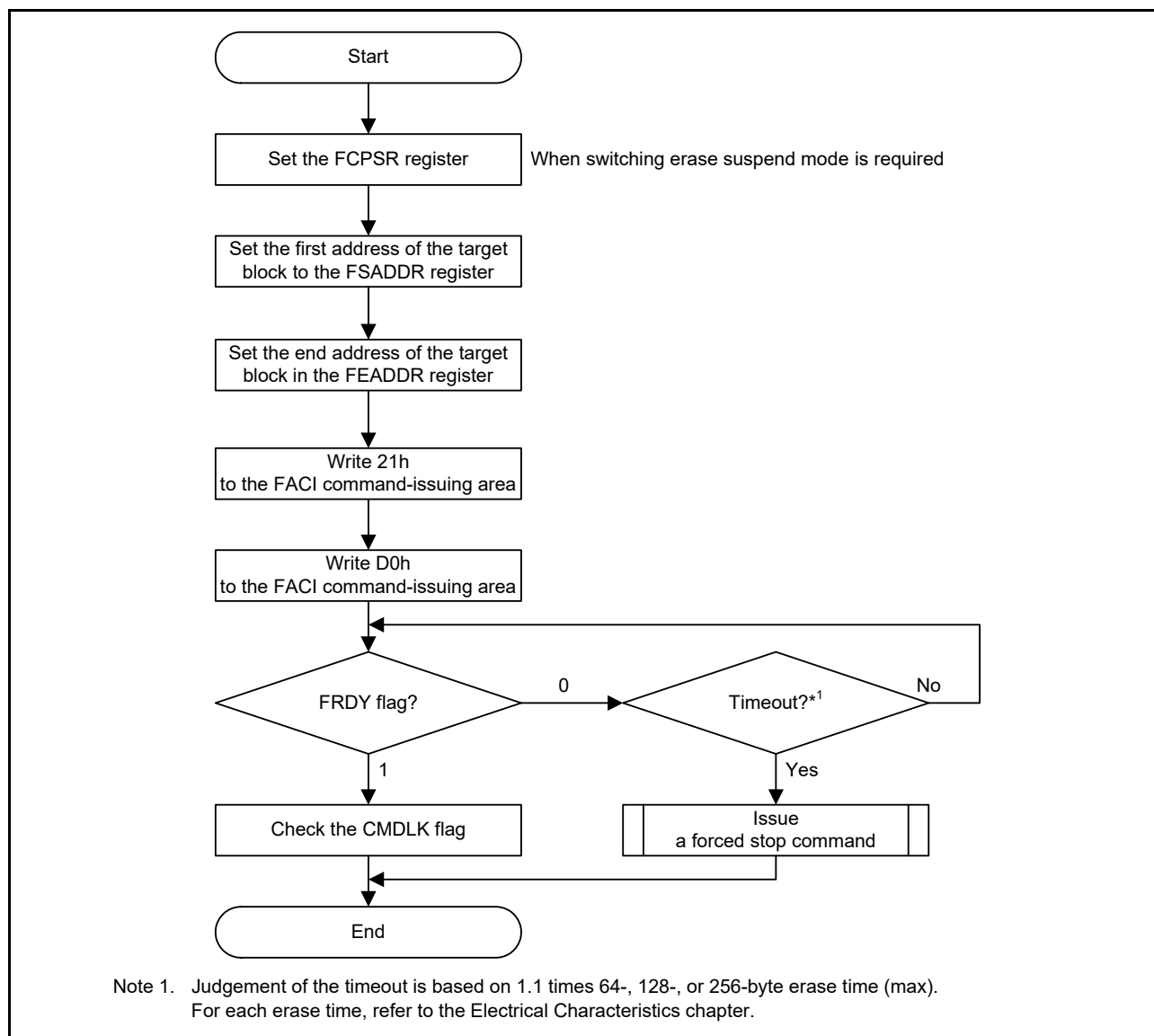


Figure 62.16 Usage of the Multi-Block Erase Command

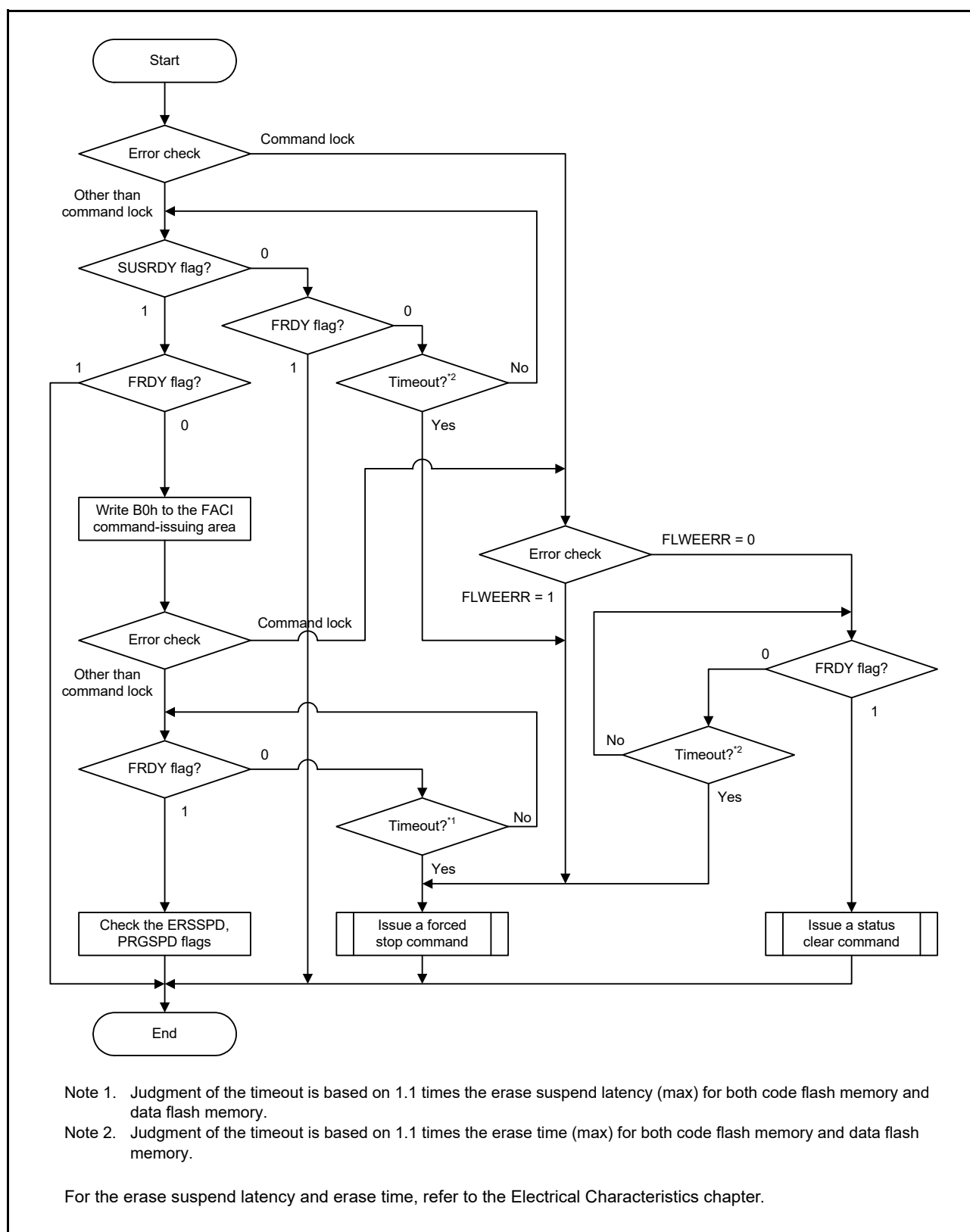
62.8.3.9 P/E Suspend Command

The P/E suspend command is used to suspend programming or erasure. Before issuing a P/E suspend command, check that the FASTAT.CMDLK flag is 0, and the execution of programming/erasure is normally performed. To confirm that the P/E suspend command can be received, also check that the FSTATR.SUSRDY flag is 1. After issuing a P/E suspend command, read the FASTAT.CMDLK flag to confirm that its value is not 1 (the flash sequencer is not in the command-locked state).

If an error occurs during programming or erasure processing, the FASTAT.CMDLK flag is set to 1. When P/E processing is completed between the FSTATR.SUSRDY flag having been confirmed to be 1 and acceptance of the P/E suspend command, the P/E suspend command is ignored and the flash sequencer does not enter the suspended state (the FSTATR.FR DY flag is 1 and the ERSSPD and PRGSPD flags in the FSTATR register are 0).

When a P/E suspend command is received and then the programming/erasure suspend processing finishes normally, the flash sequencer enters the suspended state, the FSTATR.FR DY flag is set to 1, and the ERSSPD or PRGSPD flag in the FSTATR register is 1. After issuing a P/E suspend command, check that the ERSSPD or PRGSPD flag in the FSTATR register is 1 and the suspended state is entered, and then decide the subsequent flow. If a P/E resume command is issued in the subsequent flow although the suspended state is not entered, an illegal command error occurs and the flash sequencer shifts to the command-locked state (see section 62.10.2, Error Protection).

If the erase-suspended state is entered, programming to blocks other than the one currently suspended can be performed. In addition, the program- and erase-suspended states can move to read mode by clearing the FENTRYR register.



(1) Suspension during Programming

When issuing a P/E suspend command during the flash memory programming, the flash sequencer suspends programming processing. Figure 62.18 shows the suspend operation of programming. When receiving a program or P/E resume command, the flash sequencer sets the FSTATR.FRDY flag to 0 to start programming. When the flash sequencer is ready to accept the P/E suspend command after starting programming, it sets the FSTATR.SUSRDY flag to 1. When a P/E suspend command is issued, the flash sequencer receives the command and sets the FSTATR.SUSRDY flag to 0. When the flash sequencer receives a P/E suspend command while a program pulse is being applied, the flash sequencer continues applying the pulse. After the specified pulse application time, the flash sequencer finishes pulse application, and starts the programming suspend processing and sets the FSTATR.PRGSPPD flag to 1.

When the suspend processing finishes, the flash sequencer sets the FSTATR.FRDY flag to 1 to enter the program-suspended state. When receiving a P/E resume command in the program-suspended state, the flash sequencer sets the FSTATR.FRDY and FSTATR.PRGSPPD flags to 0 and resumes programming.

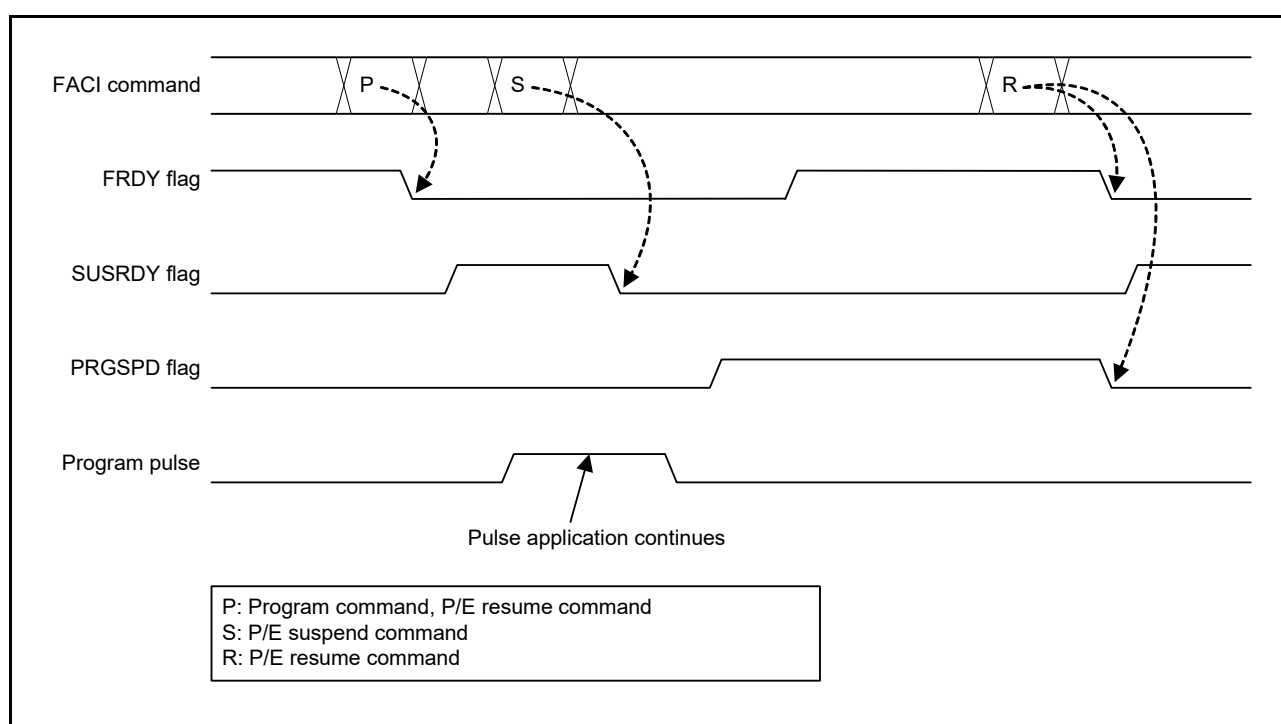


Figure 62.18 Suspension during Programming

(2) Suspension during Erasure (Suspend Priority Mode)

This MCU has a suspend priority mode for the suspension of erasure. Figure 62.19 shows the suspend operation of erasure in suspend priority mode (the FCPSR.ESUSPMD bit is 0).

When receiving a block erase, multi-block erase, or P/E resume command, the flash sequencer sets the FSTATR.FRDY flag to 0 to start erasure. When the flash sequencer is ready to accept the P/E suspend command after starting erasure, it sets the FSTATR.SUSRDY flag to 1. When a P/E suspend command is issued, the flash sequencer receives the command and sets the FSTATR.SUSRDY flag to 0. When receiving a P/E suspend command during erasure, the flash sequencer starts the suspend processing and sets the FSTATR.ERSSPD flag to 1 even if it is applying an erase pulse. When the suspend processing is completed, the flash sequencer sets the FSTATR.FRDY flag to 1 to enter the erase-suspended state. When receiving a P/E resume command in the erase-suspended state, the flash sequencer sets the FRDY and ERSSPD flags in the FSTATR register to 0 and resumes erasure. Operations of the FRDY, SUSRDY, and ERSSPD flags in the FSTATR register at the suspension and resumption of erasure are the same, regardless of the erase suspend mode.

The setting of the erase suspend mode affects the control method of erase pulses. In suspend priority mode, when receiving a P/E suspend command while erase pulse A that has never been interrupted in the past is being applied, the flash sequencer suspends the application of erase pulse A and enters the erase-suspended state. When receiving a P/E suspend command while reapplying erase pulse A after erasure is resumed by a P/E resume command, the flash sequencer continues applying erase pulse A. After the specified pulse application time, the flash sequencer finishes erase pulse application and enters the erase-suspended state. When the flash sequencer receives a P/E resume command next and erase pulse B starts to be newly applied, and then the flash sequencer receives a P/E suspend command again, the application of erase pulse B is interrupted. In suspend priority mode, delay due to suspension can be minimized because the application of an erase pulse is interrupted once per pulse and priority is given to the suspend processing.

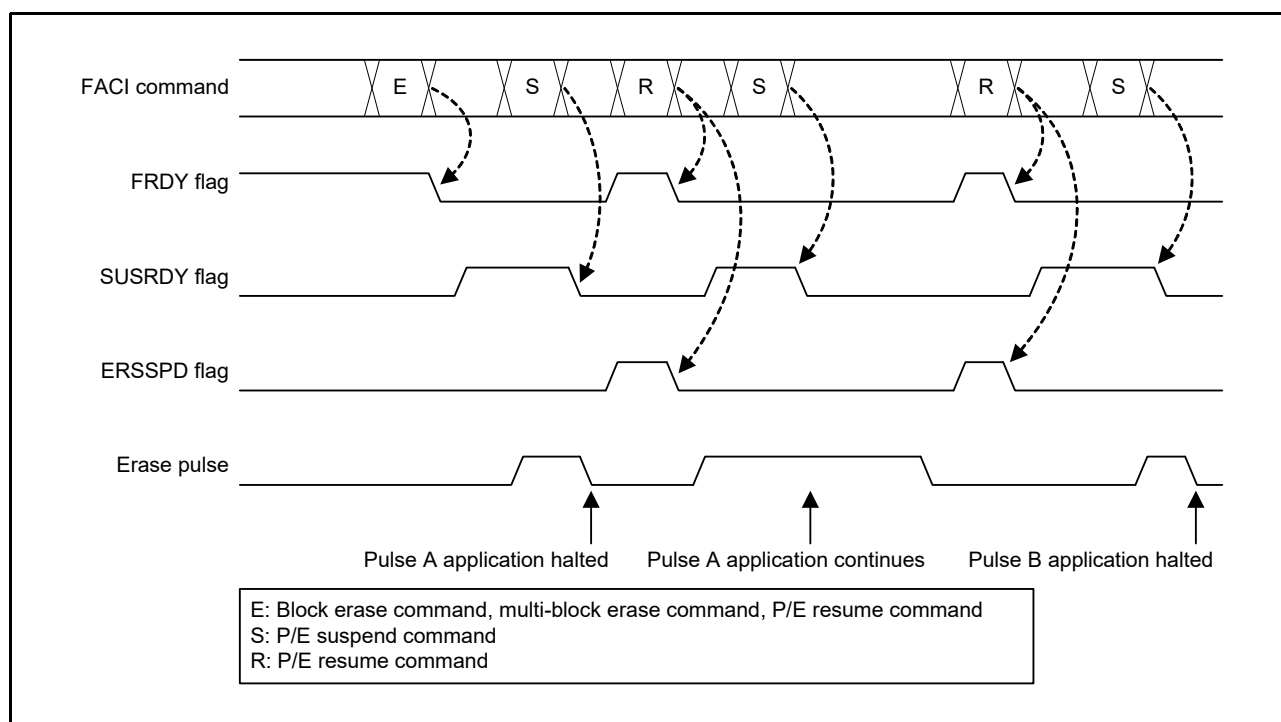


Figure 62.19 Suspension during Erasure (Suspend Priority Mode)

(3) Suspension during Erasure (Erase Priority Mode)

This MCU has an erase priority mode for the suspension of erasure.

Figure 62.20 shows the suspend operation of erasure when the erase suspend mode is set to the erase priority mode (the FCPSR.ESUSPMD bit is 1). The control method of erase pulses in erase priority mode is the same as that of program pulses for the program suspend processing.

When the flash sequencer receives a P/E suspend command while an erase pulse is being applied, the flash sequencer definitely continues applying the pulse. In this mode, the required time for the whole erasure processing can be reduced as compared with the suspend priority mode because the reapplication of erase pulses does not occur when a P/E resume command is issued.

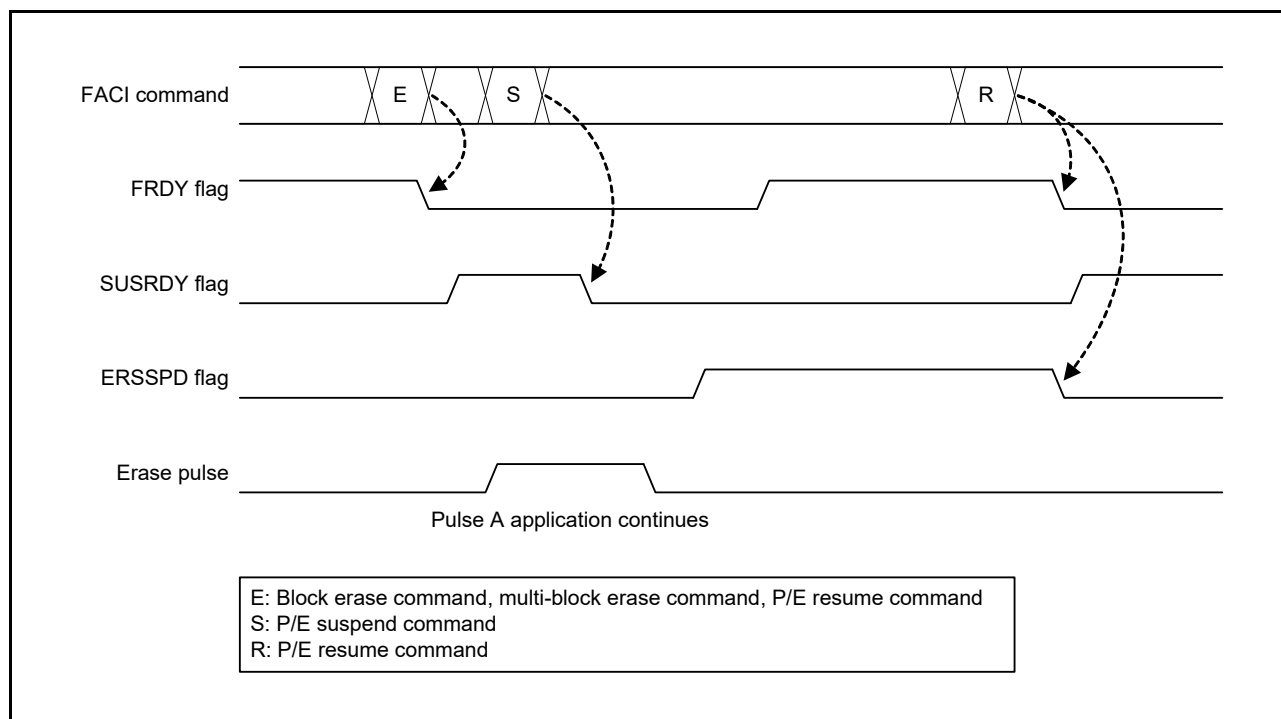


Figure 62.20 Suspension during Erasure (Erase Priority Mode)

62.8.3.10 P/E Resume Command

To resume suspended programming or erasure, use the P/E resume command. When the settings of the FENTRYR register are changed during suspend, reset the setting of the FENTRYR register to the value immediately before the P/E suspend command was issued, and then issue a P/E resume command. Completion of processing of the resumed command can be confirmed by reading the FSTATR.FRDY flag.

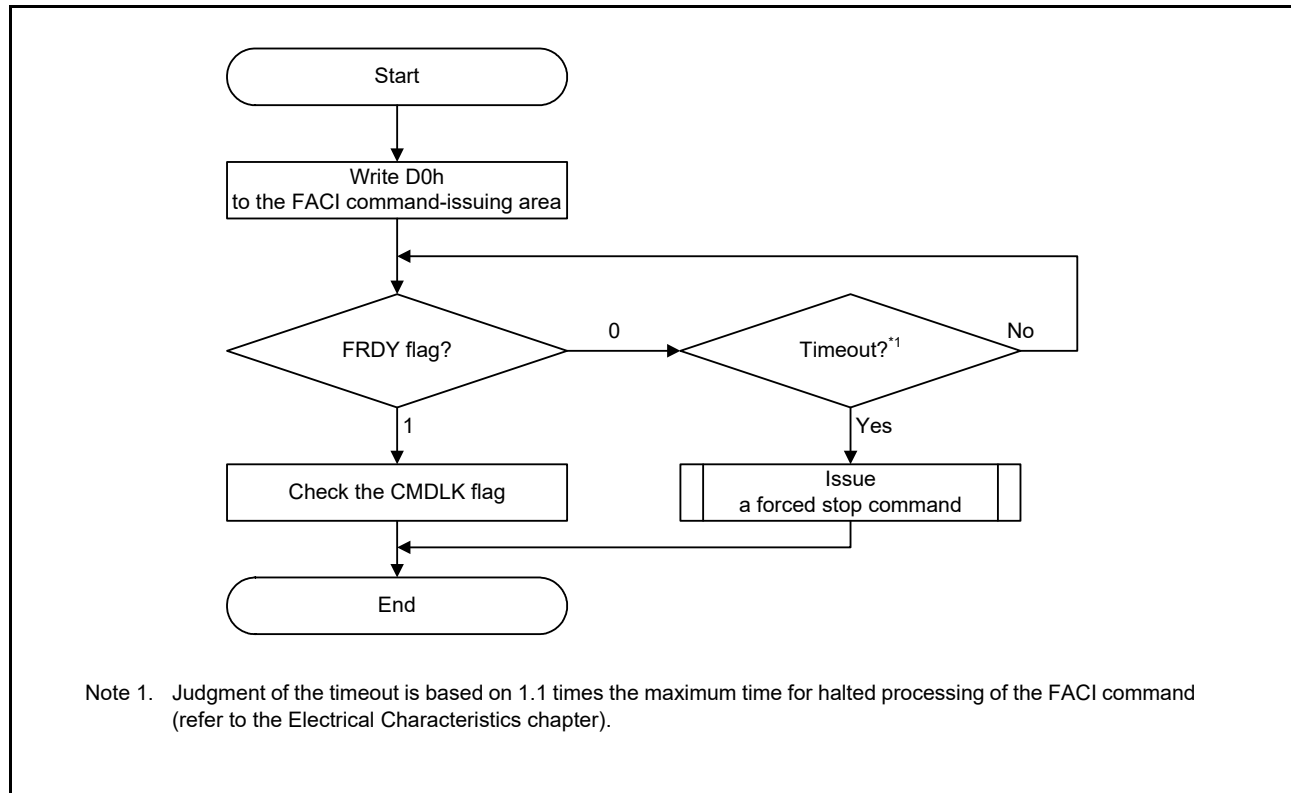


Figure 62.21 Usage of the P/E Resume Command

62.8.3.11 Status Clear Command

When one of the ILGLERR, ILGCOMERR, FESETERR, SECERR, OTERR, ERSERR, PRGERR, and FLWEERR flag bits in the FSTATR register is set to 1, the flash sequencer enters the command-locked state. When one of the CFAE and DFAE flags in the FASTAT register is set to 1, the flash sequencer also enters the command lock state. In the command-locked state, the flash sequencer can accept only status clear command or forced stop command.

The status clear command is used to clear the command-locked state (see section 62.8.3.5, **Recovery from the Command-Locked State**). To clear the CFAE, DFAE, and CMDLK flags in the FASTAT register, and the ILGLERR, ILGCOMERR, FESETERR, SECERR, OTEERR, ERSERR, and PRGERR flags in the FSTATR register in the command-locked state, the status clear command is available.

The FLWEERR flag cannot only be cleared by the status clear command, but can only be cleared by the forced stop command.

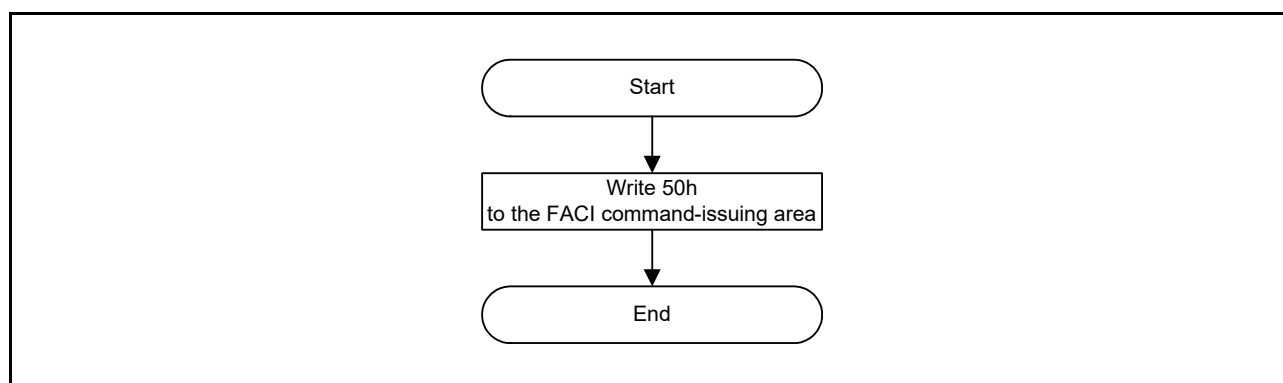


Figure 62.22 Usage of the Status Clear Command

62.8.3.12 Forced Stop Command

The forced stop command forcibly terminates command processing by the flash sequencer. Although this command halts command processing faster than the P/E suspend command, values of the area being programmed or erased are not guaranteed. Furthermore, resumption of processing is not possible. Processing of programming or erasure terminated by the forced stop command is also defined as one program/erase cycle.

Executing a forced stop command also initializes the entire FCU and part of the FACL, and the FASTAT and FSTATR registers. Accordingly, this command can be used in the procedure for recovery from the command-locked state and in processing in response to a time-out of the flash sequencer (see section 62.8.3.5, Recovery from the Command-Locked State).

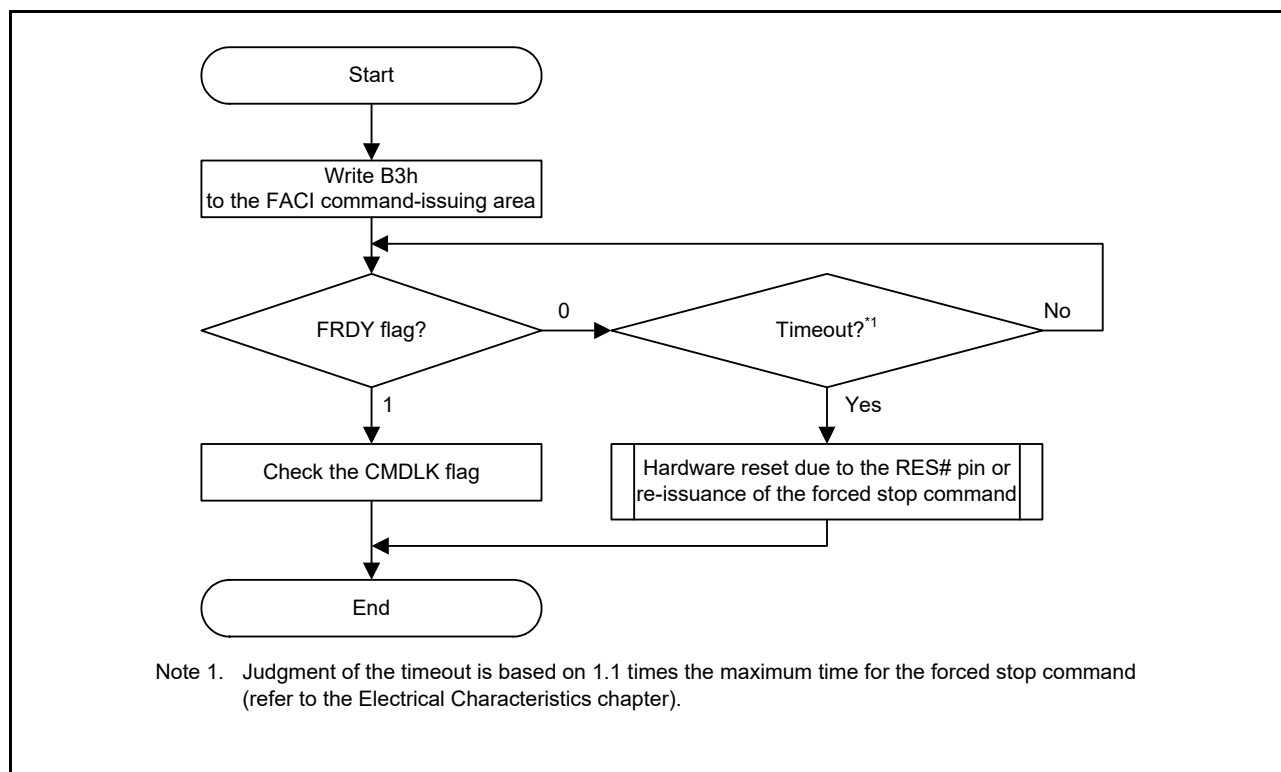


Figure 62.23 Usage of the Forced Stop Command

How to Use the Forced Stop Command when a Command is being issued

When processing is terminated by the forced stop command while a timeout is generated based on the DBFULL bit judgment of the program command, writing to the FACL command issuing area may be handled as the data written by the program command. In this case, read the FACL command issuing area and generate a command lock intentionally, then issue the forced stop command according to the method of returning from the command lock state. A command lock can be generated although the access size for reading the FACL command issuing area is in 8-, 16-, or 32-bit units.

62.8.3.13 Blank Check Command

Values read from data flash memory that has been erased but not yet been programming again (i.e. that is in the non-programmed state) are undefined. Use the blank check command when you need to confirm that an area is in the non-programmed state.

Before issuing a blank check command, set addressing mode, start and end addresses of the target area for blank checking to the FBCCNT, FSADDR, and FEADDR registers.

When the FBCCNT.BCDIR bit is 1, the value specified in the FSADDR register must be set at least the value specified in the FEADDR register.

When the FBCCNT.BCDIR bit is 0, the value specified in the FSADDR register must be the value specified in the FEADDR register or less.

When the settings of the FBCCNT.BCDIR bit, FSADDR register, and FEADDR register are inconsistent, the flash sequencer enters the command-locked state. The size of the target area for blank checking is in the range from 4 bytes to 64 Kbytes and is set in units of 4 bytes.

Write 71h and D0h to the FACL command-issuing area to start blank checking. Completion of processing can be confirmed by the FSTATR.FRDY flag. At the end of processing, the result of blank checking is stored in the FBCSTAT.BCST flag. If the target area for blank checking includes areas where programming has been completed, the flash sequencer stores the address of the programmed data that it first detected in the FPSADDR register.

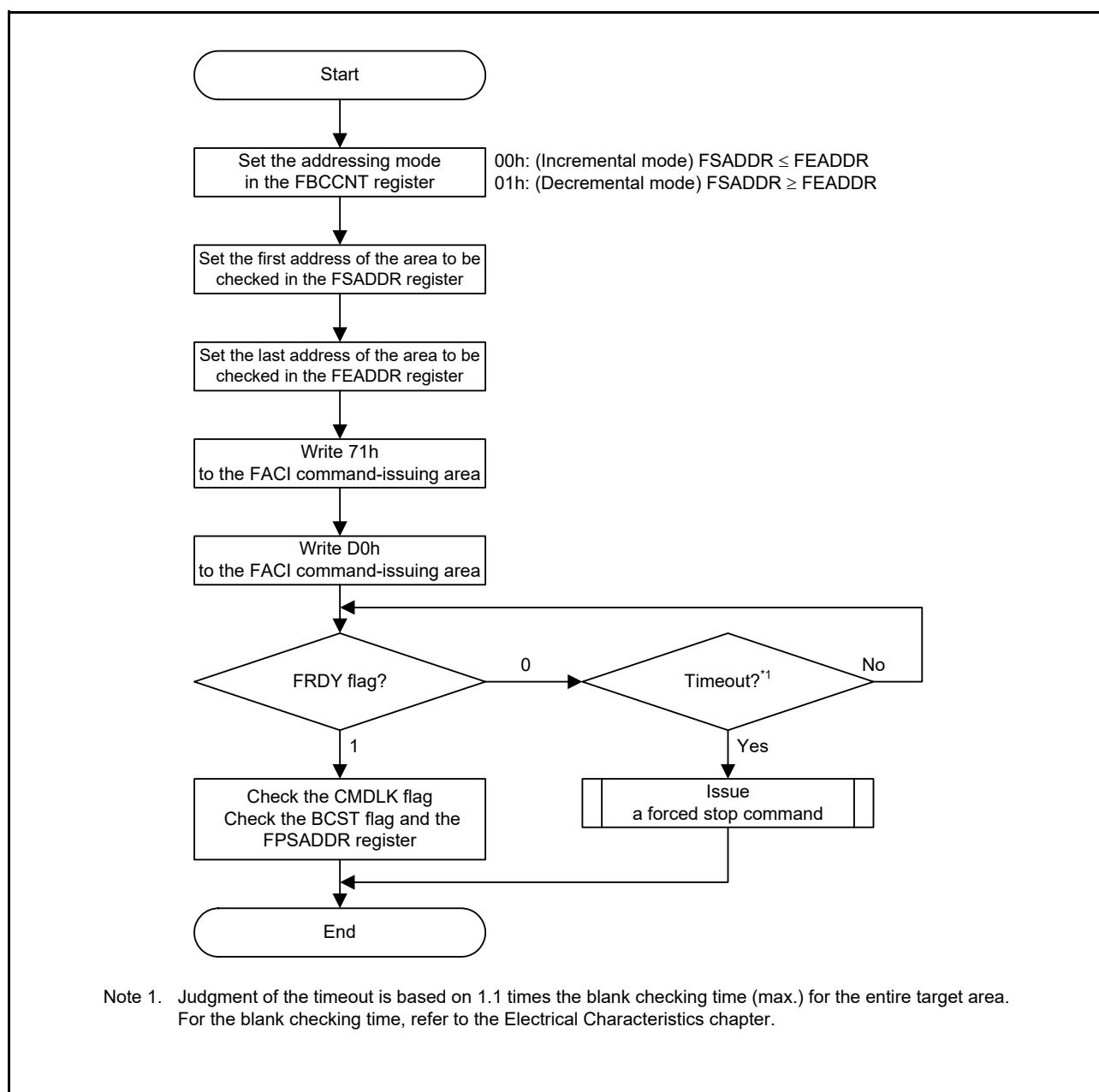


Figure 62.24 Usage of the Blank Check Command

62.8.3.14 Configuration Set Command

The configuration set command is used to set the option-setting memory (configuration setting area).

Before issuing a configuration set command, set the specified address (shown in Table 62.15) in the FSADDR register.

Writing D0h to the FACL command-issuing area in the final access for issuing the FACL command starts processing of the configuration set command.

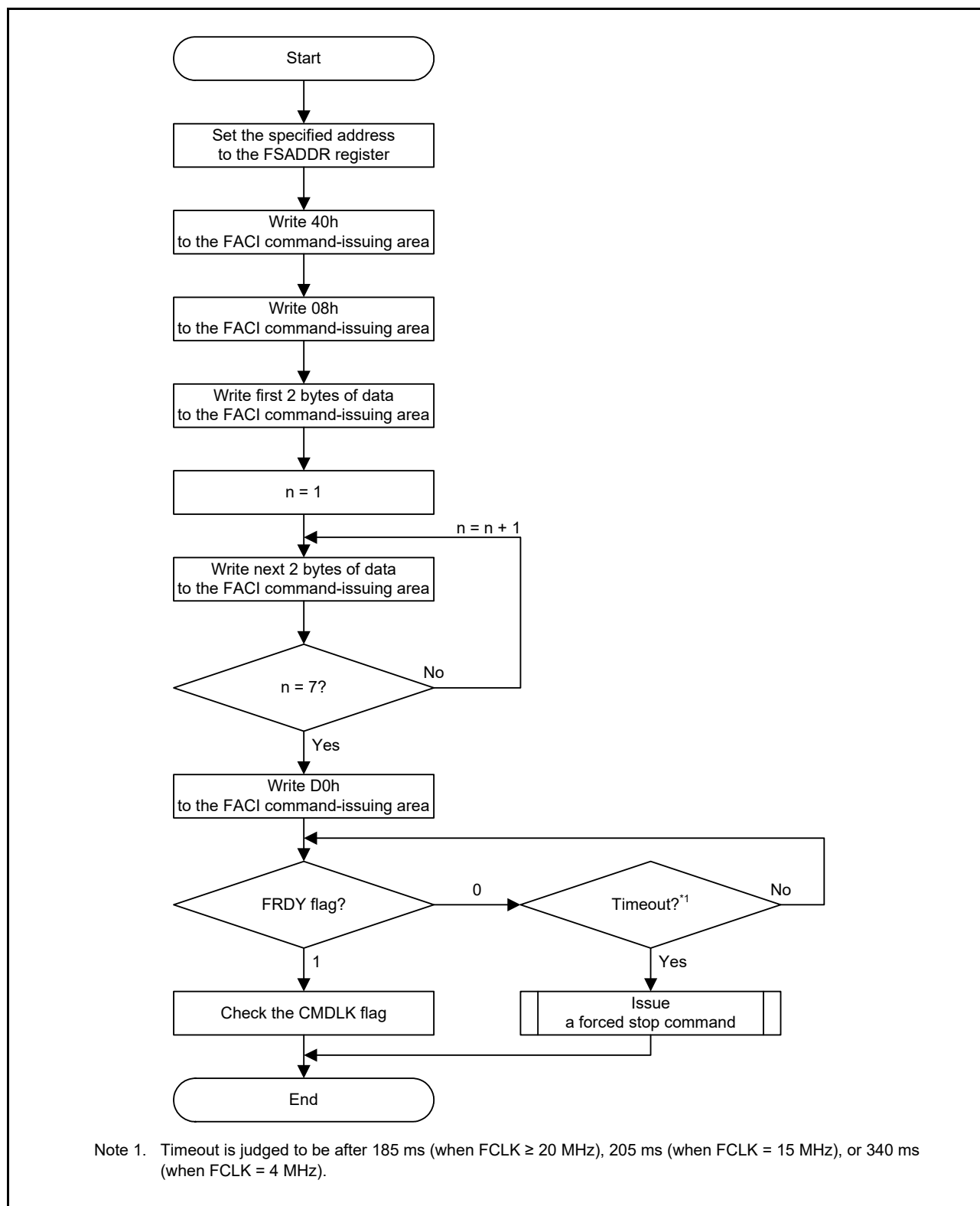


Figure 62.25 Usage of the Configuration Set Command

The correspondence between the possible target data for configuration setting and the address value set in the FSADDR register is shown in Table 62.15.

Table 62.15 Address Used by Configuration Set Command

Address	FSADDR Register Value	Setting Data	Operation of additional writing		Timing when the Setting is Enabled
			FAW.FSPR bit is 1	FAW.FSPR bit is 0	
FE7F 5D00h	00FF 5D00h	• Endian select register (MDE) • Option function select register 0 (OFS0) • Option function select register 1 (OFS1)	Writable	Writable	After a reset
FE7F 5D10h	00FF 5D10h	• TM identification data register (TMINF)	Writable	Writable	After a reset
FE7F 5D20h	00FF 5D20h	• Bank select register (BANKSEL)	Writable	Writable	After a reset
FE7F 5D40h	00FF 5D40h	• Serial programmer command control register (SPCC) • TM enable flag register (TMEF)	Writable* ¹ (from 1 to 0 only)	Writable* ¹ (from 1 to 0 only)	After a reset or when command is executed* ³
FE7F 5D50h	00FF 5D50h	• OCD/serial programmer ID setting register (OSIS)	Writable	Writable	After a reset
FE7F 5D64h	00FF 5D60h	• Flash access window setting register (FAW)*²	Writable	Not writable* ²	After a reset or when command is executed
FE7F 5D70h	00FF 5D70h	• ROM code protect register (ROMCODE)	Writable	Writable	After a reset

Note 1. Once these bits are set to 0, the bits cannot be restored to 1 by using the configuration set command.

Note 2. The FAW.FSPR bit cannot be restored to 1 once it is set to 0. Therefore, setting the access window and start-up area select bits again becomes impossible. (when the configuration set command is issued to the address of FE7F 5D64h, the flash sequencer enters the command-locked state.) Exercise extra caution when handling the FAW.FSPR bit.

Note 3. The setting in the serial programmer command control register (SPCC) is enabled after a reset. The setting of the TM enable flag register (TMEF) is enabled after a reset or when command is executed.

62.9 Suspend Operation

Reading from the flash memory is not possible during programming or erasure if the conditions for background operation given in Table 62.25 are not satisfied. When a P/E suspend command is issued to suspend the programming or erasure of the flash memory, reading from the flash memory is enabled. Regarding P/E suspend commands, there are one suspend command mode for programming and two suspend command modes for erasure (suspend priority mode and erase priority mode). To resume suspended programming or erasure, the P/E resume command is available.

62.10 Protection Functions

62.10.1 Software Protection

Software protection disables programming and erasure for the flash memory through the settings of control registers. If an attempt is made to issue an FACI command against software protection, the flash sequencer enters the command-locked state.

62.10.1.1 Protection through FWEPROR

Programming cannot proceed in any mode unless the FWEPROR.FLWE[1:0] bits are set to 01b.

62.10.1.2 Protection through FENTRYR

When the FENTRYR register is set to 0000h, the flash sequencer enters read mode. In read mode, FACI commands cannot be accepted. If an attempt is made to issue an FACI command in read mode, the flash sequencer enters the command-locked state.

62.10.2 Error Protection

Error protection detects erroneous issuance of FACI commands, unauthorized access, and flash sequencer malfunction. FACI command acceptance is disabled (command-locked state) in response to the detection of these errors. The flash memory cannot be programmed or erased while the flash sequencer is in the command-locked state. For release from the command-locked state, issue a status clear or forced stop command. The status clear command can only be used while the FSTATR.FRDY flag is 1. The forced stop command can be used regardless of the value of the FSTATR.FRDY flag. Generation of a flash access error (FIFERR) interrupt detects malfunction. An FIFERR interrupt is generated under any of the following conditions.

- When the flash sequencer enters the command-locked state (when the FASTAT.CMDLK flag is 1) while the FAEINT.CMDLKIE bit is 1
- When violation in access to the data flash memory occurred (the FASTAT.DFAE flag is 1) while the FAEINT.DFAEIE bit is 1
- When violation in access to the code flash memory occurred (when the FASTAT.CFAE flag is 1) while the FAEINT.CFAEIE bit is 1

When the flash sequencer enters the command-locked state in response to a command other than the P/E suspend command during program or erase processing, the flash sequencer continues the processing for programming or erasure. In this state, the P/E suspend command cannot be used to suspend the processing for programming or erasure. If a command is issued in the command-locked state, the ILGLERR and ILGCOMERR flags in the FSTATR register becomes 1 and the other flags retain the values set due to previous error detection.

Table 62.16 shows error protection types and status bit values after error detection.

Table 62.16 Error Protection Type (1/2)

Error Type	Description	ILGOMERR	FESETERR	SECERR	OTERR	ILGLERR	ERSERR	PRGERR	FLWEERR	CFAE	DFAE
FENTRYR setting error	AA81h was written to the FENTRYR register.	0	1	0	0	1	0	0	0	0	0
	The FENTRYR register value at suspension disagrees with that at resumption.	0	1	0	0	1	0	0	0	0	0
Illegal command error	Data was written in 16- or 32-bit units on the first access of an FAcI command.	1	0	0	0	1	0	0	0	0	0
	An undefined code was written on the first access of an FAcI command.	1	0	0	0	1	0	0	0	0	0
	A value other than D0h was written on the last access of an FAcI command with multiple access cycles.	1	0	0	0	1	0	0	0	0	0
	The value "N" (refer to Table 62.11) specified in the second access of the FAcI command in the program or configuration set command is illegal.	1	0	0	0	1	0	0	0	0	0
	A blank check command was issued with one of the following settings. • The FBCCNT.BCDIR bit is 0, and the value in the FSADDR register > that in the FEADDR register • The FBCCNT.BCDIR bit is 1, and the value in the FEADDR register > that in the FSADDR register • The value of bits b16 to b0 of the FEADDR register is set in the range from 0 8000h to 1 FFFFh	1	0	0	0	1	0	0	0	0	0/1 *1
	A multi-block erase command was issued with one of the following settings. • The value in the FSADDR register > that in the FEADDR register • The value of bits b16 to b0 of the FEADDR register is set in the range from 0 8000h to 1 FFFFh	1	0	0	0	1	0	0	0	0	0/1 *1
	An FAcI command not acceptable in each mode was issued (refer to Table 62.12).	1	0	0	0	1	0	0	0	0	0
	A program or block erase command was issued to an area protected by area protection.	1	0	0	0	1	0	0	0	0	0
	A program command was issued to the area being erased during erase-suspended state.	1	0	0	0	1	0	0	0	0	0
	An FAcI command was issued when command acceptance conditions are not satisfied (refer to Table 62.13).	0/1 *2	0/1 *2	0/1 *2	0/1 *2	1	0/1 *2	0/1 *2	0/1 *2	0/1 *2	0/1 *2
Erase error	An error occurs during erase operation.	0	0	0	0	0	1	0	0	0	0
Program error	An error occurs during program operation.	0	0	0	0	0	0	1	0	0	0
Code flash memory access violation	A program command or block erase command was issued under the following settings in code flash memory P/E mode. • The value of bits b23 to b0 of the FSADDR register is set in the range from 00 0000h to BF FFFFh	0	0	0	0	1	0	0	0	1	0
	A configuration set command was issued with one of the following settings in code flash memory P/E mode. • The value of bits b23 to b0 of the FSADDR register is set in the range from 00 0000h to BF FFFFh • The value of bits b12 to b0 of the FSADDR register is set in the range from 0000h to 1CFFh or from 1D80h to 1FFFh	0	0	0	0	1	0	0	0	1	0

Table 62.16 Error Protection Type (2/2)

Error Type	Description	ILGCOMERR	FESETERR	SECERR	OTERR	ILGLERR	ERSERR	PRGERR	FLWEERR	CFAE	DFAE
Data flash memory access violation	A program command or block erase command was issued under the following settings in data flash memory P/E mode. • The value of bits b16 to b0 of the FSADDR register is set in the range from 0 8000h to 1 FFFFh	0	0	0	0	1	0	0	0	0	1
	A multi-block erase command or blank check command was issued under the following settings in data flash memory P/E mode. • The value of bits b16 to b0 of the FSADDR register is set in the range from 0 8000h to 1 FFFFh	1	0	0	0	1	0	0	0	0	1
Security error	A configuration set command was issued to the FAW register while the FAW.FSPR bit was 0.	0	0	1	0	1	0	0	0	0	0
Others	The FACI command-issuing area was accessed in read mode	0	0	0	1	1	0	0	0	0	0
	The FACI command-issuing area was read in code flash memory P/E mode or data flash memory P/E mode	0	0	0	1	1	0	0	0	0	0
Flash P/E protection error	A flash memory program/erase protection error for the setting of the FWEPROR register was detected during command processing by the flash sequencer.	0	0	0	0	0	0/1	0/1	1	0	0

Note 1. If the setting value in the FSADDR register matches the condition for violations in access to the data flash memory, the FASTAT.DFAE bit is set to 1.

Note 2. This is the value when a command is executed.

62.10.3 Start-Up Program Protection

Protection of the startup program is for protection of the program to be started after a reset (the startup program). This function provides a way to safely update the startup program when rewriting is suspended during a reset.

The startup area is 8 Kbytes in size and is assigned to the user area in the code flash memory. This function uses the values of the FAW.BTFLG bit and the FSUACR.SAS[1:0] bits to change the area where the startup program is stored in block units (see Figure 62.26 to Figure 62.29).

In protection of the startup program, the state of the selection of the startup area can be fixed by the access window protection bit (FAW.FSPR). However, the FAW.FSPR bit never be restored to 1 once the flag is set to 0. Exercise extra caution when handling the FAW.FSPR bit.

In addition, this protection cannot be used when dual mode is selected by the bank mode switching function (when the MDE.BANKMD[2:0] bits are 000b).

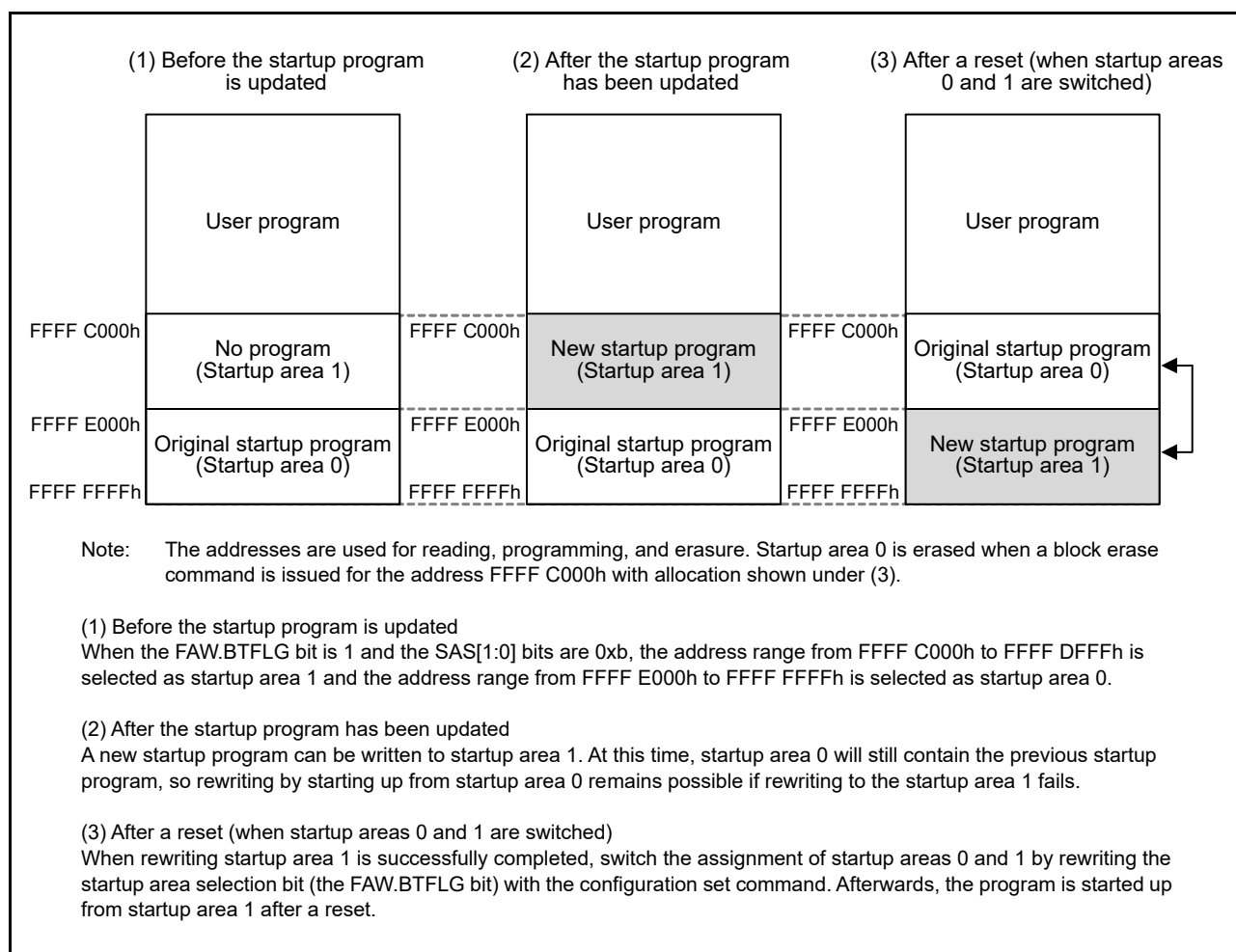


Figure 62.26 Concept of Protection of the Startup Program

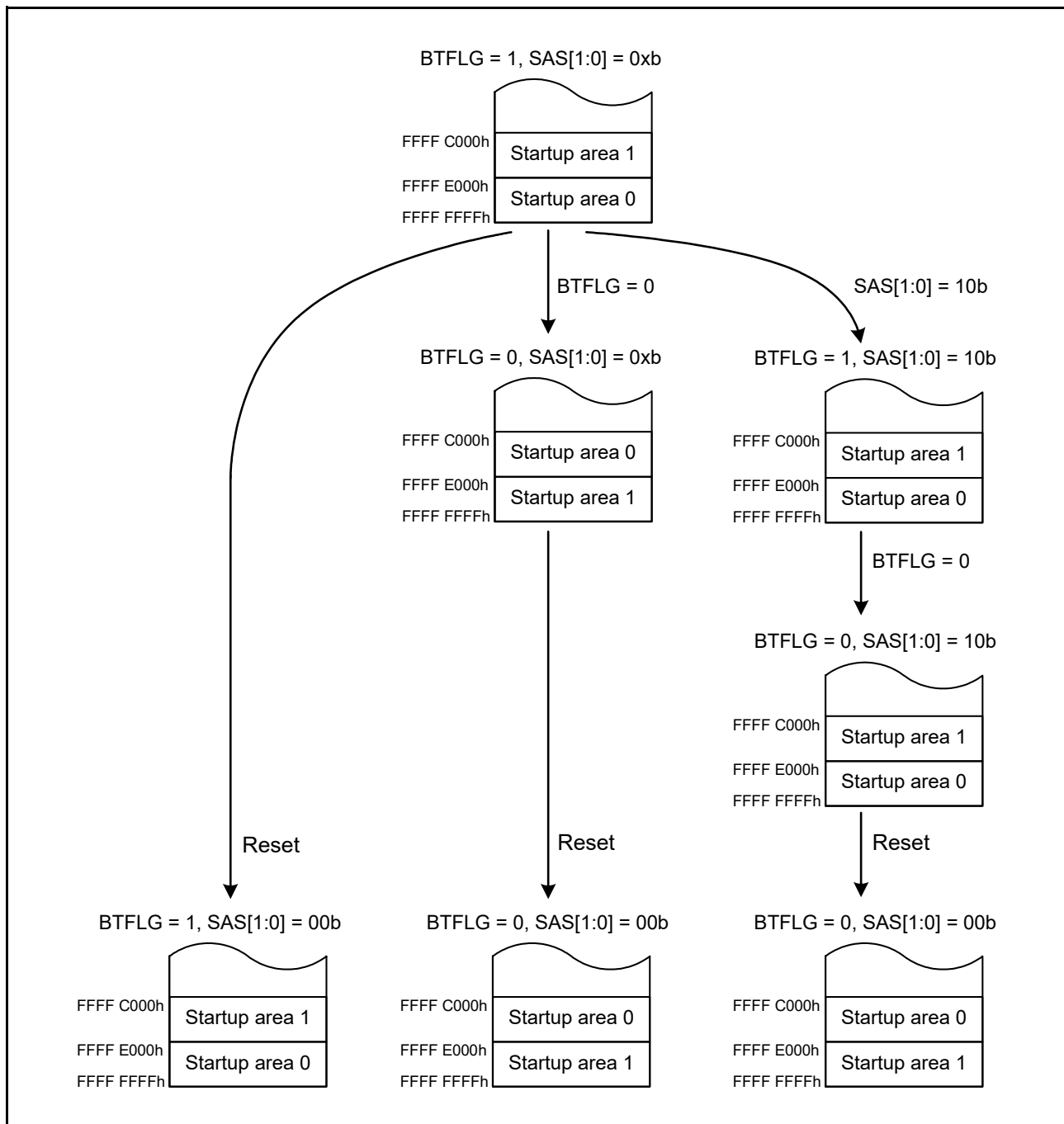


Figure 62.27 Example 1 of Transitions for Startup Program Protection Settings

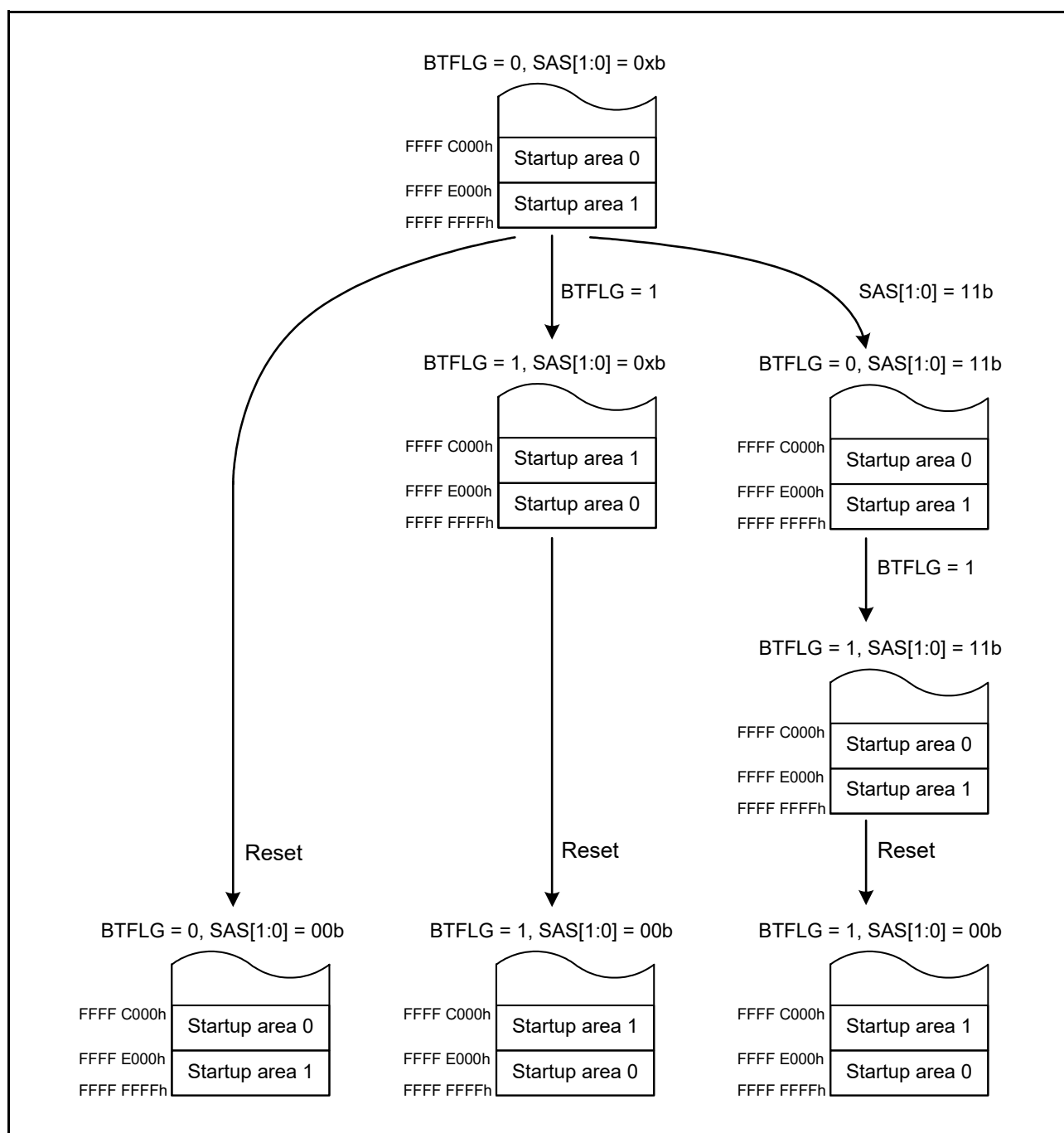


Figure 62.28 Example 2 of Transitions for Startup Program Protection Settings

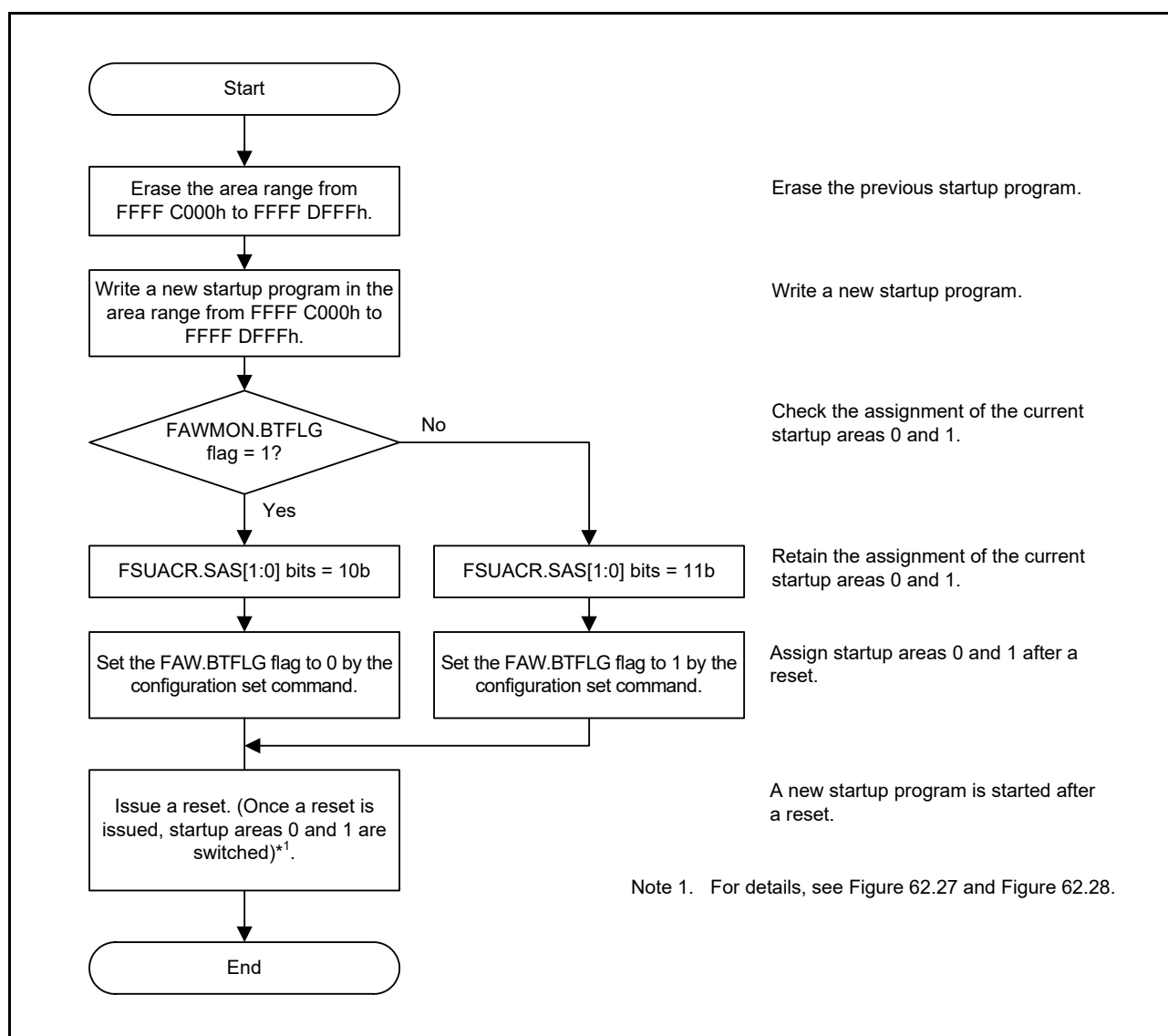


Figure 62.29 Example of Flowchart for Selecting Startup Area

62.10.4 Protection by Area Protection

Issuing an FACL command that programs or erases the area set outside the access window leads to the command-locked state. The access window is valid only in the user area of the code flash memory. The access window is valid in self-programming mode or serial programming mode.

The access window can be set in block units.

Figure 62.30 shows the Area Protection Overview.

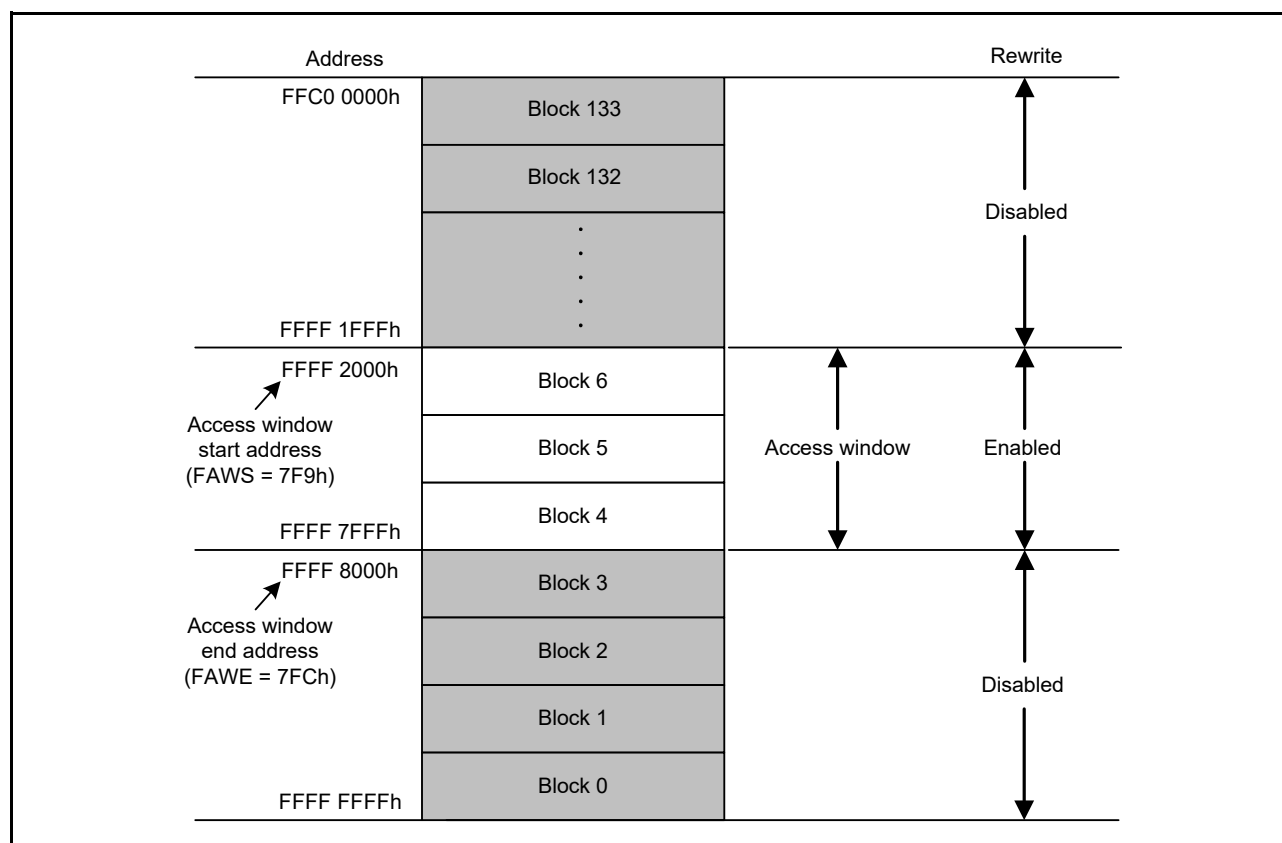


Figure 62.30 Area Protection Overview (When Blocks 4 to 6 are Set as the Access Window in Products with 4 Mbytes of Code Flash Memory)

62.10.5 Dual Bank Function

This protection uses the functions of bank mode switching and startup bank selection to update a program while a user program is running and to provide a safe method of updating in cases where programming is suspended during a reset.

62.10.5.1 Switching Bank Modes

The bank mode switching function selects either linear mode in which the user area in the code flash memory is used as one area, or dual mode in which the user area is divided into two bank areas. Figure 62.31 shows an example of flow of switching bank modes. A reset after setting the MDE.BANKMD[2:0] bits in the option setting memory determines the mode of the bank mode switching function. Selecting dual mode enables the startup bank selection function.

When dual mode is selected by bank mode switching function (the MDE.BANKMD[2:0] bits are 000b), start-up program protection function cannot be used.

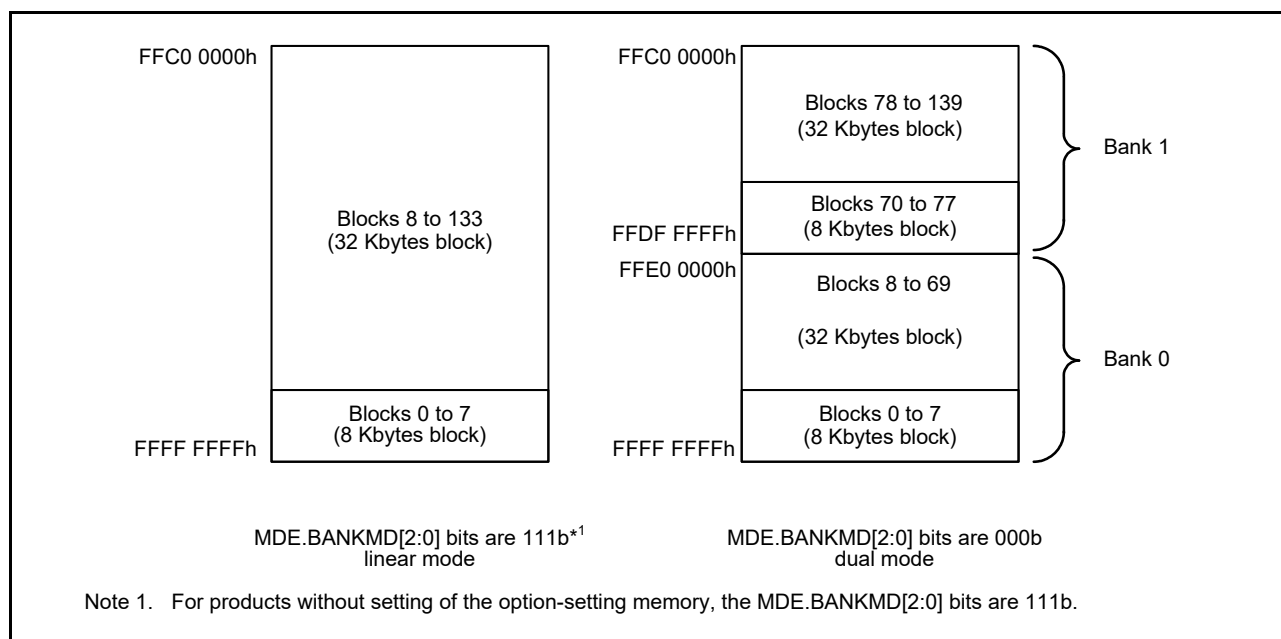


Figure 62.31 Example of Flow of Switching Bank Modes (For Products with 4 Mbytes of Code Flash Memory)

62.10.5.2 Selecting the Startup Bank

Startup bank selection provides a way to safely update the program by selecting a bank area to be started in dual mode (when the MDE.BANKMD[2:0] bits are 000b) when programming is suspended during a reset. Figure 62.32 is a schematic view of startup bank selection and Figure 62.33 shows an example of the flow of startup bank selection. A reset after setting the value of the BANKSEL.BANKSWP[2:0] bits in the option-setting memory changes the addresses of banks 0 and 1 and booting up a program proceeds from the updated area. When the address is switched by using startup bank selection, the P/E target for the FOCI commands is also switched. This function is invalid in linear mode.

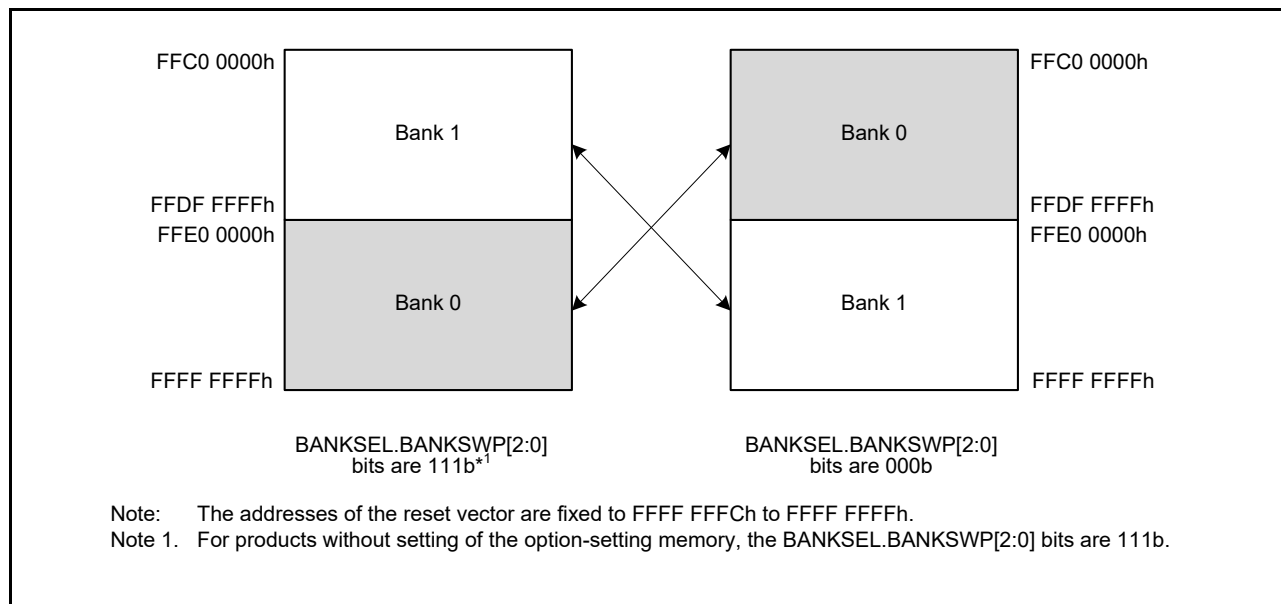


Figure 62.32 Example of Startup Bank Selection (For Products with 4 Mbytes of Code Flash Memory)

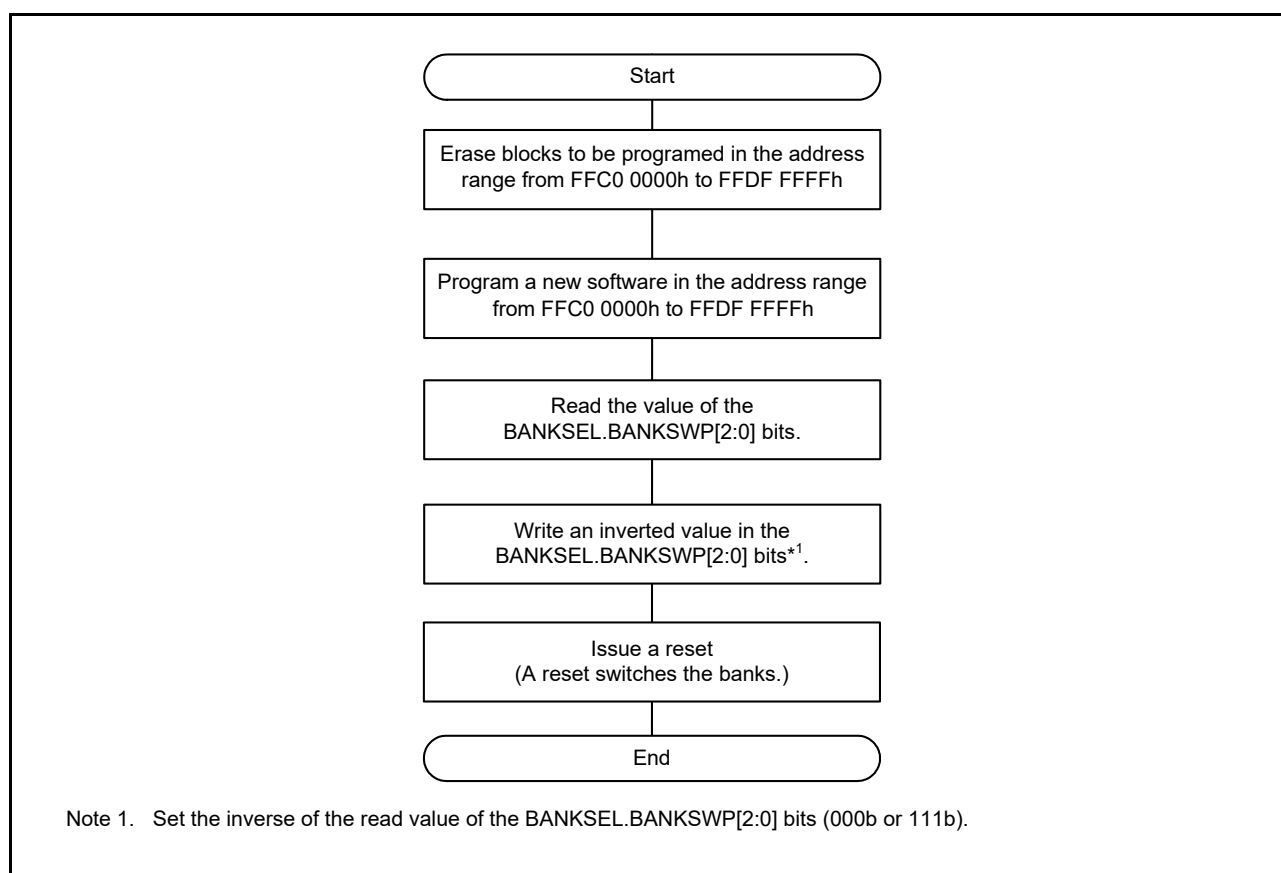


Figure 62.33 Example of Startup Bank Selection Flow (For Products with 4 Mbytes of Code Flash Memory)

62.11 Boot Mode

There are three serial programming modes; the boot mode (SCI interface) with SCI, the boot mode (USB interface) with USB, and the boot mode (FINE interface) with FINE. Table 62.17 lists the I/O pins used in boot mode.

Table 62.17 I/O Pins Used in Boot Mode

Pin Name	I/O	Mode to be Used	Use
MD	Input	Boot mode (SCI interface) Boot mode (USB interface)	Selection of operating mode
PC7/UB	Input		Selection of boot mode (SCI interface) or boot mode (USB interface)
PF2/RXD1 (224- and 176-pin products) P30/RXD1 (145-, 144-, and 100-pin products)	Input	Boot mode (SCI interface)	For host communication (to receive data through SCI)
PF0/TXD1 (224- and 176-pin products) P26/TXD1 (145-, 144-, and 100-pin products)	Output		For host communication (to transmit data through SCI)
USB0_DP, USB0_DM	I/O	Boot mode (USB interface)	Data input/output of USB
P16/USB0_VBUS	Input		Detection of connection and disconnection of USB cables
P35/UPSEL	Input		Selection of USB bus-powered mode or self-powered mode
MD/FINED	I/O	Boot mode (FINE interface)	Selection of operating mode, FINE data I/O

62.11.1 Boot Mode (SCI Interface)

In boot mode (SCI interface), the host sends control commands and data for programming, and the flash memory is programmed or erased accordingly. An on-chip SCI handles transfer between the host and this MCU in asynchronous mode. Tools for transmission of control commands and the data for programming must be prepared in the host. When this MCU is activated in boot mode (SCI interface), the program on the dedicated area the MCU is executed. The boot program automatically adjusts the bit rate of the SCI and controls programming/erasure by receiving control commands from the host.

Figure 62.34 shows the system configuration for operations in boot mode (SCI interface).

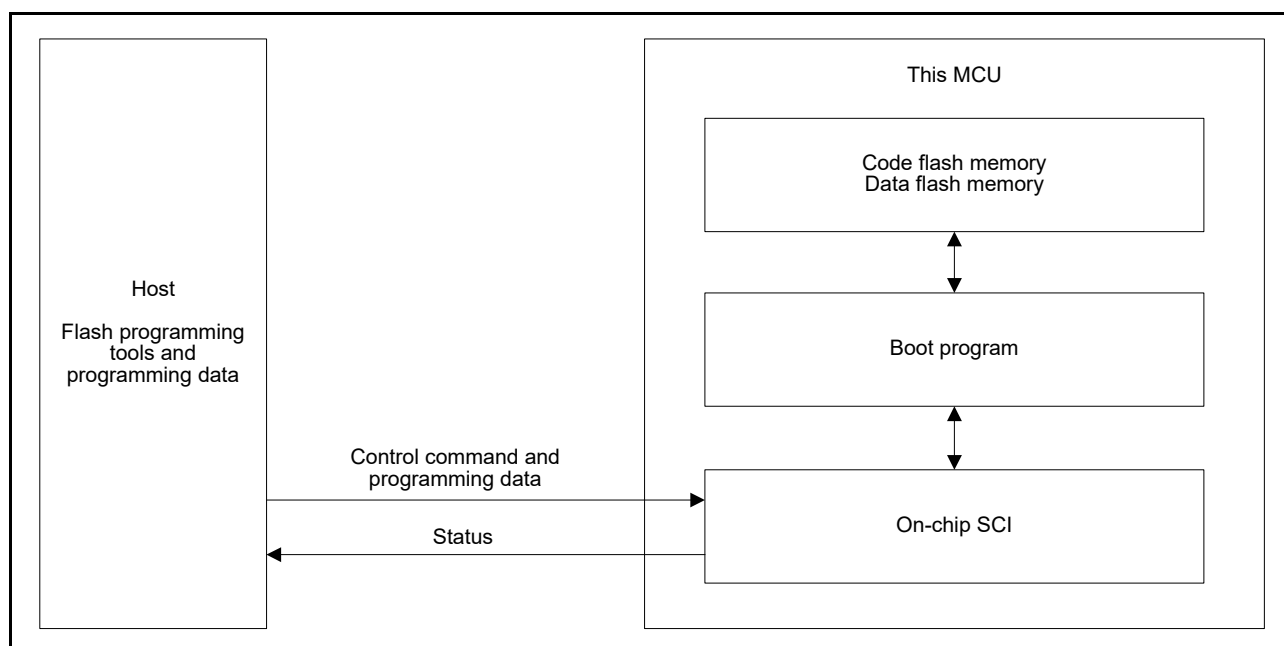


Figure 62.34 System Configuration for Operations in Boot Mode (SCI Interface)

62.11.2 Boot Mode (USB Interface)

In boot mode (USB interface), the flash memory can be programmed or erased by sending control commands and program data from the host. An on-chip USB is used for communications between the host and this MCU. The host requires tools for sending control commands and data for programming. Figure 62.35 shows the configuration of a system for use in boot mode (USB interface).

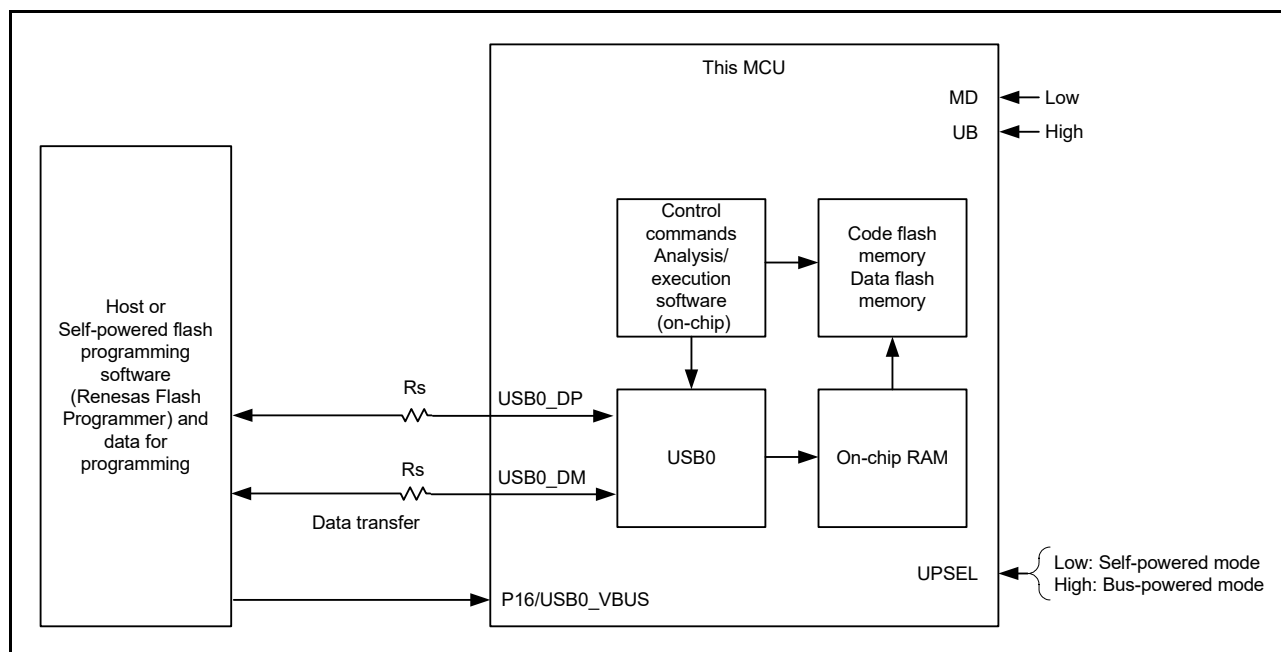


Figure 62.35 System Configuration in Boot Mode (USB Interface)

In boot mode (USB interface), self-powered mode or bus-powered mode is selected for operation. Release from the reset state while the low level is being applied to the MD and UPSEL pins and the high level is being applied to the UB pin selects self-powered mode. Release from the reset state while the low is being applied to the MD pin and the high level is being applied to the UB and UPSEL pins are high selects bus-powered mode.

Table 62.18 shows the enumeration information for each mode, and Table 62.19 shows the device class information.

Table 62.18 Enumeration Information

USB Specification	Ver.2.0 (Full-Speed)	
Maximum current	Self-powered mode (pins P35 and UPSEL = Low)	100 mA
	Bus-powered mode (pins P35 and UPSEL = High)	500 mA

Table 62.19 Device Class Information

Device Class	Communication Device Class (CDC)
Sub class	Abstract Control Model (ACM)

62.11.3 Boot Mode (FINE Interface)

The flash memory can be programmed and erased using the FINE in boot mode (FINE interface). The flash memory can be rewritten.

62.11.3.1 Operating Conditions in Boot Mode (FINE Interface)

FINE is used to communicate with the serial programmer in boot mode (FINE Interface).

Figure 62.36 shows an Example of Pin Connections in Boot Mode (FINE Interface). Table 62.20 lists Pin Handling in Boot Mode (FINE Interface).

The example of pin connections shown in Figure 62.36 is a simplified circuit. Operations are not guaranteed in all systems.

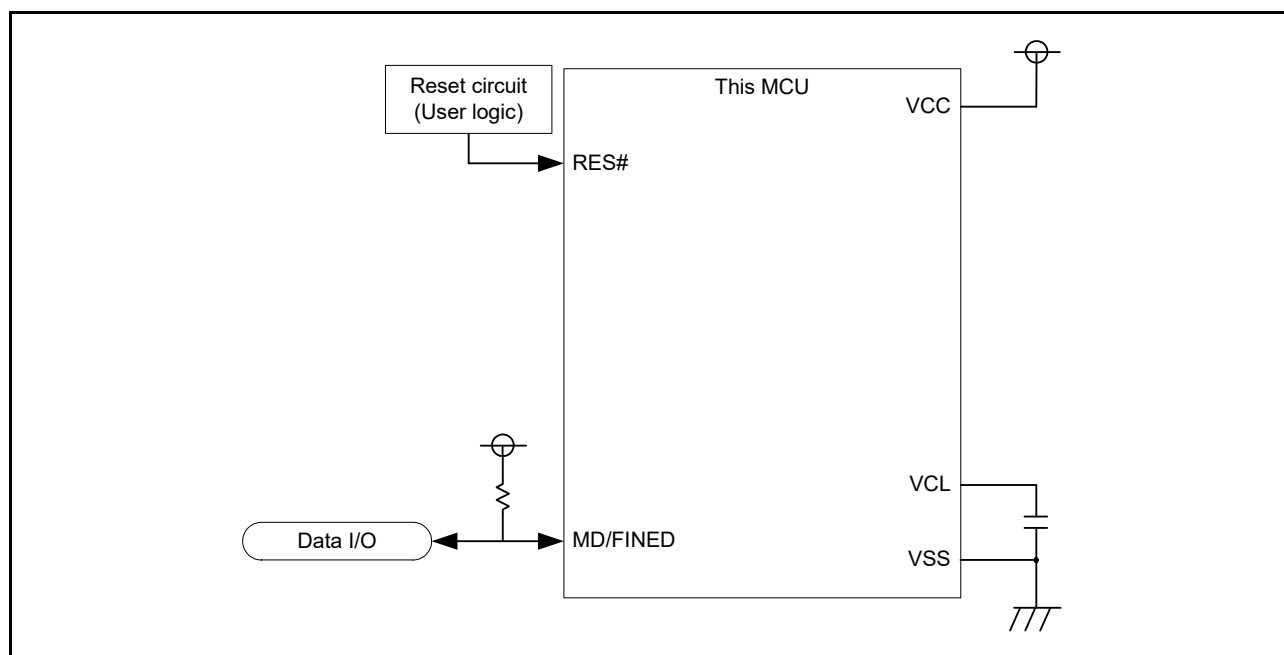


Figure 62.36 Example of Pin Connections in Boot Mode (FINE Interface)

Table 62.20 Pin Handling in Boot Mode (FINE Interface)

Pin Name	Name	I/O	Function
VCC, VSS	Power supply input	Input	Input 2.7 V or higher to the VCC pin. Input 0 V to the VSS pin.
VCL	Decoupling capacitor connect pin	—	Connect to the VSS pin via a 0.22-μF multilayer ceramic capacitor for stabilizing the internal voltage.
MD	Operating mode control/data I/O	I/O	Connect the VCC pin via a resistor (pull up).
RES#	Reset input	Input	Reset pin. Connect to the reset circuit.

62.12 On-Chip Debugger ID Code Protection

This function is used to prohibit connection to the on-chip debugger (OCD). When an emulator is connected, the ID code of the OSIS register, which is written on the option-setting memory, is used for judging the ID codes.

The ID code sent from the emulator is judged if it matches the ID code in the OSIS register. If the codes match, connection to the OCD proceeds. If they do not match, connection to the OCD is prohibited.

62.13 Serial Programmer ID Code Protection

This function is used to prohibit connection with the serial programmer. When a serial programmer is connected, the control and ID codes of the OSIS register, which are written on the option-setting memory, are used for judgment of the ID codes.

Control or ID codes sent from serial programmers are checked to see if they match the control or ID code in the OSIS register. If the codes match, connection to the serial programmer proceeds. If they do not match, connection with the serial programmer does not proceed. If the control code is 45h and the codes do not match three times in a row, all data in the flash memory are erased.*1

Note 1. When the FAW.FSPR bit is 0, data in the flash memory are not erased.

62.14 ROM Code Protection

ROM code protection is a function to prohibit reading, programming, or erasing of the flash memory when the parallel programmer is used.

For details, see section 7.2.10, ROM Code Protection Register (ROMCODE).

62.15 Boot Mode Communications Protocol

This section describes the communications protocol for use in boot mode. When developing a serial programmer, use this communications protocol to control it.

62.15.1 How to Start the Chip Up in Boot Mode

(1) How to Start the Chip Up in Boot Mode (SCI Interface)

The chip starts up in boot mode (SCI interface) if both the MD and UB pins are at the low level on release from the reset state (i.e. when the level on the RES# pin changes from low to high). A waiting time of at least 400 ms is required while the RES# pin is held at the high level after the chip starts up in boot mode (SCI interface) until communications with the MCU can proceed.

Figure 62.37 shows the states of pins up to communications in boot mode (SCI interface) becoming possible.

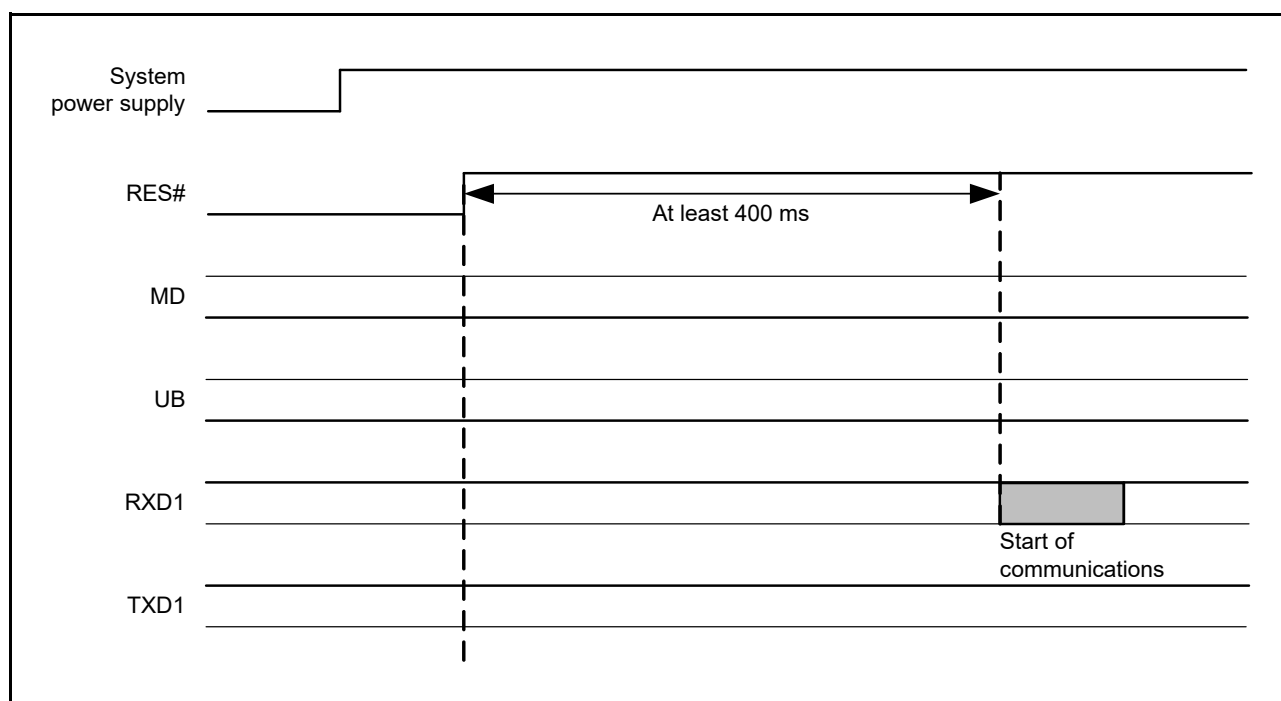


Figure 62.37 States of Pins Up to Communications in Boot Mode (SCI Interface) Becoming Possible

(2) How to Start the Chip Up in Boot Mode (USB Interface)

The chip starts up in boot mode (USB interface) if the MD pin is at the low level and UB pin is at the high level on release from the reset state (i.e. when the level on the RES# pin changes from low to high).

62.15.2 State Transitions in Boot Mode

62.15.2.1 State Transitions in Boot Mode (SCI Interface)

Figure 62.38 shows a flowchart for transition to boot mode (SCI interface).

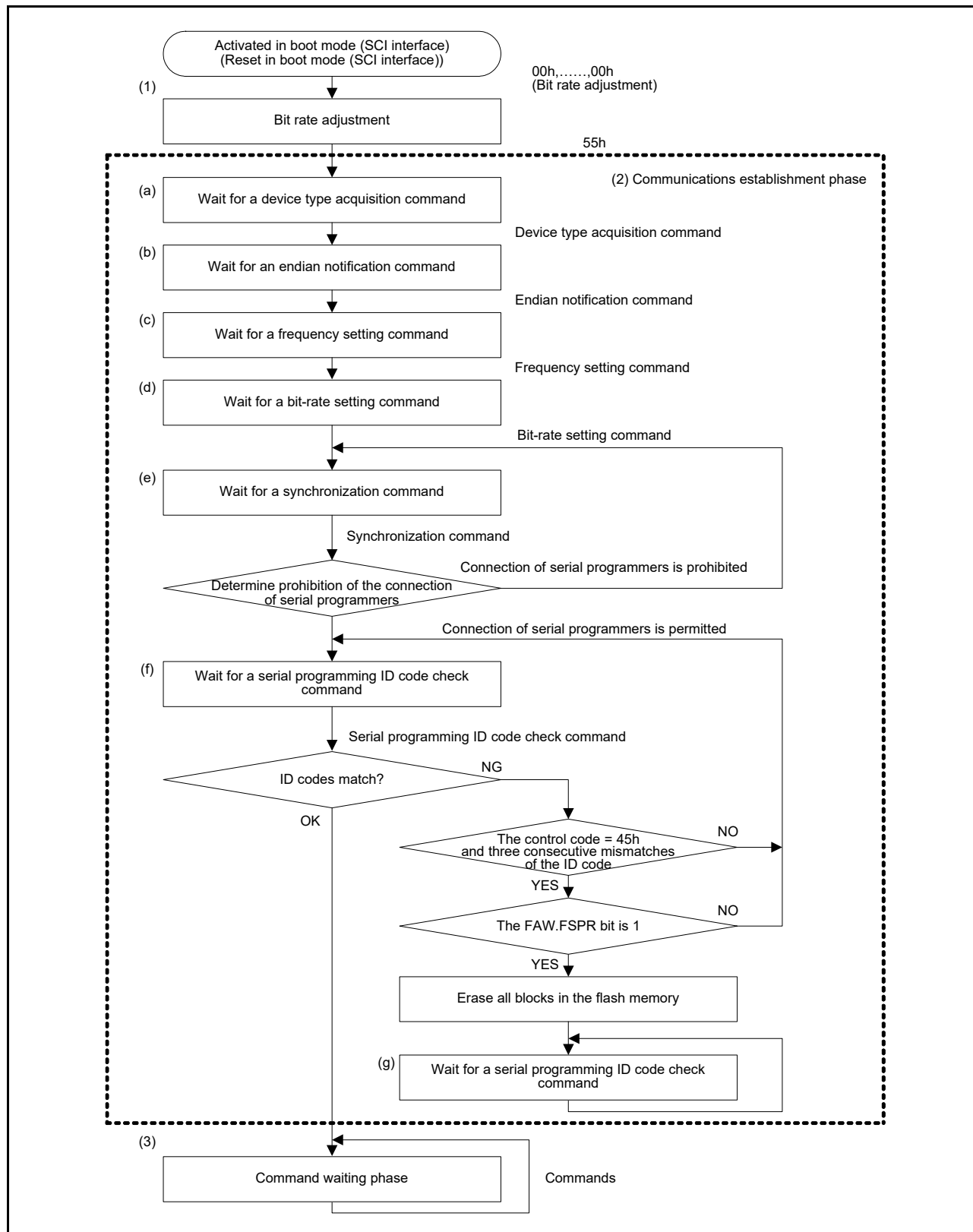


Figure 62.38 Flowchart for Transition to Boot Mode (SCI Interface)

(1) Matching the bit rates

When this MCU is activated in boot mode, the bit rate of the SCI is automatically adjusted to match that of the host. On completion of this automatic bit rate adjustment, this MCU transmits the value 00h to the host. On subsequent correct reception of the value 55h sent from the host, this MCU enters the communications establishment phase. For details on matching of the bit rates, see section 62.15.3, Automatic Adjustment of the Bit Rate.

(2) Communications establishment phase

The device, endian, frequency, and bit rate are selected in this phase. In this phase, the serial programmer ID code protection is judged. For the commands to use in the communications establishment phase, see section 62.15.5, Communications Establishment Phase.

(a) Waiting for a device type acquisition command

In this state, the MCU is waiting for a device type acquisition command to be sent from the host. When it receives a device type acquisition command, the state shifts to waiting for an endian notification command. For details of the device type acquisition command, see section 62.15.9, Device Type Acquisition Command.

(b) Waiting for an endian notification command

In this state, the MCU is waiting for an endian notification command to be sent from the host. When it receives an endian notification command, the state shifts to waiting for a frequency setting command. For details of the endian notification command, see section 62.15.10, Endian Notification Command.

(c) Waiting for a frequency setting command

In this state, the MCU is waiting for a frequency setting command to be sent from the host. When it receives a frequency setting command, the state shifts to waiting for a bit-rate setting command. For details of the frequency setting command, see section 62.15.11, Frequency Setting Command.

(d) Waiting for a bit-rate setting command

In this state, the MCU is waiting for a bit-rate setting command to be sent from the host. When it receives a bit-rate setting command, the state shifts to waiting for a synchronization command. For details of the bit-rate setting command, see section 62.15.12, Bit-Rate Setting Command.

(e) Waiting for a synchronization command

In this state, the MCU is waiting for a synchronization command to be sent from the host. When it receives a synchronization command, it transitions to the waiting state of the serial programming ID code check command. If the MCU has been set to prohibit the connection of a serial programmer, this MCU transmits an error code to indicate that connecting a serial programmer is prohibited and remains in the state of waiting for a synchronization command. For details of the synchronization command, see section 62.15.13, Synchronization Command.

(f) Waiting for a serial programming ID code check command

Wait for the serial programming ID code check command to be sent from the host. The control code or ID code that was sent is compared with the control code or ID code in the OSIS register. The device enters the command waiting phase if they match. If they do not match, the next transition is back to the state of waiting for a serial programming ID code check command.

If the control code is 45h and the codes do not match three times in a row, all data in the flash memory are erased.*1
For details of the ID code check command, see section 62.15.15, Serial Programming ID Code Check Command.

Note 1. When the FAW.FSPR bit is 0, the blocks in those areas are not erased.

(g) Waiting for a serial programming ID code check command (after erasure)

After all blocks in the flash memory are erased, reboot the MCU in boot mode.

(3) Phase of waiting for commands

In this state, programming and erasure proceed in accord with commands from the host. For details of the commands that can be issued in the command waiting phase, see [section 62.15.6, Command Waiting Phase](#).

62.15.2.2 State Transitions in Boot Mode (USB Interface)

Figure 62.39 shows a flowchart for transition to boot mode (USB interface).

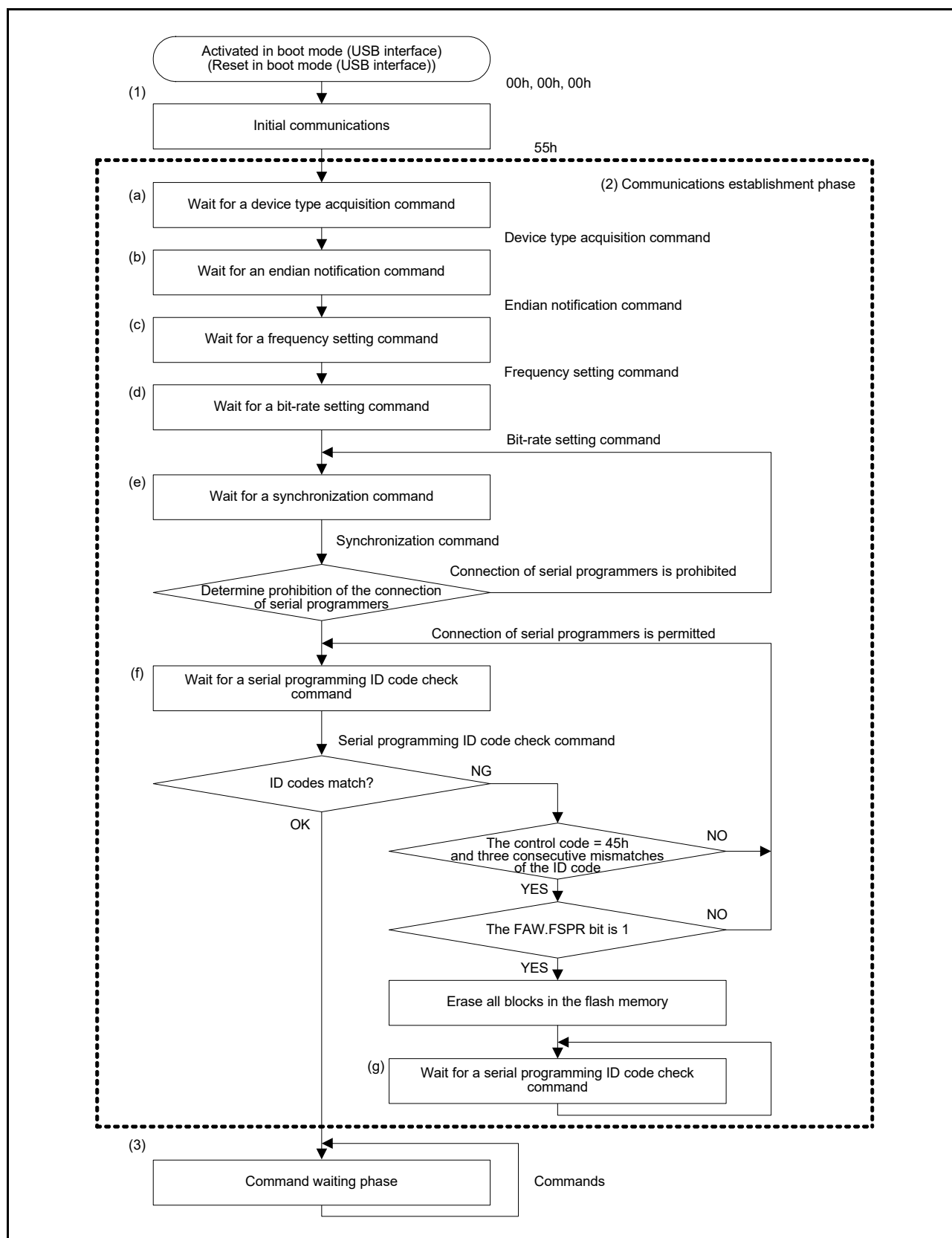


Figure 62.39 Flowchart for Transition to Boot Mode (USB Interface)

(1) Initial Communications

Enumeration starts when this MCU is activated in boot mode (USB interface).

Once the process of enumeration between the host and this MCU is completed, send the value 00h from the host to this MCU three times.

This MCU ignores transmission of the value 00h from the host from the fourth time and onward.

This MCU returns 00h to the host when receiving the value 00h three times. On subsequent correct reception of the value 55h sent from the host, this MCU enters the communications establishment phase.

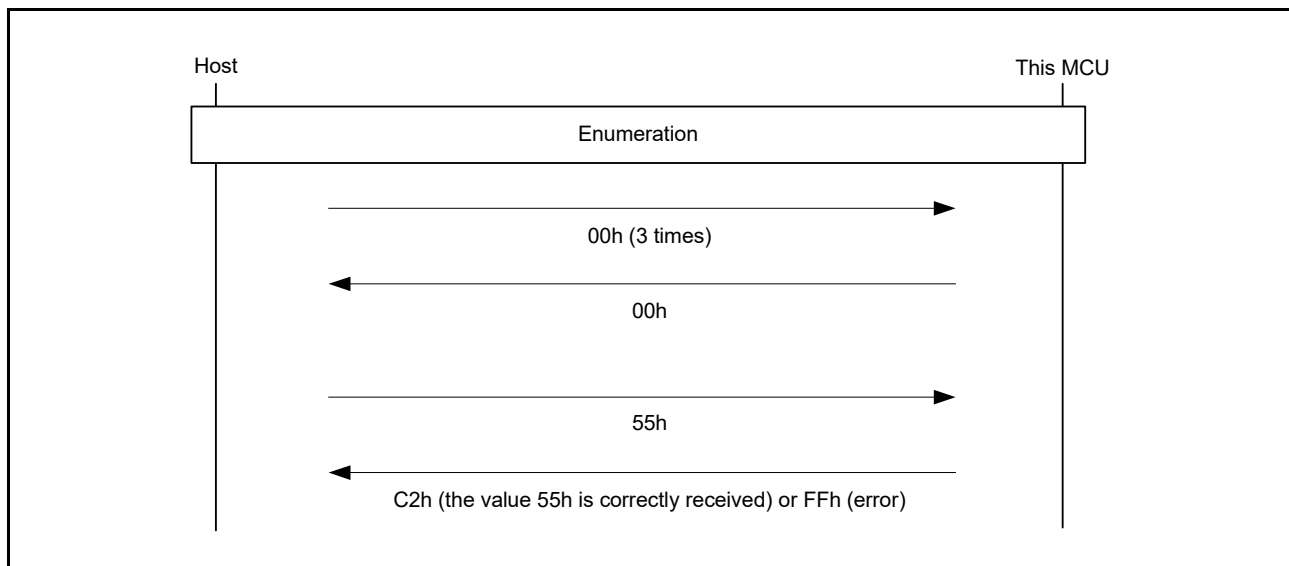


Figure 62.40 Initial Communications Between the Host and This MCU

(2) Communications establishment phase

The device, endian, frequency, and bit rate are selected in this phase. In this phase, the serial programmer ID code protection is judged. For the commands to use in the communications establishment phase, see section 62.15.5, Communications Establishment Phase.

(a) Waiting for a device type acquisition command

In this state, the MCU is waiting for a device type acquisition command to be sent from the host. When it receives a device type acquisition command, the state shifts to waiting for an endian notification command. For details of the device type acquisition command, see section 62.15.9, Device Type Acquisition Command.

(b) Waiting for an endian notification command

In this state, the MCU is waiting for an endian notification command to be sent from the host. When it receives an endian notification command, the state shifts to waiting for a frequency setting command. For details of the endian notification command, see section 62.15.10, Endian Notification Command.

(c) Waiting for a frequency setting command

In this state, the MCU is waiting for a frequency setting command to be sent from the host. When it receives a frequency setting command, the state shifts to waiting for a bit-rate setting command. For details of the frequency setting command, see section 62.15.11, Frequency Setting Command.

(d) Waiting for a bit-rate setting command

In this state, the MCU is waiting for a bit-rate setting command to be sent from the host. When it receives a bit-rate setting command, the state shifts to waiting for a synchronization command. For details of the bit-rate setting command, see section 62.15.12, Bit-Rate Setting Command.

(e) Waiting for a synchronization command

In this state, the MCU is waiting for a synchronization command to be sent from the host. When it receives a synchronization command, it transitions to the waiting state of the serial programming ID code check command. If the MCU has been set to prohibit the connection of a serial programmer, this MCU transmits an error code to indicate that connecting a serial programmer is prohibited and remains in the state of waiting for a synchronization command. For details of the synchronization command, see section 62.15.13, Synchronization Command.

(f) Waiting for a serial programming ID code check command

In this state, the MCU is waiting for a serial programming ID code check command to be sent from the host. The control code or ID code that was sent is compared with the control code or ID code in the OSIS register. The device enters the command waiting phase if they match. If they do not match, the next transition is back to the state of waiting for a serial programming ID code check command.

If the control code is 45h and the codes do not match three times in a row, all data in the flash memory are erased.*1

For details of the ID code check command, see section 62.15.15, Serial Programming ID Code Check Command.

Note 1. When the FAW.FSPR bit is 0, the blocks in those areas are not erased.

(g) Waiting for a serial programming ID code check command (after erasure)

After all blocks in the flash memory are erased, reboot the MCU in boot mode.

(3) Phase of waiting for commands

In this state, programming and erasure proceed in accord with commands from the host. For details of the commands that can be issued in the command waiting phase, see section 62.15.6, Command Waiting Phase.

62.15.3 Automatic Adjustment of the Bit Rate

When this MCU is booted up in boot mode (SCI interface), asynchronous transfer by the SCI is used to measure the periods at low level of consecutive bytes with value 00h that are sent from the host. While the period at low level is being measured, set the host's SCI transfer format to 8-bit data, one stop bit, no parity, and a transfer rate of 9,600 bps. This MCU measures the periods at low level in the signal from the host, adjusts the bit rate of its SCI, and then sends the value 00h to the host.

If reception of the value 00h by the host is successful, the host responds by sending the value 55h to this MCU. If successful reception of 00h by the host is not possible, reboot this MCU in boot mode, and then repeat the process of automatically adjusting the bit rate. If reception of the value 55h by this MCU is successful, it responds by sending C2h to the host, and if successful reception of 55h by this MCU is not possible, it responds by sending FFh to the host. If this MCU is started up in boot mode (USB interface), the bit rate is not automatically adjusted.

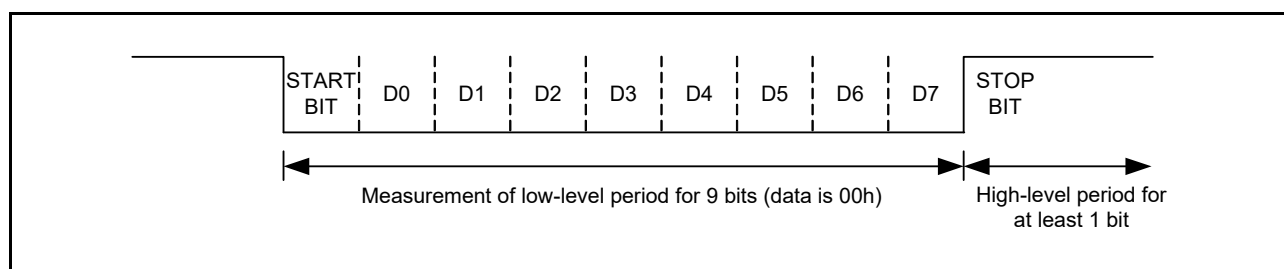


Figure 62.41 Transfer Format Used by SCI in Automatic Adjustment of Bit Rate

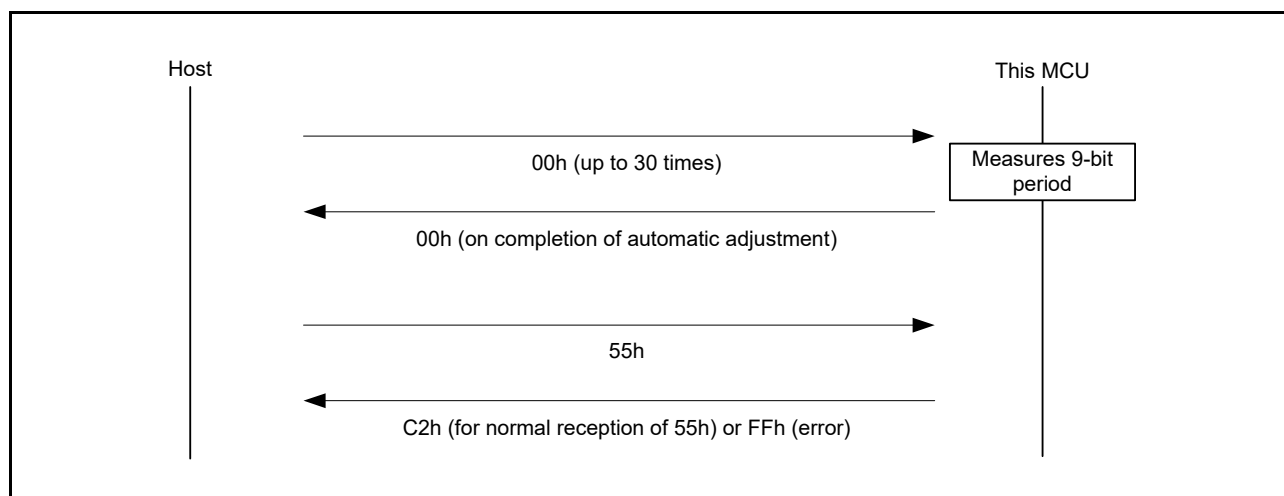


Figure 62.42 Sequence of Transfer between Host and This MCU

For the host's SCI bit rate, ensure that SCI communications proceed under the conditions given in Table 62.21.

Table 62.21 Conditions for Automatic Bit-Rate Adjustment

Bit Rate of SCI in Host
9,600 bps

62.15.4 Packet Format

(1) Command packet

The host sends commands to this MCU in the format below.

S	L	L	C	Command information	S	E
O	N	N	O	(variable length)	U	T
H	H	L	M	(up to 255 bytes)	M	X

Symbol	Code	Description
SOH	01h	Start of a packet (1 byte)
LNH	—	Packet length (length of COM and the command information) (8 to 15 bits) (1 byte)
LNL	—	Packet length (length of COM and the command information) (0 to 7 bits) (1 byte)
COM	—	Command code (1 byte)
Command information	—	Command information (up to 255 bytes)
SUM*1	—	Two's complement of the sum of values of LNH, LNL, COM, and the command information (1 byte)
ETX	03h	End of a packet (1 byte)

Note 1. SUM indicates the 1 byte of data that produces 00h as the result of adding LNH, LNL, COM, the command information, and SUM itself.

(2) Status packet and data packet

Data transmission proceeds between the host and this MCU in the format below.

S	L	L	R	Data	S	E	E
O	N	N	E	(variable length)	U	T	or T
D	H	L	S	(up to 1024 bytes)	M	B	X

Symbol	Code	Description
SOD	81h	Start of a packet (1 byte)
LNH	—	Packet length (length of RES and the data) (8 to 15 bits) (1 byte)
LNL	—	Packet length (length of RES and the data) (0 to 7 bits) (1 byte)
RES	—	Response code (1 byte)
Data	—	Data (up to 1024 bytes)
SUM*1	—	Two's complement of the sum of values of LNH, LNL, RES, and the data (1 byte)
ETB	17h	End of a packet (1 byte)
ETX	03h	End of the last packet (1 byte)

Note 1. SUM indicates the 1 byte of data that produces 00h as the result of adding LNH, LNL, RES, the data, and SUM itself.

62.15.5 Communications Establishment Phase

Table 62.22 lists the commands available in the commands establishment phase.

The synchronization command and ID authentication mode acquisition command can also be used in the command waiting phase.

Table 62.22 Commands Available in the Communications Establishment Phase

Command Name	Function
Device type acquisition	Transmits the oscillation frequency and CPU operating frequency (in Hz) supported by boot mode to the host.
Endian notification	Indicates whether big-endian or little-endian is to be used.
Frequency setting	Sets the values of the oscillation frequency and CPU operating frequency (in Hz).
Bit-rate setting	Changes the bit rate.
Synchronization	This command is used in processing for communications synchronization. It is also used when confirming whether the MCU can accept commands.
ID authentication mode acquisition	This command sends the enabled state of serial programmer ID code protection to the host.
Serial programming ID code check	Determines whether the control code and ID code written in the option-setting memory matches the control code and ID code sent by the host.

In the communications establishment phase, send commands from the host in the order of the device type acquisition, endian notification, frequency setting, bit-rate setting, and synchronization commands according to the responses to commands. Send the ID authentication mode acquisition or serial programming ID code check command following the synchronization command.

If commands are issued in an incorrect order or other commands are issued, this MCU returns a response indicating a flow error.

62.15.6 Command Waiting Phase

Table 62.23 lists the commands available in the command waiting phase.

The synchronization command and ID authentication mode acquisition command can also be used in the communications establishment phase.

Table 62.23 Commands Available in the Command Waiting Phase

Command Name	Function
Synchronization	See Table 62.22.
Blank check	Check that a selected area is blank.
Block erase	Erases a selected single block.
Area erase	Erases the specified area.
Program	Programs the selected area.
Read	Reads data from a selected area.
ID authentication mode acquisition	See Table 62.22.
Simple addition checksum	Calculates the sum of values in a selected area.
Configuration clear	Erases the set value of the option-setting memory and the TM target areas.
Configuration program	Programs the set value in the option-setting memory.
Configuration read	Reads the set value of the option-setting memory.
Acquisition of the number of the area information	Obtains the number of area information of the flash memory.
Acquisition of the area information	Obtains the area information of the flash memory.

If the host has sent an undefined command, this MCU returns a response indicating an error in the form of a unsupported command.

62.15.7 Command Transfer Sequence

Though the sequence of transfer differs from command to command, common transfer sequences are used for the commands that only make settings for this MCU and for the commands that obtain information on the settings in this MCU. However, as the contents of the command packet, status packet, and data packet differ for each command, see the sections on the individual commands for details.

(1) Common transfer sequence for the commands that only make settings

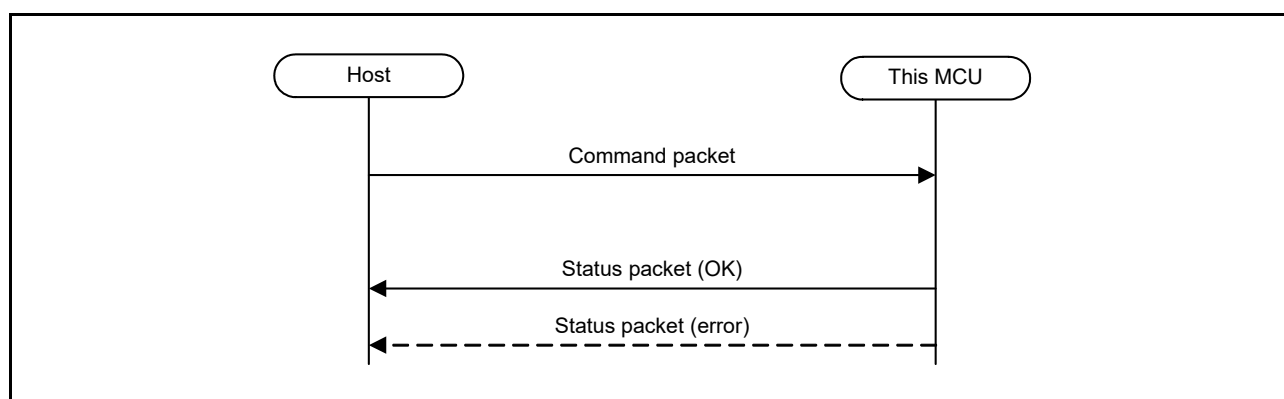


Figure 62.43 Common Transfer Sequence for the Commands that Only Make Settings

(2) Common transfer sequence for the commands that obtain information on settings

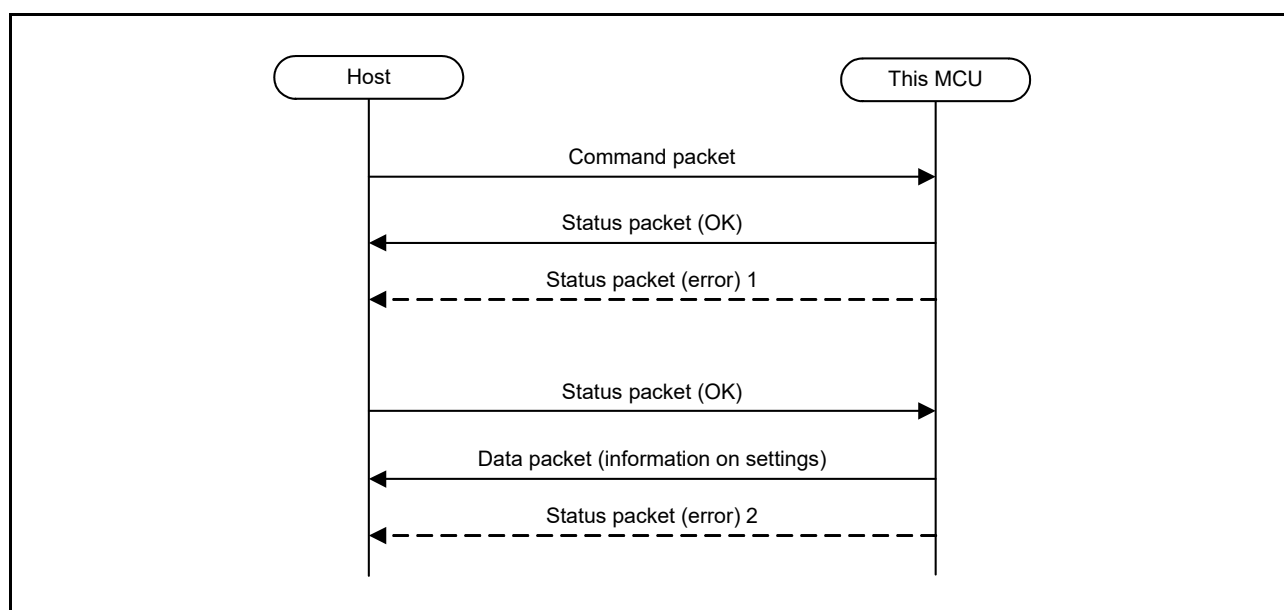


Figure 62.44 Common Transfer Sequence for the Commands that Obtain Information on Settings

Table 62.24 Common Transfer Sequence

Command Name	Common Transfer Sequence Type
Device type acquisition	Command that obtains information on settings
Endian notification	Command that only makes a setting
Frequency setting	Command that obtains information on settings
Bit rate setting	Not in a common transfer sequence
Synchronization	Command that only makes a setting
ID authentication mode acquisition	Command that obtains information on settings
Serial programming ID code check	Command that only makes a setting
Blank check	Command that only makes a setting
Block erase	Command that only makes a setting
Area erase	Command that only makes a setting
Program	Not in a common transfer sequence
Read	Not in a common transfer sequence
Simple addition checksum	Command that obtains information on settings
Configuration clear	Command that only makes a setting
Configuration program	Not in a common transfer sequence
Configuration read	Command that obtains information on settings
Acquisition of the number of the area information	Command that obtains information on settings
Acquisition of the area information	Command that obtains information on settings

For the command transfer sequences that are not in a common transfer sequence, see the sections on the individual commands.

62.15.8 Unsupported Commands

When this MCU receives an undefined command packet, it returns an unsupported error (C0h) and enters the command waiting state.

(1) Command packet structure

S	L	L	C	S	E
O	N	N	O	U	T
H	H	L	M	M	X

SOH: 01h
 LNH: Packet length (8 to 15 bits)
 LNL: Packet length (0 to 7 bits)
 COM: Command code*1
 SUM: Sum of values
 ETX: 03h

Note 1. Command code other than those specified in Table 62.23.

(2) Status packet structure

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 80h | COM (command code)
 ERR: Error code
 C0h ("unsupported" error)
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 SUM: Sum of values
 ETX: 03h

62.15.9 Device Type Acquisition Command

This command is used to make the MCU send the input frequency and system clock frequency (in Hz) supported by boot mode (SCI interface).

This command can only be accepted in the communications establishment phase.

(1) Command packet structure

S	L	L	C	S	E
O	N	N	O	U	T
H	H	L	M	M	X

SOH: 01h
 LNH: 00h
 LNL: 01h
 COM: 38h
 SUM: C7h
 ETX: 03h

(2) Data packet structure

S	L	L	R	T	O	O	C	C	S	E
O	N	N	E	Y	S	S	P	P	U	T
D	H	L	S	P	A	I	A	I	M	X

SOD: 81h
 LNH: 00h
 LNL: 19h
 RES: 38h (OK)
 TYP: Type code (8 bytes)*1
 OSA: Maximum input frequency (4 bytes)
 OSI: Minimum input frequency (4 bytes)
 CPA: Maximum system clock frequency (4 bytes)
 CPI: Minimum system clock frequency (4 bytes)
 SUM: Sum of values
 ETX: 03h

An example of the values sent is given below.

Maximum input frequency = 16,000,000 Hz

OSA (1st byte): 00h

OSA (2nd byte): F4h

OSA (3rd byte): 24h

OSA (4th byte): 00h

Minimum input frequency = 16,000,000 Hz

OSI (1st byte): 00h

OSI (2nd byte): F4h

OSI (3rd byte): 24h

OSI (4th byte): 00h

Maximum system clock (ICLK) = 120,000,000 Hz

CPA (1st byte): 07h

CPA (2nd byte): 27h

CPA (3rd byte): 0Eh

CPA (4th byte): 00h

Minimum system clock: (ICLK) = 120,000,000 Hz

CPI (1st byte): 07h

CPI (2nd byte): 27h

CPI (3rd byte): 0Eh

CPI (4th byte): 00h

In boot mode (USB interface), transfer is at the input frequency when the value of OSA or OSI is in use (20 MHz or 24 MHz).

Note 1. Reserved data

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 38h (OK)
 SUM: C7h
 ETX: 03h

(4) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: B8h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(5) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: B8h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.10 Endian Notification Command

This command is used to inform the MCU of the endian (big or little).

Specify either endian as the endian information according to the data to be programmed.

This command can only be accepted in the communications establishment phase.

(1) Command packet structure

S	L	L	C	E	S	E
O	N	N	O	N	U	T
H	H	L	M	D	M	X

SOH: 01h
 LNH: 00h
 LNL: 02h
 COM: 36h
 END: Endian information
 00h (big-endian)
 01h (little-endian)
 SUM: Sum of values
 ETX: 03h

(2) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 36h (OK)
 SUM: C9h
 ETX: 03h

(3) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: B6h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 D7h (endian error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.11 Frequency Setting Command

This command is used to set the values of oscillation frequency and CPU operating frequency (in Hz).

In boot mode (SCI or FINE interface), the HOCO runs at 16 MHz and the ICLK runs at 120 MHz, so set the input frequency to 16 MHz and the system clock frequency to 120 MHz. Additionally, in boot mode (SCI or FINE interface), the FCLK and PCLKB run at 60 MHz, so the MCU returns the hexadecimal value corresponding to 60,000,000 to indicate the frequency of the peripheral module clock.

In boot mode (USB interface), the input frequency in use is 20 MHz or 24 MHz and the ICLK runs at 120 MHz, so set the input frequency to 20 MHz or 24 MHz and the system clock frequency to 120 MHz. Additionally, in boot mode (USB interface), the FCLK runs at 60 MHz and the UCLK runs at 48 MHz, so the MCU returns the hexadecimal value corresponding to 48,000,000 to indicate the frequency of the peripheral module clock.

This command can only be accepted in the communications establishment phase.

(1) Command packet structure

S	L	L	C	O	O	O	O	C	C	C	C	S	E
O	N	N	O	C	C	C	C	C	C	C	C	U	T
H	H	L	M	1	2	3	4	1	2	3	4	M	X

SOH: 01h

LNH: 00h

LNL: 09h

COM: 32h

Ex. When the input frequency is 16,000,000 Hz and the system clock frequency is 120,000,000 Hz, send the values as below.

OC1: 00h CC1: 07h

OC2: F4h CC2: 27h

OC3: 24h CC3: 0Eh

OC4: 00h CC4: 00h

OC1: Input frequency

OC2: Input frequency

OC3: Input frequency

OC4: Input frequency

CC1: System clock frequency

CC2: System clock frequency

CC3: System clock frequency

CC4: System clock frequency

SUM: Sum of values

ETX: 03h

(2) Data packet structure

S	L	L	R	F	F	F	F	P	P	P	P	S	E
O	N	N	E	Q	Q	Q	Q	F	F	F	F	U	T
D	H	L	S	1	2	3	4	1	2	3	4	M	X

SOD: 81h

LNH: 00h

LNL: 09h

RES: 32h

Ex. An example of the values sent is given below.

System clock frequency = 120,000,000 Hz

Peripheral clock frequency = 60,000,000 Hz

FQ1: 07h PF1: 03h

FQ2: 27h PF2: 93h

FQ3: 0Eh PF3: 87h

FQ4: 00h PF4: 00h

FQ1: System clock frequency

FQ2: System clock frequency

FQ3: System clock frequency

FQ4: System clock frequency

PF1: Peripheral clock frequency

PF2: Peripheral clock frequency

PF3: Peripheral clock frequency

PF4: Peripheral clock frequency

SUM: Sum of values

ETX: 03h

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h

LNH: 00h

LNL: 01h

RES: 32h (OK)

SUM: CDh

ETX: 03h

(4) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: B2h (error)

ERR: Error code

C1h (packet error)

C2h (checksum error)

C3h (flow error)

D1h (input frequency error)

D2h (system clock (ICLK) frequency error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.12 Bit-Rate Setting Command

This command is used to change the bit rate after receiving bit rate data (in bps).

If an error occurs, the bit rate is not changed.

This command can only be accepted in the communications establishment phase.

In boot mode (USB or FINE interface), the bit rate is not changed. The bit rate should be any desired value.

(1) Procedure

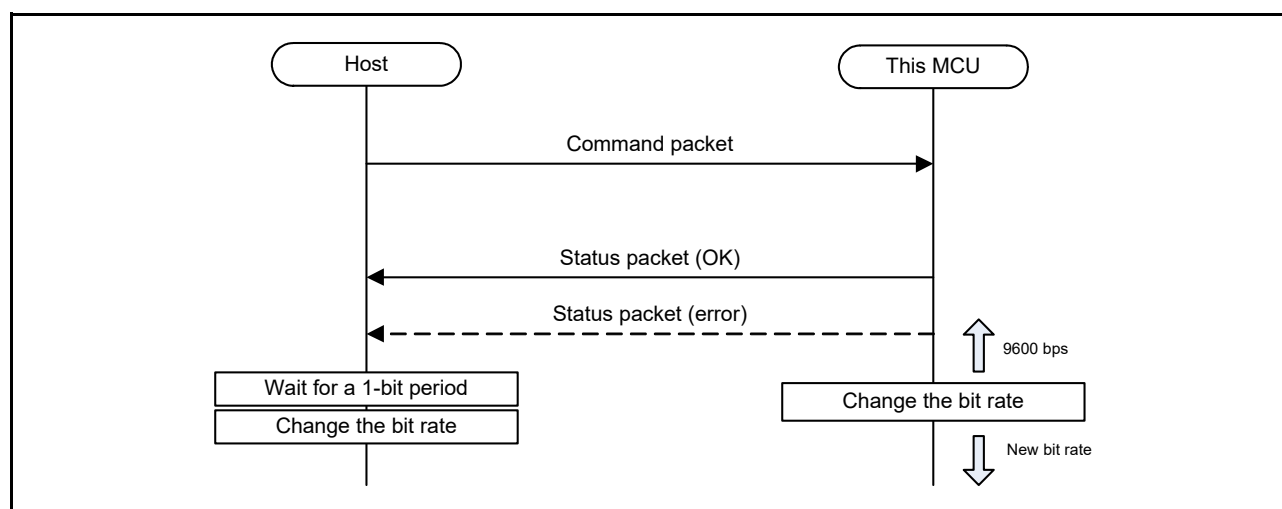


Figure 62.45 Bit-Rate Setting Command Transfer Sequence

(2) Command packet structure

S	L	L	C	B	B	B	B	S	E
O	N	N	O	R	R	R	R	U	T
H	H	L	M	1	2	3	4	M	X

When the bit rate is 2,000,000 bps, send the values as below.

BR1: 00h

BR2: 1Eh

BR3: 84h

BR4: 80h

SOH: 01h

LNH: 00h

LNL: 05h

COM: 34h

BR1: Bit rate

BR2: Bit rate

BR3: Bit rate

BR4: Bit rate

SUM: Sum of values

ETX: 03h

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h

LNH: 00h

LNL: 01h

RES: 34h (OK)

SUM: CBh

ETX: 03h

(4) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: B4h (error)

ERR: Error code

C1h (packet error)

C2h (checksum error)

C3h (flow error)

D4h (bit rate error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.13 Synchronization Command

This command is used in processing to synchronize communications.

It is also used when checking whether the MCU is ready to accept commands. If a serial programmer connection is prohibited, a serial programmer connection prohibition error is returned.

This command can be accepted in both the communications establishment and command waiting phases.

(1) Command packet structure

S	L	L	C	S	E
O	N	N	O	U	T
H	H	L	M	M	X

SOH: 01h
 LNH: 00h
 LNL: 01h
 COM: 00h
 SUM: FFh
 ETX: 03h

(2) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 00h (OK)
 SUM: FFh
 ETX: 03h

(3) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 80h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 DCh (serial programmer connection prohibition error)
 SUM: Sum of values
 ETX: 03h

62.15.14 ID Authentication Mode Acquisition Command

This command sends the enabled state of serial programmer ID code protection to the host.

This command can be accepted in both the communications establishment and command waiting phases.

(1) Command packet structure

S	L	L	C	S	E
O	N	N	O	U	T
H	H	L	M	M	X

SOH: 01h
 LNH: 00h
 LNL: 01h
 COM: 2Ch
 SUM: D3h
 ETX: 03h

(2) Data packet structure

S	L	L	R	M	S	E
O	N	N	E	O	U	T
D	H	L	S	D	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 2Ch (OK)
 MOD: ID authentication information (1 byte)
 00h (serial programmer ID code protection is enabled)
 SUM: Sum of values
 ETX: 03h

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 2Ch (OK)
 SUM: D3h
 ETX: 03h

(4) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: ACh (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(5) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: ACh (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.15 Serial Programming ID Code Check Command

This command sends the result when the control and ID codes set in the OSIS register match those received from the host.

This command can be accepted in the communications establishment phase. When the serial programmer ID code protection is enabled, the MCU does not enter the command waiting phase unless processing in response to this command ends normally.

If the control code is 45h, the codes do not match three times in a row, all data in the flash memory are erased.*1

Note 1. When the FAW.FSPR bit is 0, data in the flash memory are not erased.

(1) Command packet structure

S	L	L	C	I	S	E
O	N	N	O	D	U	T
H	H	L	M	C	M	X

SOH: 01h

LNH: 00h

LNL: 11h

COM: 30h

IDC: Control code and ID code (16 bytes)*1

SUM: Sum of values

ETX: 03h

Note 1. Send the values as below.

<ID code>

ID = 0F0E0D0C0B0A09080706050403020100h

(Control code: 00h, ID code 2: 01h, ID code 3: 02h, ... , ID code 16: 0Fh)

<Data for transmission>

1st byte	2nd byte	3rd byte	4th byte	5th byte	6th byte	7th byte	8th byte	...
00h	01h	02h	03h	04h	05h	06h	07h	...

(2) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h

LNH: 00h

LNL: 01h

RES: 30h (OK)

SUM: CFh

ETX: 03h

(3) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: B0h (error)

ERR: Error code

C1h (packet error)

C2h (checksum error)

C3h (flow error)

DBh (ID code mismatch error)

E1h (Error of erasure)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.16 Blank Check Command

This command checks whether the specified area is blank.

Specify addresses aligned with 128-byte boundaries for the code flash memory and with 4-byte boundaries for the data flash memory. The option-setting memory cannot be specified. When trusted memory (TM) is enabled, an error occurs if the area for blank-checking include any part of the area being handled as TM.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	S	S	S	S	E	E	E	E	S	E	SOH: 01h
O	N	N	O	H	H	L	L	H	H	L	L	U	T	LNH: 00h
H	H	L	M	H	L	H	L	H	L	H	L	M	X	LNL: 09h
COM: 10h														SHH: Blank check start address (24 to 31 bits)
														SHL: Blank check start address (16 to 23 bits)
														SLH: Blank check start address (8 to 15 bits)
														SLL: Blank check start address (0 to 7 bits)
														EHH: Blank check end address (24 to 31 bits)
														EHL: Blank check end address (16 to 23 bits)
														ELH: Blank check end address (8 to 15 bits)
														ELL: Blank check end address (0 to 7 bits)
														SUM: Sum of values
														ETX: 03h

(2) Status packet structure, normal termination

S	L	L	R	S	E	SOD: 81h
O	N	N	E	U	T	LNH: 00h
D	H	L	S	M	X	LNL: 01h
						RES: 10h (OK)
						SUM: EFh
						ETX: 03h

(3) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
LNH: 00h
LNL: 02h
RES: 90h (error)
ERR: Error code
C1h (packet error)
C2h (checksum error)
C3h (flow error)
D0h (address error)
E0h (non-blank error)
SUM: Sum of values
ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.17 Block Erase Command

This command is used to erase a specified single block.

Specify the block for erasure as the first address in the block.

When the TM function is enabled, blocks targeted for TM target area cannot be erased. The option-setting memory cannot be erased.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	S	S	S	S	S	E
O	N	N	O	H	H	L	L	U	T
H	H	L	M	H	L	H	L	M	X

SOH: 01h

LNH: 00h

LNL: 05h

COM: 12h

SHH: First address in the block for erasure (24 to 31 bits)

SHL: First address in the block for erasure (16 to 23 bits)

SLH: First address in the block for erasure (8 to 15 bits)

SLL: First address in the block for erasure (0 to 7 bits)

SUM: Sum of values

ETX: 03h

(2) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h

LNH: 00h

LNL: 01h

RES: 12h (OK)

SUM: EDh

ETX: 03h

(3) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: 92h (error)

ERR: Error code

C1h (packet error)

C2h (checksum error)

C3h (flow error)

D0h (address error)

DAh (protection error)

E1h (erase error)

E7h (flash sequencer error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.18 Area Erase Command

The area erase command erases the specified area successively, block by block, in ascending order of addresses.

The target area can be selected from the code flash memory or data flash memory.

When the TM function is enabled, blocks other than those being handled as TM are erased.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	A	S	E
O	N	N	O	R	U	T
H	H	L	M	E	M	X

SOH: 01h
 LNH: 00h
 LNL: 02h
 COM: 50h
 ARE: Area
 00h (code flash memory)
 20h (data flash memory)
 SUM: Sum of values
 ETX: 03h

(2) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 50h (OK)
 SUM: AFh
 ETX: 03h

(3) Status packet structure, error occurrence 1

S	L	L	R	S	E	S	E
O	N	N	E	U	R	U	T
D	H	L	S	M	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: D0h (Error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 D5h (area error)
 DAh (protection error)
 E1h (erase error)
 E7h (flash sequencer error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.19 Program Command

This command sets up the MCU for data to be programmed to its flash memory and specifies the area where the data are to be programmed.

Specify the data length in 128-byte units for the code flash memory and 4-byte units for the data flash memory. In addition, the program start address must be specified in 128-byte aligned for the code flash memory and 4-byte aligned for data flash memory. When the TM function is enabled, areas including the TM target area cannot be programmed. This command can only be accepted in the command waiting phase.

(1) Procedure

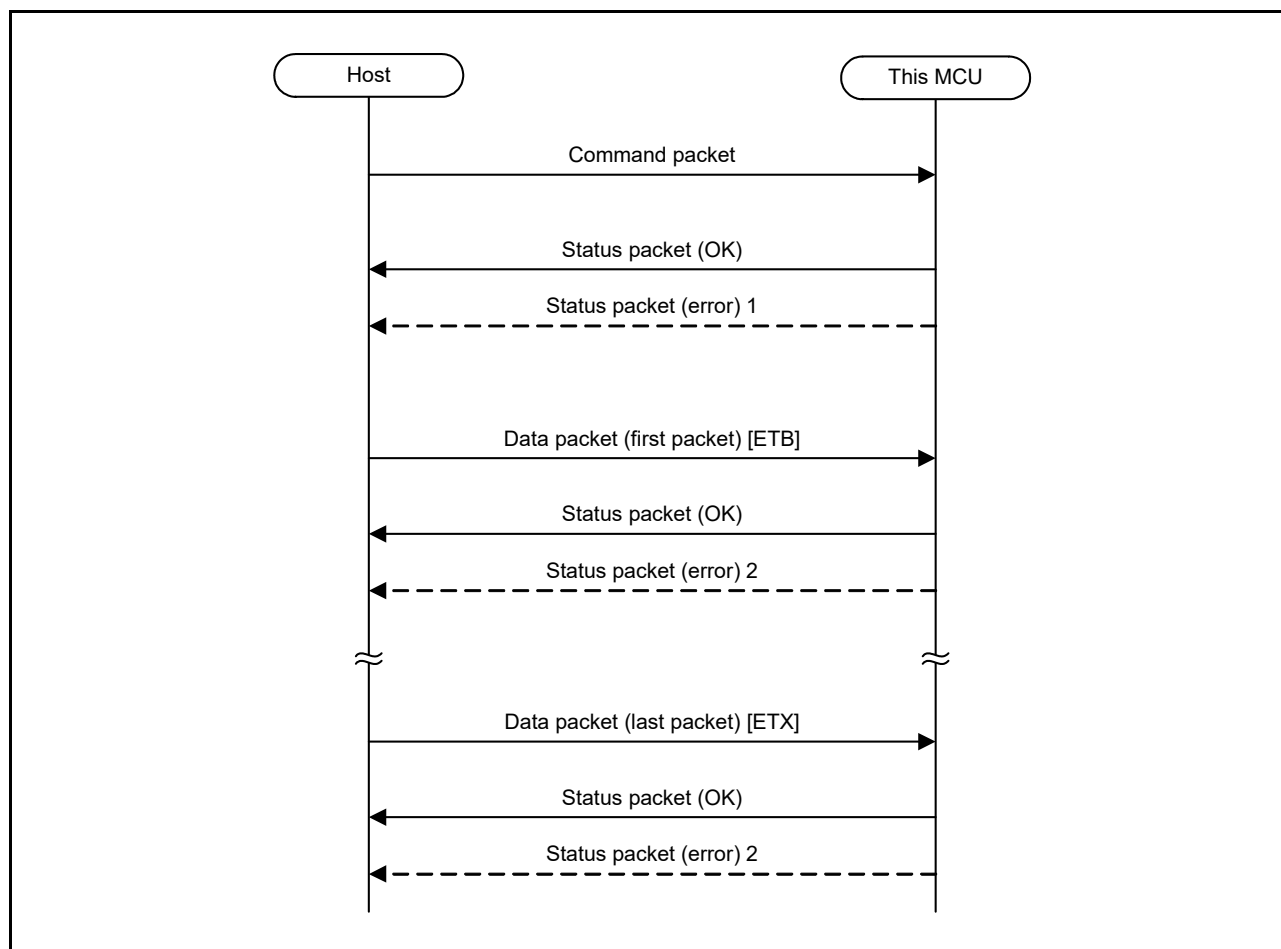


Figure 62.46 Program Command Transfer Sequence

(2) Command packet structure

S	L	L	C	S	S	S	S	E	E	E	E	S	E
O	N	N	O	H	H	L	L	H	H	L	L	U	T
H	H	L	M	H	L	H	L	H	L	H	L	M	X

SOH: 01h
 LNH: 00h
 LNL: 09h
 COM: 13h
 SHH: Program start address (24 to 31 bits)
 SHL: Program start address (16 to 23 bits)
 SLH: Program start address (8 to 15 bits)
 SLL: Program start address (0 to 7 bits)
 EHH: Program end address (24 to 31 bits)
 EHL: Program end address (16 to 23 bits)
 ELH: Program end address (8 to 15 bits)
 ELL: Program end address (0 to 7 bits)
 SUM: Sum of values
 ETX: 03h

(3) Data packet structure

S	L	L	R	Data				S	E	E
O	N	N	E					U	T	T
D	H	L	S					M	B	r
										X

SOD: 81h
 LNH: Data length + 1 (8 to 15 bits)
 LNL: Data length + 1 (0 to 7 bits)
 RES: 13h (OK)
 Data: Program data
 SUM: Sum of values
 ETB: 17h
 ETX: 03h

(4) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 13h (OK)
 SUM: ECh
 ETX: 03h

(5) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 93h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 D0h (address error)
 DAh (protection error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(6) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: 93h (error)

ERR: Error code

C1h (packet error)

C2h (checksum error)

E2h (program error)

E7h (flash sequencer error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.20 Read Command

This command is used to read data from the specified area in the flash memory and send it to the host.

The minimum unit for reading is 1 byte. When the TM target area is read while the TM function is enabled, 0 is read. The option-setting memory cannot be specified.

This command can only be accepted in the command waiting phase.

(1) Procedure

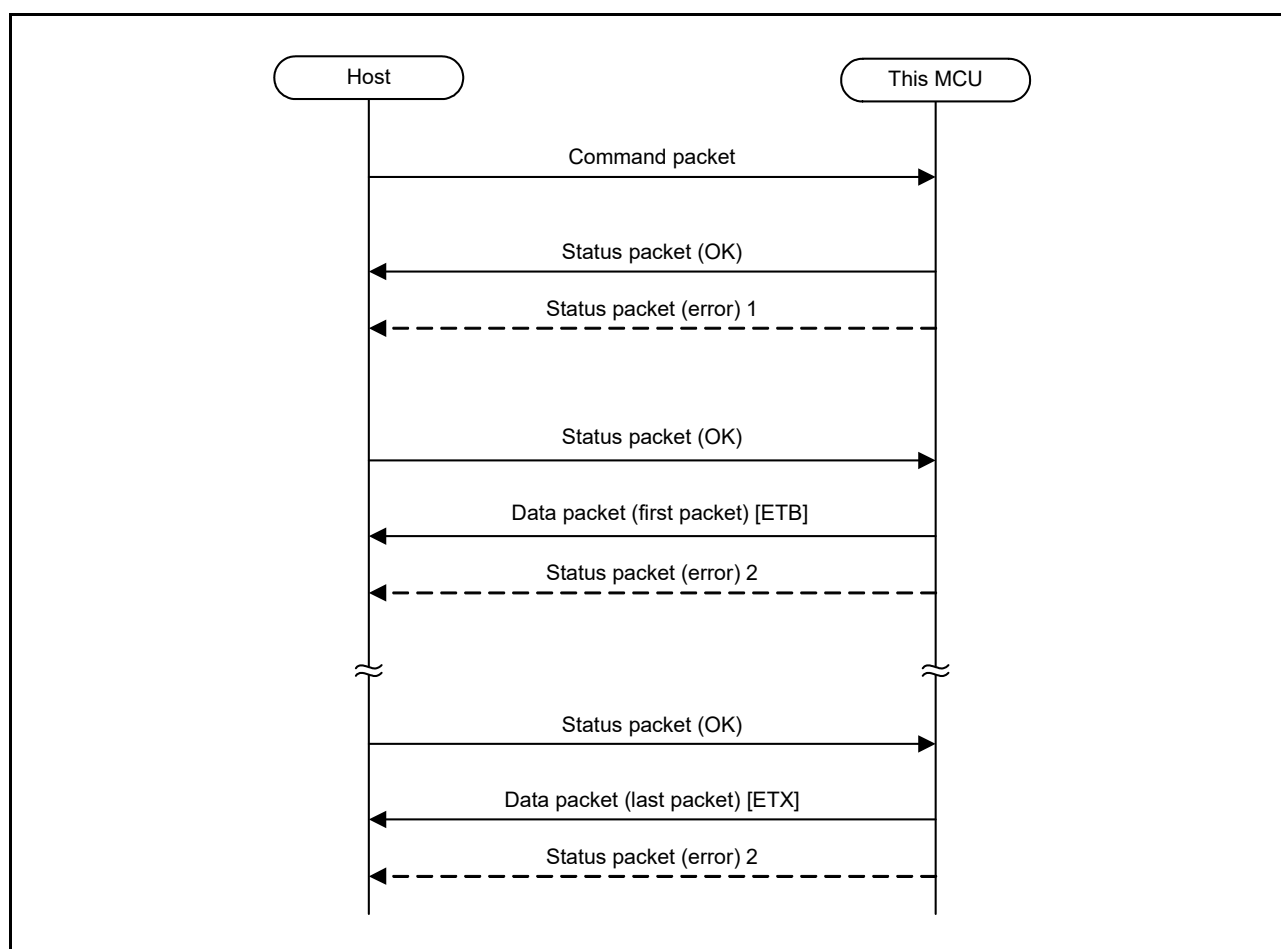


Figure 62.47 Read Command Transfer Sequence

(2) Command packet structure

S	L	L	C	S	S	S	S	E	E	E	E	S	E
O	N	N	O	H	H	L	L	H	H	L	L	U	T
H	H	L	M	H	L	H	L	H	L	H	L	M	X

SOH: 01h
 LNH: 00h
 LNL: 09h
 COM: 15h
 SHH: Read start address (24 to 31 bits)
 SHL: Read start address (16 to 23 bits)
 SLH: Read start address (8 to 15 bits)
 SLL: Read start address (0 to 7 bits)
 EHH: Read end address (24 to 31 bits)
 EHL: Read end address (16 to 23 bits)
 ELH: Read end address (8 to 15 bits)
 ELL: Read end address (0 to 7 bits)
 SUM: Sum of values
 ETX: 03h

(3) Data packet structure

S	L	L	R	Data				S	E	E
O	N	N	E					U	T	or T
D	H	L	S					M	B	X

SOD: 81h
 LNH: Data length + 1 (8 to 15 bits)
 LNL: Data length + 1 (0 to 7 bits)
 RES: 15h (OK)
 Data: Read data
 SUM: Sum of values
 ETB: 17h
 ETX: 03h

(4) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 15h (OK)
 SUM: EAh
 ETX: 03h

(5) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 95h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 D0h (address error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(6) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 95h (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.21 Configuration Clear Command

This command is used to clear the option-setting memory (configuration setting area).

This command also erases the TM target areas. Therefore, when this command is in use, do not set the TM target areas outside the access window when setting an access window. The TM function is disabled after a reset.

While the FAW.FSPR bit is valid, a protection error occurs.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	S	E
O	N	N	O	U	T
H	H	L	M	M	X

SOH: 01h
 LNH: 00h
 LNL: 01h
 COM: 1Ch
 SUM: E3h
 ETX: 03h

(2) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 1Ch (OK)
 SUM: E3h
 ETX: 03h

(3) Status packet structure, error occurrence

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 9Ch (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 DAh (protection error)
 E0h (blank error)
 E1h (erase error)
 E2h (program error)
 E7h (flash sequencer error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.22 Simple Addition Checksum Command

This command is used to calculate the sum of values in the specified area and send the result to the host. While the TM function is enabled, however, the values in the TM target areas are not included in the calculation.

The target area of this command can be selected from the code flash memory or data flash memory. Calculation is by simple addition. If the size of each area is less than 8×2^n Kbytes, the area range from the final address of each area to 8×2^n Kbyte area is supplemented by FFh. The initial value is 0 and the sum of values in the specified area is obtained by adding the values of all bytes. If this command is issued for the data flash memory that includes erased blocks, the result is undefined. When a simple addition checksum is to be executed for the data flash memory, make sure that data have been written throughout the specified area.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	A	S	E
O	N	N	O	R	U	T
H	H	L	M	E	M	X

SOH: 01h
 LNH: 00h
 LNL: 02h
 COM: 4Dh
 ARE: Area information
 00h (code flash memory)
 20h (data flash memory)
 SUM: Sum of values
 ETX: 03h

(2) Data packet structure

S	L	L	R	S	S	S	S	S	E
O	N	N	E	D	D	D	D	U	T
D	H	L	S	1	2	3	4	M	X

SOD: 81h
 LNH: 00h
 LNL: 05h
 RES: 4Dh (OK)
 SD1: Sum of values
 SD2: Sum of values
 SD3: Sum of values
 SD4: Sum of values
 If the sum of values is 01234567h, the settings are as follows.
 SD1 = 01h
 SD2 = 23h
 SD3 = 45h
 SD4 = 67h
 SUM: Sum of values
 ETX: 03h

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 4Dh (OK)
 SUM: B2h
 ETX: 03h

(4) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: CDh (error)
 ERR: Error code
 C1h (packet error)
 C2h (checksum error)
 C3h (flow error)
 D5h (area error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(5) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: CDh (error)

ERR: Error code

C1h (packet error)

C2h (checksum error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.23 Configuration Program Command

This command is used to receive data to be written to the option-setting memory and to program the data in the specified area.

Programming must be specified in the unit acquired by using the area information acquisition command.

This command can only be accepted in the command waiting phase.

Programming in the area including the FAW register while the FAW.FSPR bit is enabled leads to a protection error.

Once the FAW.FSPR bit has been enabled, the bit cannot be disabled.

For details of the FAW.FSPR bit, refer to [section 7.2.9, Flash Access Window Setting Register \(FAW\)](#) in [section 7, Option-Setting Memory \(OFSM\)](#).

Rewriting the SPCC.SPE bit while the connection of serial programmer is prohibited leads to normal termination, but programming does not proceed.

In addition, this MCU cannot be connected to a serial programmer at a reset while connection to a serial programmer is disabled.

Writing to the TMEF.TMEF[2:0] bits, TMINF register, and MDE.BANKMD[2:0] bits while the TMEF.TMEF[2:0] bits are 000b in linear mode leads to normal termination, but programming does not proceed.

Writing to the TMEF.TMEF[2:0] bits, TMINF register, and MDE.BANKMD[2:0] bits while the TMEF.TMEF[2:0] bits or the TMEF.TMEFDB[2:0] bits are 000b in dual mode leads to normal termination, but programming does not proceed.

(1) Procedure

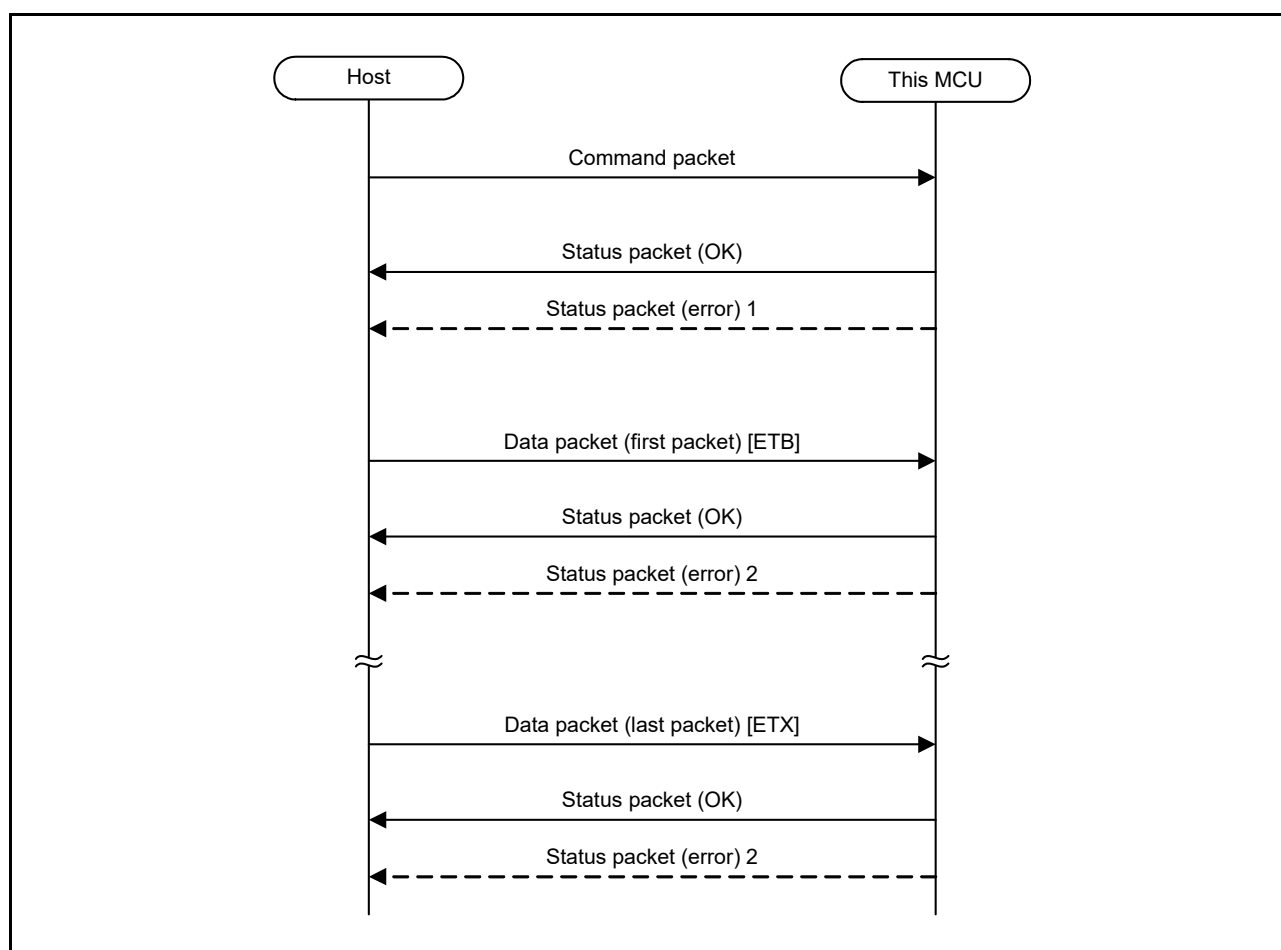


Figure 62.48 Configuration Program Command Transfer Sequence

(2) Command packet structure

S	L	L	C	S	S	S	S	E	E	E	E	S	E	
O	N	N	O	H	H	L	L	H	H	L	L	U	T	
H	H	L	M	H	L	H	L	H	L	H	L	M	X	

SOH: 01h
 LNH: 00h
 LNL: 09h
 COM: 51h
 SHH: Start address (24 to 31 bits)
 SHL: Start address (16 to 23 bits)
 SLH: Start address (8 to 15 bits)
 SLL: Start address (0 to 7 bits)
 EHH: End address (24 to 31 bits)
 EHL: End address (16 to 23 bits)
 ELH: End address (8 to 15 bits)
 ELL: End address (0 to 7 bits)
 SUM: Sum of values
 ETX: 03h

(3) Data packet structure

S	L	L	R		S	E	E	
O	N	N	E	Data	U	T	or	T
D	H	L	S		M	B		X

SOD: 81h
 LNH: Data length + 1 (8 to 15 bits)
 LNL: Data length + 1 (0 to 7 bits)
 RES: 51h (OK)
 Data: Program data*1
 SUM: Sum of values
 ETB: 17h
 ETX: 03h

Note 1. The data for programming must be sent from the host in order from that for the lowest address to that for the highest address.

<Write data>

Address	0h	1h	2h	3h	4h	5h	6h	7h	...
Data	00h	01h	02h	03h	04h	05h	06h	07h	...

<Program data>

1st byte	2nd byte	3rd byte	4th byte	5th byte	6th byte	7th byte	8th byte	...
00h	01h	02h	03h	04h	05h	06h	07h	...

(4) Status packet structure, normal termination

S	L	L	R	S	E	
O	N	N	E	U	T	
D	H	L	S	M	X	

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 51h (OK)
 SUM: AEh
 ETX: 03h

(5) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: D1h (Error)
 ERR: Error code
 C1h (Packet error)
 C2h (Checksum error)
 C3h (Flow error)
 D0h (address error)
 DAh (protection error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(6) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: D1h (Error)

ERR: Error code

C1h (Packet error)

C2h (Checksum error)

E1h (Erase error)

E2h (Program error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.24 Configuration Read Command

This command is used to read data from the area specified by the option-setting memory and to transmit the data to the host.

The minimum unit for reading is 4 bytes.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	S	S	S	S	E	E	E	E	S	E	
O	N	N	O	H	H	L	L	H	H	L	L	U	T	
H	H	L	M	H	L	H	L	H	L	H	L	M	X	

SOH: 01h
LNH: 00h
LNL: 09h
COM: 52h
SHH: Start address (24 to 31 bits)
SHL: Start address (16 to 23 bits)
SLH: Start address (8 to 15 bits)
SLL: Start address (0 to 7 bits)
EHH: End address (24 to 31 bits)
EHL: End address (16 to 23 bits)
ELH: End address (8 to 15 bits)
ELL: End address (0 to 7 bits)
SUM: Sum of values
ETX: 03h

(2) Data packet structure

S	L	L	R		S	E	E	
O	N	N	E	Data	U	T	or	T
D	H	L	S		M	B		X

SOD: 81h
LNH: Data length + 1 (8 to 15 bits)
LNL: Data length + 1 (0 to 7 bits)
RES: 52h (OK)
Data: Read data*1
SUM: Sum of values
ETB: 17h
ETX: 03h

Note 1. The data is read in the order from the lower address to the higher address.

<Program data>

Address	0h	1h	2h	3h	4h	5h	6h	7h	...
Data	00h	01h	02h	03h	04h	05h	06h	07h	...

<Read data>

1st byte	2nd byte	3rd byte	4th byte	5th byte	6th byte	7th byte	8th byte	...
00h	01h	02h	03h	04h	05h	06h	07h	...

(3) Status packet structure, normal termination

S	L	L	R	S	E	
O	N	N	E	U	T	
D	H	L	S	M	X	

SOD: 81h
LNH: 00h
LNL: 01h
RES: 52h (OK)
SUM: ADh
ETX: 03h

(4) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: D2h (Error)
 ERR: Error code
 C1h (Packet error)
 C2h (Checksum error)
 C3h (Flow error)
 D0h (address error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(5) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: D2h (Error)
 ERR: Error code
 C1h (Packet error)
 C2h (Checksum error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.25 Acquisition of the Number of the Area Information Command

This command is used to transmit the number of area that the MCU has to the host.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	S	E
O	N	N	O	U	T
H	H	L	M	M	X

SOH: 01h
 LNH: 00h
 LNL: 01h
 COM: 53h
 SUM: ACh
 ETX: 03h

(2) Data packet structure

S	L	L	R	N	S	E
O	N	N	E	O	U	T
D	H	L	S	A	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: 53h (OK)
 NOA: Number of area information (1 byte)
 05h (in linear mode)
 08h (in dual mode)
 SUM: Sum of values
 ETX: 03h

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
 LNH: 00h
 LNL: 01h
 RES: 53h (OK)
 SUM: ACh
 ETX: 03h

(4) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: D3h (Error)
 ERR: Error code
 C1h (Packet error)
 C2h (Checksum error)
 C3h (Flow error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(5) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
 LNH: 00h
 LNL: 02h
 RES: D3h (Error)
 ERR: Error code
 C1h (Packet error)
 C2h (Checksum error)
 SUM: Sum of values
 ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.26 Area Information Acquisition Command

This command is used to transmit information of the specified area to the host.

The address of the erase command, program command, or configuration program command must be specified according to the range and unit (alignment) returned by this command.

This command can only be accepted in the command waiting phase.

(1) Command packet structure

S	L	L	C	N	S	E
O	N	N	O	U	U	T
H	H	L	M	M	M	X

SOH: 01h
 LNH: 00h
 LNL: 02h
 COM: 54h
 NUM: Area number [0 to NOA-1]
 SUM: C5h
 ETX: 03h

(2) Data packet structure

S	L	L	R	K	S	E	E	W	S	E
O	N	N	E	O	A	A	A	A	U	T
D	H	L	S	A	D	D	U	U	M	X

SOD: 81h
 LNH: 00h
 LNL: 12h
 RES: 54h (OK)
 KOA: Kind of the area
 00h (code flash memory)
 20h (data flash memory)
 30h (option-setting memory)
 40h (Trusted Memory disabled)
 41h (Trusted Memory enabled)
 SAD: Start address
 EAD: End address
 EAU: Erase access unit (alignment) [Byte]
 WAU: Write access unit (alignment) [Byte]
 SUM: Sum of values
 ETX: 03h

Data packet in linear mode

NUM	KOA	SAD	EAD	EAU	WAU	Description
00h	00h	FFFF 0000h	FFFF FFFFh	0000 2000h	0000 0080h	Code flash memory blocks 0 to 7 (8-Kbyte block)
01h	00h	FFC0 0000h	FFFE FFFFh	0000 8000h	0000 0080h	Code flash memory blocks 8 to 133 (32-Kbyte block)
02h	20h	0010 0000h	0010 7FFFh	0000 0040h	0000 0004h	Data flash memory
03h	30h	FE7F 5D00h	FE7F 5D7Fh	0000 0080h	0000 0010h	Option-setting memory
04h	40h	FFFE 0000h	FFFE FFFFh	0000 8000h	0000 0080h	Trusted Memory blocks 8 and 9 disabled
	41h	FFFE 0000h	FFFE FFFFh	0000 8000h	0000 0080h	Trusted Memory blocks 8 and 9 enabled

Settings of the TMEF.TMEF[2:0] bits and TMEF.TMEFDB[2:0] bits and setting of KOA while NUM is 04h in linear mode

TMEF.TMEF[2:0] bits	TMEF.TMEFDB[2:0] bits	KOA while NUM is 04h
111b	Don't care	40h
000b	Don't care	41h

Data packet in dual mode

NUM	KOA	SAD	EAD	EAU	WAU	Description
00h	00h	FFFF 0000h	FFFF FFFFh	0000 2000h	0000 0080h	Code flash memory blocks 0 to 7 (8-Kbyte block)
01h	00h	FFE0 0000h	FFFE FFFFh	0000 8000h	0000 0080h	Code flash memory blocks 8 to 69 (32-Kbyte block)
02h	00h	FFDF 0000h	FFDF FFFFh	0000 2000h	0000 0080h	Code flash memory blocks 70 to 77 (8-Kbyte block)
03h	00h	FFC0 0000h	FFDE FFFFh	0000 8000h	0000 0080h	Code flash memory blocks 78 to 139 (32-Kbyte block)
04h	20h	0010 0000h	0010 7FFFh	0000 0040h	0000 0004h	Data flash memory
05h	30h	FE7F 5D00h	FE7F 5D7Fh	0000 0080h	0000 0010h	Option-setting memory
06h	40h	FFFE 0000h	FFFE FFFFh	0000 8000h	0000 0080h	Trusted Memory blocks 8 and 9 disabled
	41h	FFFE 0000h	FFFE FFFFh	0000 8000h	0000 0080h	Trusted Memory blocks 8 and 9 enabled
07h	40h	FFDE 0000h	FFDE FFFFh	0000 8000h	0000 0080h	Trusted Memory blocks 78 and 79 disabled
	41h	FFDE 0000h	FFDE FFFFh	0000 8000h	0000 0080h	Trusted Memory blocks 78 and 79 enabled

Settings of the TMEF.TMEF[2:0] bits and TMEF.TMEFDB[2:0] bits and setting of KOA while NUM is 06h or 07h in dual mode

TMEF.TMEF[2:0] bits	TMEF.TMEFDB[2:0] bits	KOA while NUM is 06h or 07h
111b	111b	40h
000b	000b	41h

(3) Status packet structure, normal termination

S	L	L	R	S	E
O	N	N	E	U	T
D	H	L	S	M	X

SOD: 81h
LNH: 00h
LNL: 01h
RES: 54h (OK)
SUM: ABh
ETX: 03h

(4) Status packet structure, error occurrence 1

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h
LNH: 00h
LNL: 02h
RES: D4h (Error)
ERR: Error code
C1h (Packet error)
C2h (Checksum error)
C3h (Flow error)
SUM: Sum of values
ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

(5) Status packet structure, error occurrence 2

S	L	L	R	E	S	E
O	N	N	E	R	U	T
D	H	L	S	R	M	X

SOD: 81h

LNH: 00h

LNL: 02h

RES: D4h (Error)

ERR: Error code

C1h (Packet error)

C2h (Checksum error)

SUM: Sum of values

ETX: 03h

After the error code is returned, the chip returns to the command waiting state.

62.15.27 Usage Example

(1) Example of the Procedure for Reprogramming

Figure 62.49 shows an example of the procedure for reprogramming.

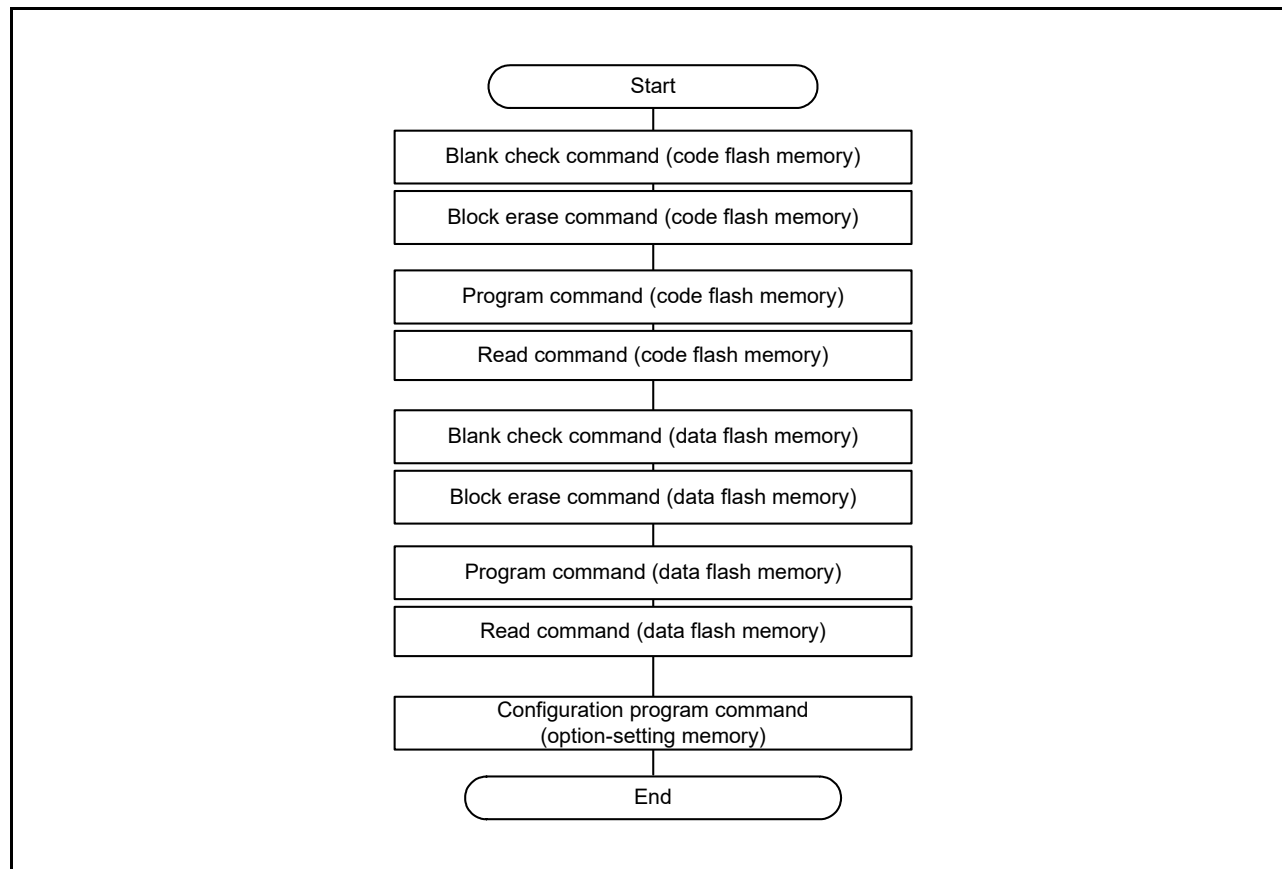


Figure 62.49 Example of Reprogramming Procedure

62.15.28 Rewriting Flash Memory When Using Dual Mode

The flash memory is erased (blank) as shipped by Renesas, so the setting of the MDE.BANKMD[2:0] bits is 111b (linear mode). To use the flash memory in dual mode, set the MDE.BANKMD[2:0] bits to 000b (dual mode).

(1) Rewriting in Dual Mode

Figure 62.50 shows an example of rewriting flash memory after switching to dual mode. The user program can be written to bank 0, bank 1, or both.

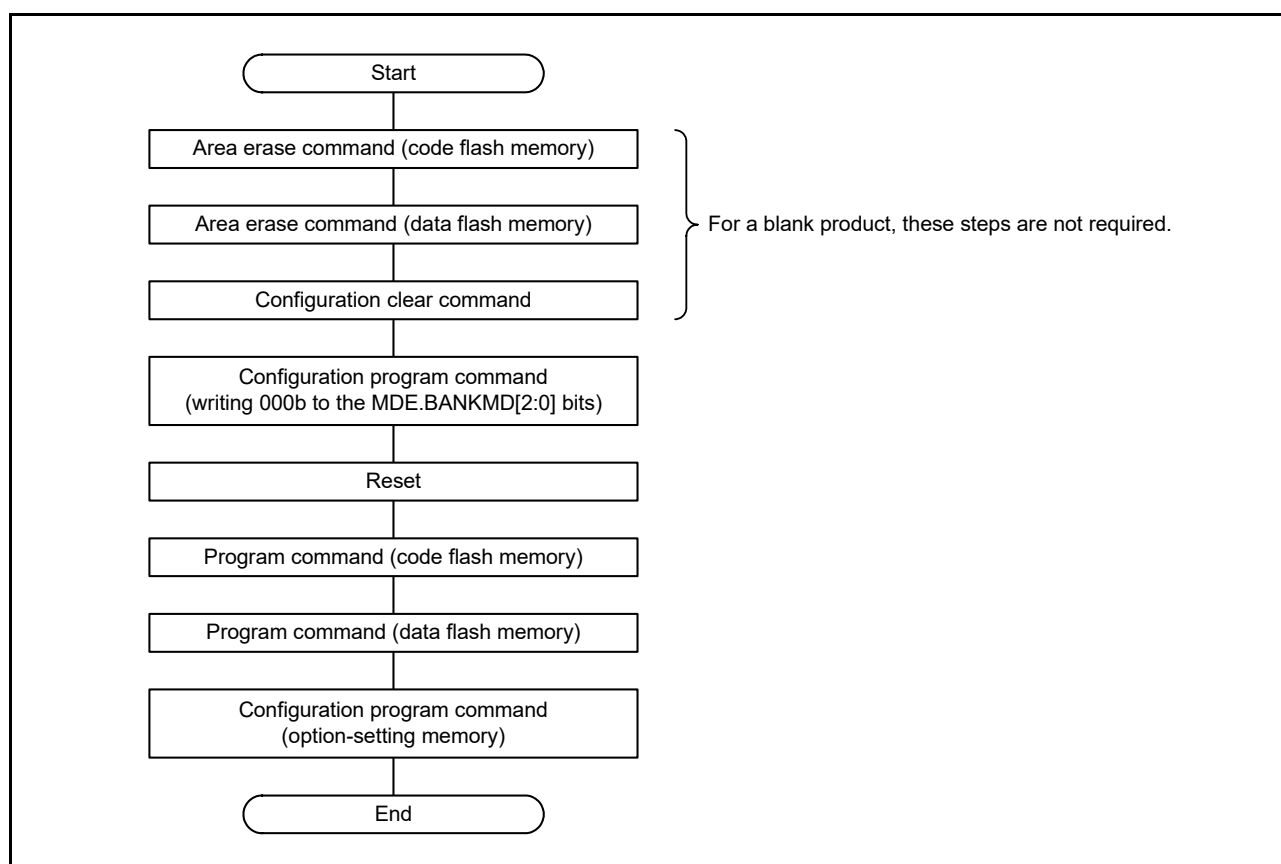


Figure 62.50 Example of Rewriting Flash Memory after Switching to Dual Mode

(2) Rewriting in Linear Mode

Figure 62.51 shows an example of rewriting flash memory in linear mode. Figure 62.52 shows the memory map before and after a reset.

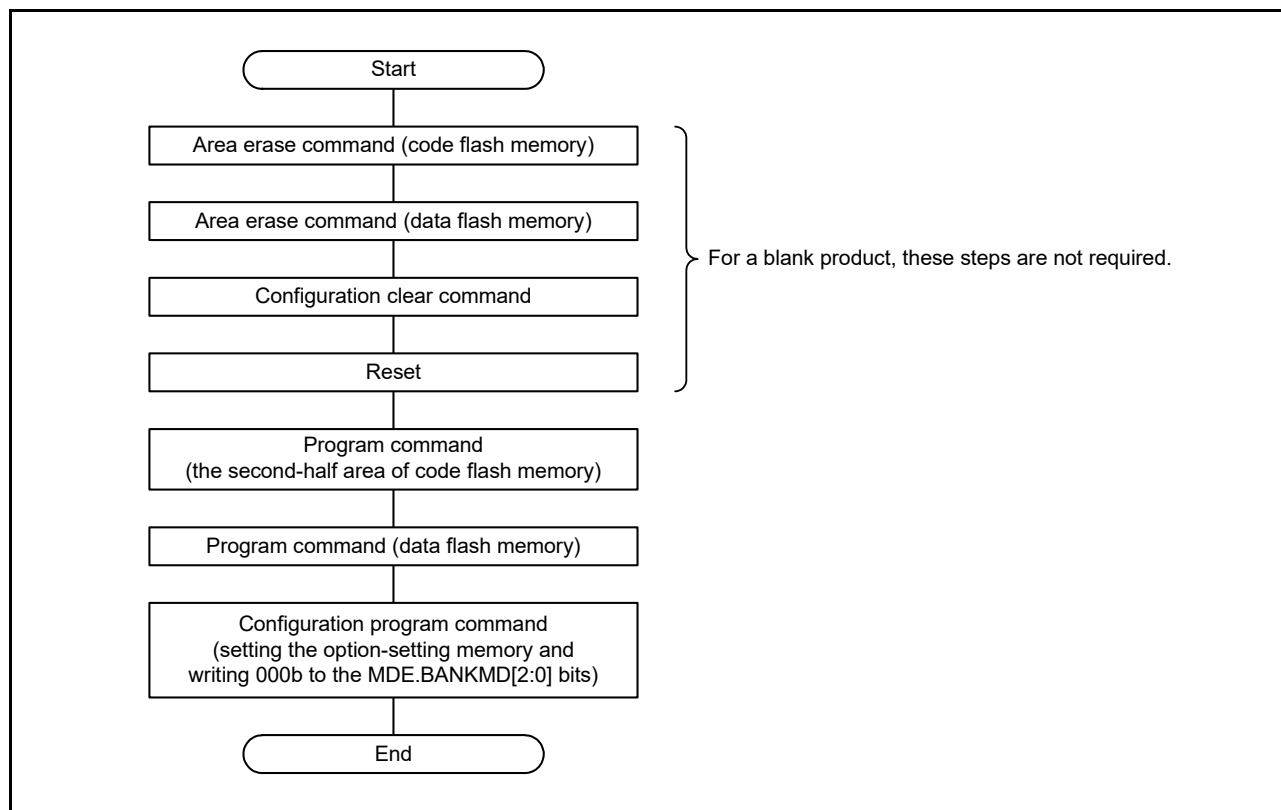


Figure 62.51 Example of Rewriting Flash Memory in Linear Mode

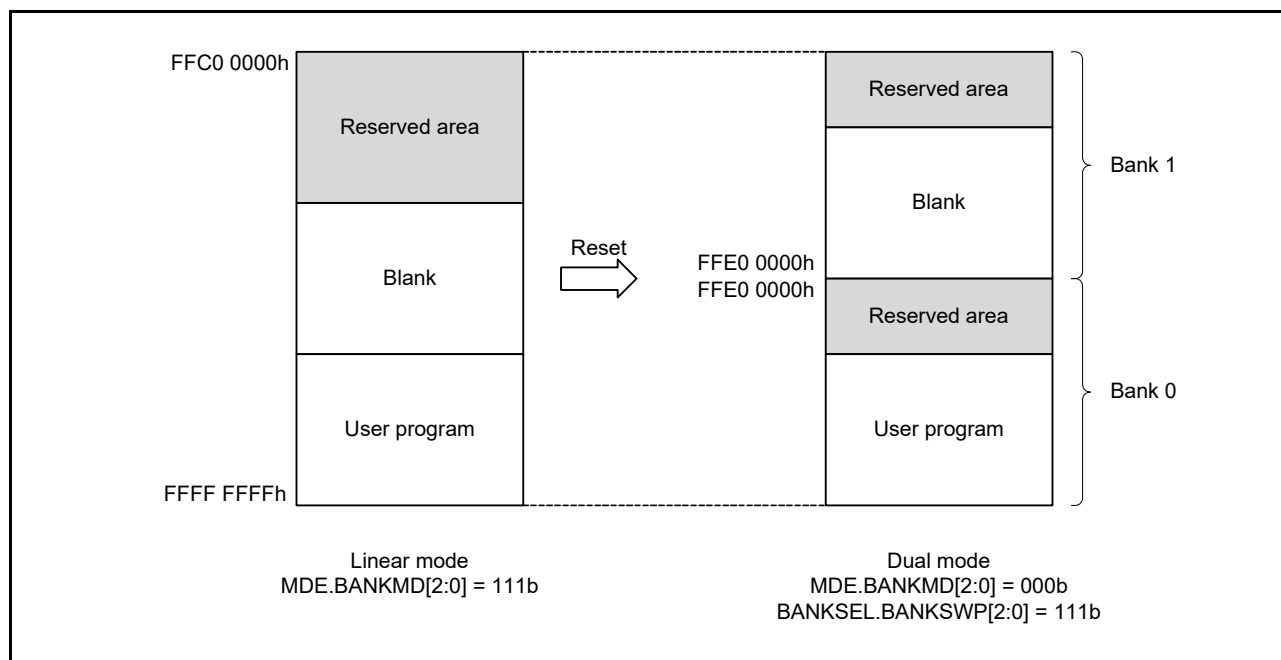


Figure 62.52 Memory Map before and after a Reset (Example for Products with Less Than 4 Mbytes of Code Flash Memory)

62.16 Using the Serial Programmer for Rewriting

A dedicated serial programmer can be used to rewrite flash memory in boot mode.

(1) Serial Programming

By providing a connector on the board, it is possible to program the MCU with a serial programmer while the MCU is mounted on the board.

62.16.1 Environments for Serial Programming

The recommended environments for rewriting the flash memory of the MCU are described below.

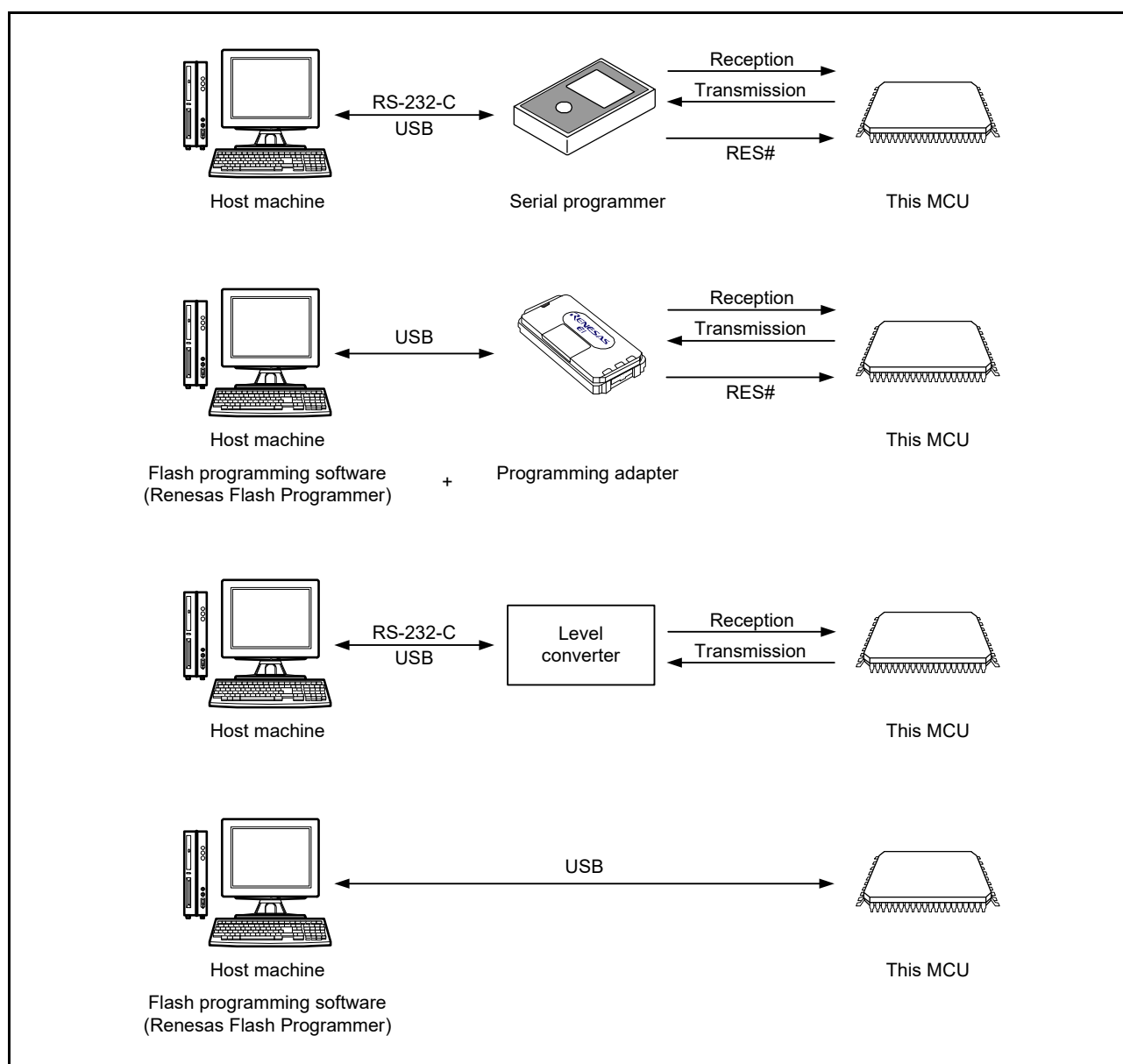


Figure 62.53 Environments for Rewriting the Flash Memory

Note: For details of the serial programmers, refer to the manual of each serial programmer; for details of the Renesas Flash Programmer flash programming software, refer to the Renesas Flash Programmer Flash Programming Software User's Manual.

62.17 Programming through Self-Programming

62.17.1 Overview

This MCU supports programming of the flash memory by the user program itself. The FACL commands can be used with user programs for writing to the flash memory. This allows upgrading of user programs and rewriting of constant data fields.

When the data flash memory is programmed, the background operation facility makes it possible to execute a rewriting program from the code flash memory to program the data flash memory. Furthermore, the rewriting program can be copied to internal RAM or external memory in advance of the programming operation, and executed from the given destination to program the data flash memory.

Background operation is also available for use when the address ranges of the area of code flash memory to be programmed and the area of code flash memory to be read satisfy particular conditions (see Table 62.25). At the time of self-programming in this case, a rewriting program in code flash memory can be used to program the code flash memory. Also, the rewriting program can be copied to internal RAM or external memory in advance of the programming operation, and executed from the given destination to program the code flash memory. This is useful when the address ranges do not satisfy the conditions for background operation.

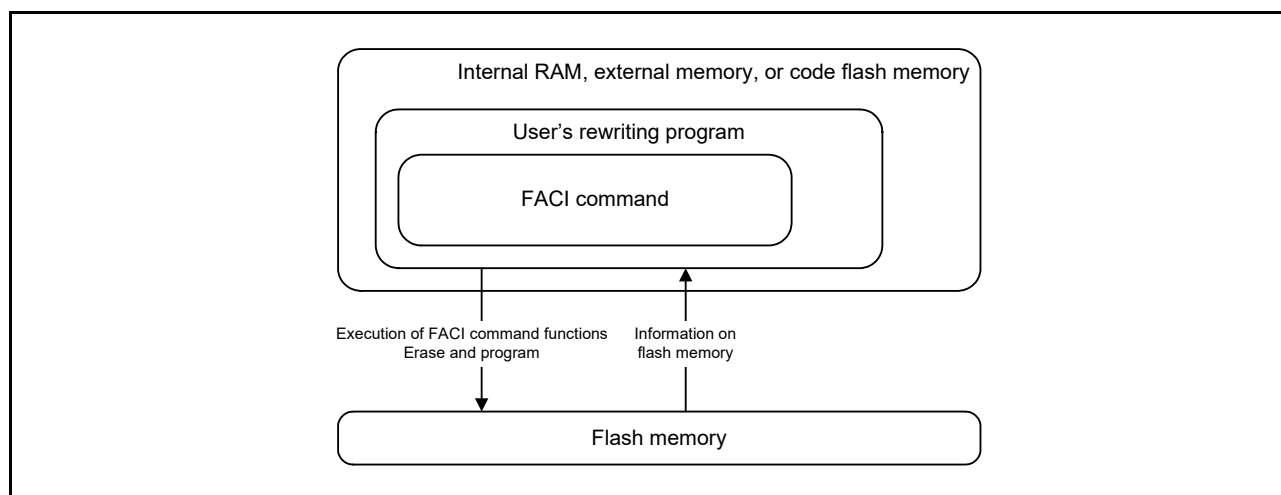


Figure 62.54 Schematic View of Self-Programming

62.17.2 Background Operation

The BGO is can be used to execute the flash rewrite routine on the code flash memory when the data flash memory or other area of code flash memory is rewritten.

Background operations can be used when the combination of the flash memory for rewriting and the flash memory for reading is any of those listed below.

Table 62.25 Conditions under which Background Operation is Usable

		Range for Rewriting	Range for Reading
Common to linear and dual modes		Data flash memory	Code flash memory
Linear mode	Products with 4 Mbytes of code flash memory	First half (2-Mbyte area) of the code flash memory (addresses FFC0 0000h to FFDF FFFFh)	- Second half (2-Mbyte area) of the code flash memory (addresses FFE0 0000h to FFFF FFFFh) - Data flash memory
		Second half (2-Mbyte area) of the code flash memory (addresses FFE0 0000h to FFFF FFFFh)	- First half (2-Mbyte area) of the code flash memory (addresses FFC0 0000h to FFDF FFFFh) - Data flash memory
Dual mode	When the BANKSEL.BANKSWP[2:0] bits are 111b:	Bank 1 area of the code flash memory	- Bank 0 area of the code flash memory - Data flash memory
	When the BANKSEL.BANKSWP[2:0] bits are 000b:	Bank 0 area of the code flash memory	- Bank 1 area of the code flash memory - Data flash memory

62.18 Reading Flash Memory

62.18.1 Reading Code Flash Memory

Special settings are not required to read code flash memory after release from the reset state. Data can simply be read out through access to addresses in the code flash memory.

When reading code flash memory that has been erased but not yet been programming again (i.e. that is in the non-programmed state), all bits are read as 1.

62.18.2 Reading Data Flash Memory

Special settings are not required to read data flash memory after release from the reset state. Data can simply be read out through access to addresses in the data flash memory.

Values read from data flash memory that has been erased but not yet been programming again (i.e. that is in the non-programmed state) are undefined. Use blank checking when you need to confirm that an area is in the non-programmed state.

62.19 Trusted Memory

This MCU has a trusted memory (hereafter called TM) facility to prevent the reading of blocks 8 and 9 and blocks 78 and 79 (in dual mode) in the code flash memory by third party software. When set as trusted memory, these areas are suitable for storing software that handles processing for encryption algorithms, device control software that is associated with proprietary or confidential know-how, purchased middleware, and so on.

Table 62.26 lists the specifications of the TM function, Table 62.27 lists access restrictions within the TM target area when the TM function is enabled, and Figure 62.55 shows the cases where the CPU is able to operate in relation to the TM target area.

Table 62.26 TM Specifications

Item	Description
TM target area	Linear mode: Blocks 8 and 9 in the code flash memory (64 Kbytes in total) Dual mode: Blocks 8 and 9 in the code flash memory (64 Kbytes in total) and blocks 78 and 79 (64 Kbytes in total)
Access restrictions when TM is enabled	See Table 62.27, Restrictions on Access to the TM Target Area while the TM function is Enabled.
How to run program code when the TM function is enabled	When the TM function is enabled, starting to run program code in an area being handled as TM is only possible with a branch instruction from program code outside the areas being handled as TM.
Interrupt processing during the execution of program code in an area being handled as TM while the TM function is enabled	Both the acceptance of requests for interrupt processing and return from interrupt processing are possible.
Security function	Enabling the TM function restricts access to program code in the areas for handling as TM to instruction fetching only
Protection functions	- Restrictions on data access to the TM target area when the TM function is enabled*¹ - Once enabled, the TM function prevents its own disabling until the areas being handled as TM are erased. - Once enabled, the TM function prevents further writing to the areas being handled as TM.

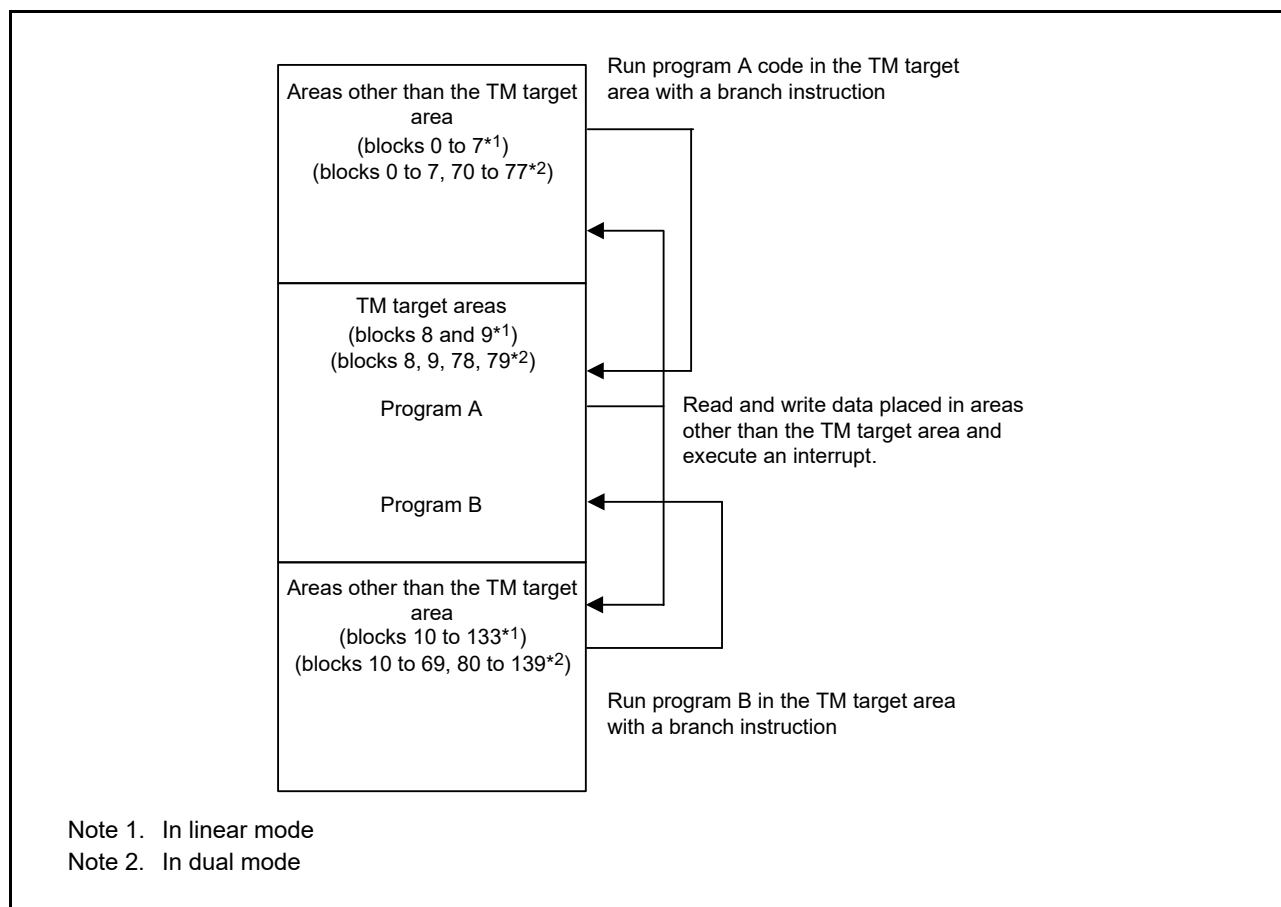
Note 1. Access to data in operations that include the borders of the TM target areas is also not allowed.

Table 62.27 Restrictions on Access to the TM Target Area while the TM function is Enabled

Type of Access	CPU	DMAC/DTC/EXDMAC/ Extended bus master
Instruction fetching	Yes	—
Access to data	No*1	No

Note: When using the on-chip debugger (OCD), the same restrictions apply as for the DMAC in the table above. For the OCD operation to the TM target area, refer to the manual of the emulator to be used.

Note 1. Place data to which access will be required in areas other than the TM target areas.

**Figure 62.55 Cases where CPU is Allowed to Operate in Relation to the TM Target Area when the TM Function is Enabled**

62.19.1 Allocating Program Code to the TM Target Area

When the TM function is enabled, implement countermeasures in the form of software for the TM target area as required as further measures to prevent the running of programs for access to consecutive addresses in the TM target areas from areas outside the TM target areas.

62.19.2 How to Enable the TM Function

62.19.2.1 By Self-Programming

After writing to blocks 8 and 9 of the code flash memory, i.e. the TM target areas (after programming blocks 8, 9, 78, and 79 in dual mode), use the configuration set command of the FACI to enable the TM function.

Figure 62.56 is a flowchart of the procedure for enabling the TM function by self-programming.

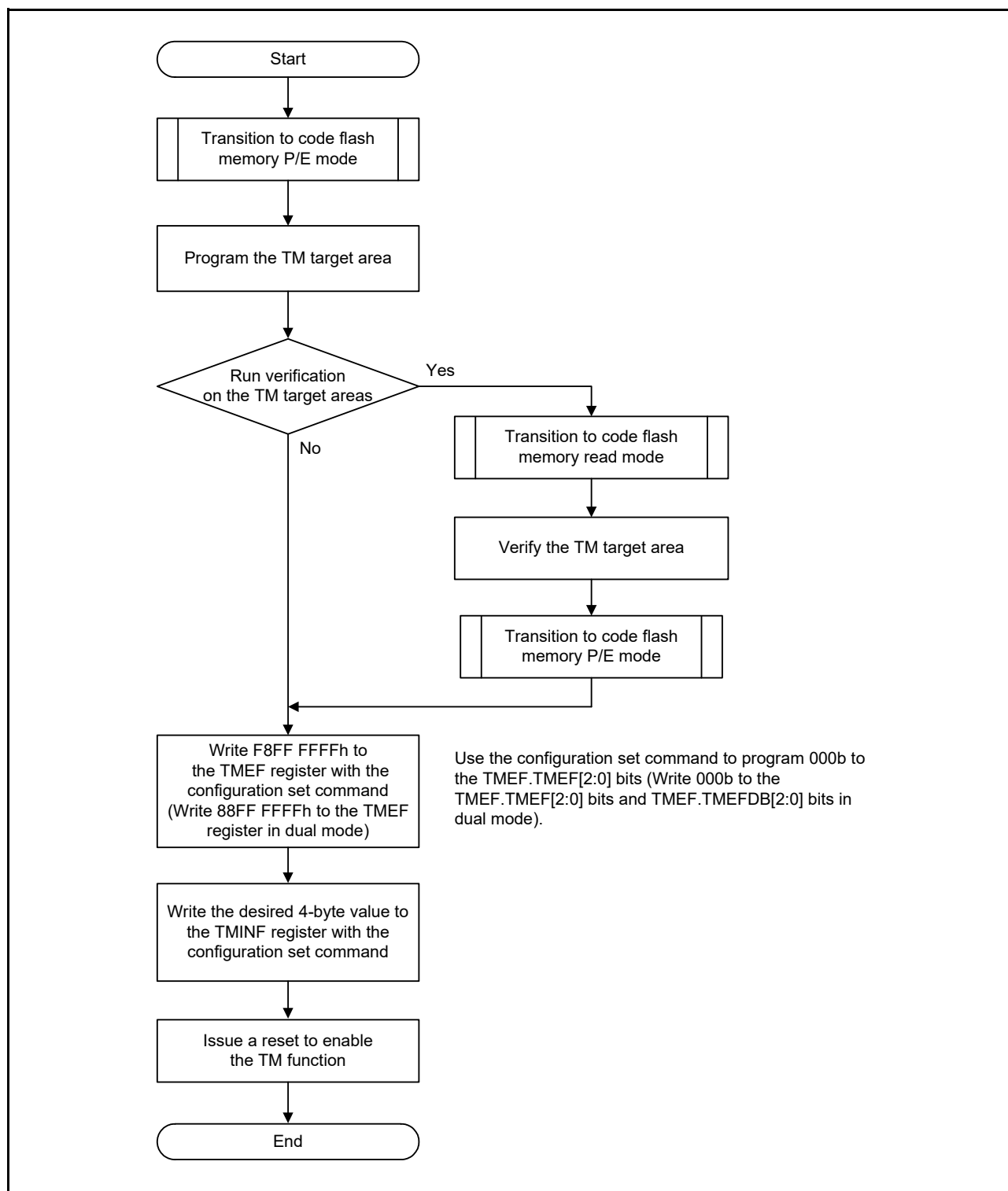


Figure 62.56 Flowchart for Enabling the TM Function by Self-Programming

62.19.2.2 By Using Boot Mode

In boot mode, use the configuration program command among the boot commands to enable the TM function after writing to blocks 8 and 9 of the code flash memory (or after writing to blocks 8, 9, 78, and 79 in dual mode).

For the configuration program command among the boot commands, see section 62.15.23, Configuration Program Command.

Figure 62.57 is a flowchart of enabling the TM function in boot mode.

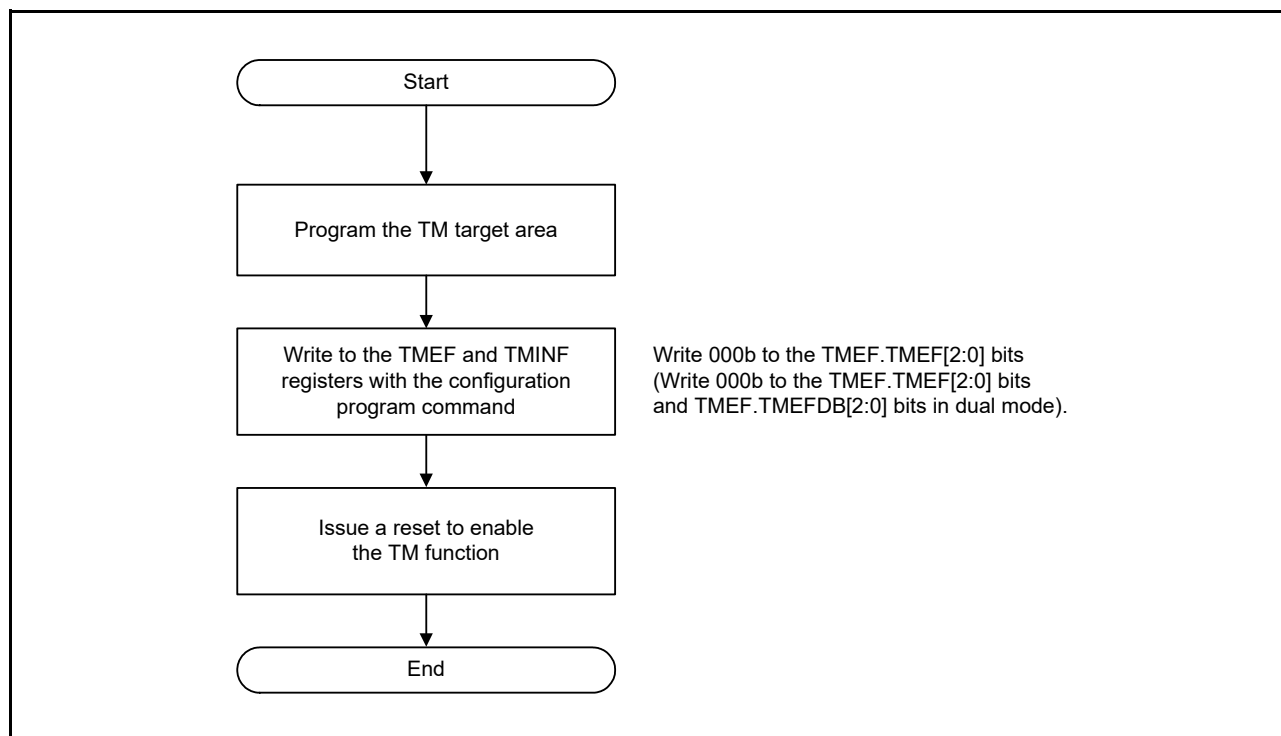


Figure 62.57 Flowchart for Enabling the TM Function by Using Boot Mode

62.19.3 How to Disable the TM function

The TM function cannot be disabled unless the TM target areas are first erased with the configuration clear command. Do not use the configuration clear command for this purpose unless the TM function is to be disabled.

Figure 62.58 is a flowchart of disabling the TM function in boot mode.

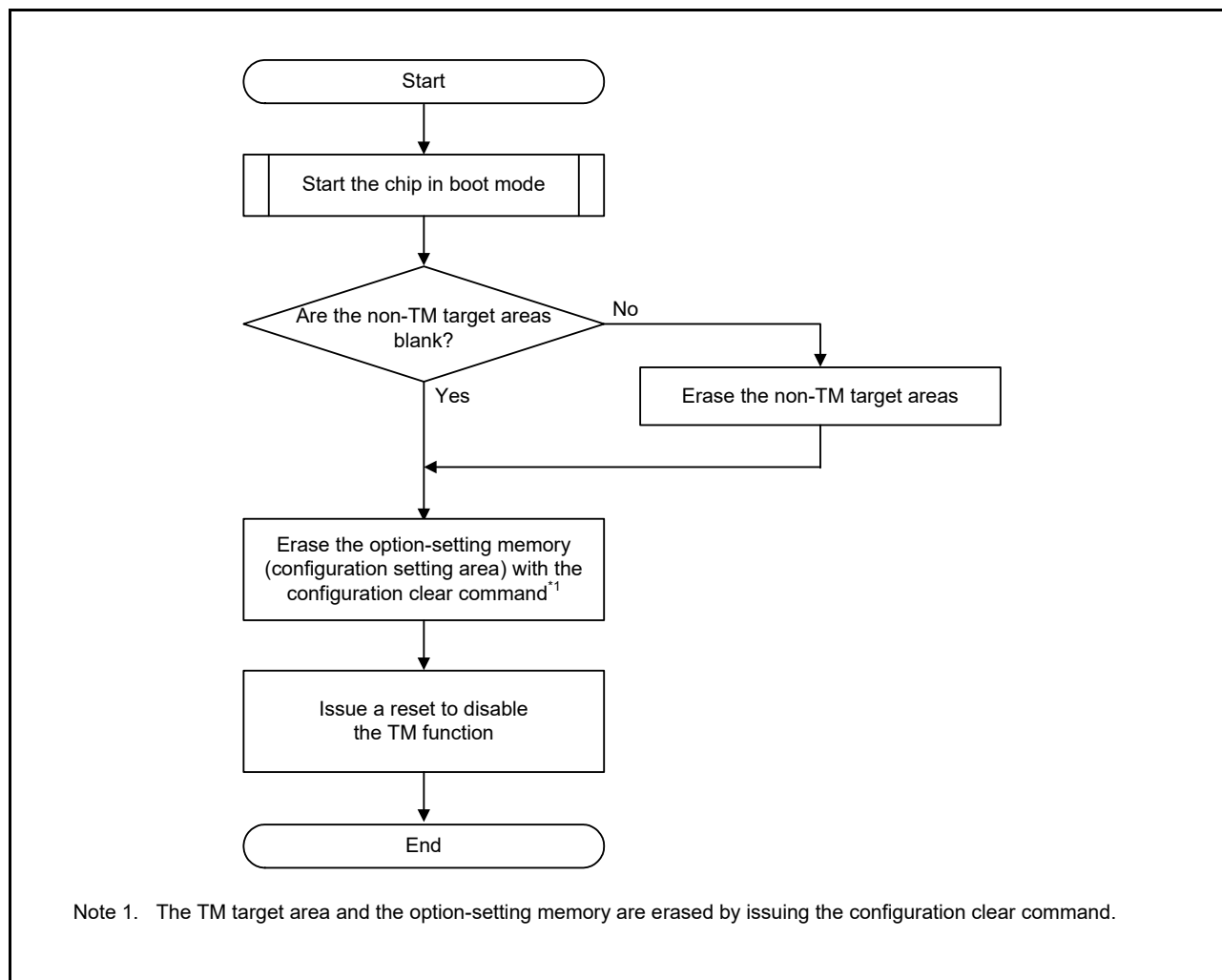


Figure 62.58 Flowchart for Disabling the TM Function in Boot Mode

62.19.4 Notes on Enabling the TM function

62.19.4.1 Protection against Access to the TM Target Areas

While the TM function is enabled, the only type of access to the TM target areas is instruction fetching by the CPU, so do not allocate data to the TM target areas.

If the CPU, DMAC, DTC, EXDMAC, extended bus master, or OCD attempt access to data in the TM target areas while the TM function is enabled, values are always read as 0 instead of the actual values.

62.19.4.2 Further Writing to the TM Target Areas

Further writing to the TM target areas is not allowed as long as the TM function is enabled.

If you do need to write further data, follow the procedure described in [section 62.19.3, How to Disable the TM function](#), to disable the TM function, write the modified or extended data to blocks 8 and 9 of the code flash memory (blocks 8, 9, 78, and 79 in dual mode), and then follow the procedure described in [section 62.19.2, How to Enable the TM Function](#), to enable the TM function.

62.19.4.3 Executing the Configuration Clear Command

Follow the procedure in [section 62.19.3, How to Disable the TM function](#), before issuing the configuration clear command.

62.19.4.4 When the MPU Setting is for Access to the TM Target Areas

When the TM function is enabled, even if the MPU is set to allow access to the TM target areas, the TM function takes priority.

62.19.4.5 FACL Block Erase Command for the TM Target Areas

There are no special restrictions on block erasure of the TM target areas with the FACL block erase command.

Accordingly, since each area corresponds to an erase block, the areas can be erased by issuing the block erase command.

62.19.4.6 Conditions for Correct Operation of the TM Function

The TM function operates normally under the conditions prescribed in [section 63, Electrical Characteristics](#).

62.20 Usage Notes

62.20.1 Reading Area Where Program or Erase Operation was Aborted or Suspended

The data stored in the area where program or erase operation has been aborted or suspended are undefined. To avoid malfunctions caused by reading undefined data, take care not to fetch instructions or read data from the area where program or erase operation was aborted or suspended.

62.20.2 Suspension During Programming/Erase

When processing of programming/erase is stopped by issuing the P/E suspend command, the programming/erase processing can be resumed by issuing the P/E resume command. If the flash sequencer enters the command-locked state for any reason and issues the forced stop command after the suspended processing is normally completed and the ERSSPD flag or PRGSPD flag is set to 1, the suspended processing cannot be resumed. In addition, the values in the area where the processing was suspended are not guaranteed. Erase that area.

62.20.3 Prohibition of Additional Programming

Programming a given area of the code flash memory or data flash memory twice is not possible. To program the code flash memory or data flash memory where has been programmed, erase the target area.

62.20.4 Resets During Programming/Erase, or Blank Checking

In the case of a reset due to the signal on the RES# pin during programming/erase, or blank checking of the flash memory, wait for at least t_{RESWF} (see section 63, Electrical Characteristics) of the reset input period once the operating voltage is within the range stipulated in the electrical characteristics, then release the device from the reset state.

62.20.5 Allocation of Vectors for Interrupts and Other Exceptions During Programming/Erase

Generation of an interrupt or other exception during programming/erase may lead to fetching of the vector from the code flash memory. Under conditions where BGO cannot be used, set the address of the vector to an address that is not in the code flash memory. Alternatively, make sure that no handling of interrupts or exceptions proceeds during programming/erase.

62.20.6 Items Prohibited During Programming/Erase, or Blank Checking

High voltage is applied to the flash memory during programming/erase, or blank checking. To prevent damage to the flash memory, do not perform the following operations.

- Have the operating voltage from the power supply go beyond the permitted range.
- Change the FWEPROR.FLWE[1:0] bits.
- Change the SYSCR0.ROME bit.
- Change the OPCCR.OPCM[2:0] bits.
- Change the SCKCR.FCK[3:0] and PCLKB[3:0] bits.
- Change the SCKCR3.CKSEL[2:0] bits.
- Change the RSTCKCR.RSTCKEN bit.
- Transition to the all module clock stop mode, software standby mode, or deep software standby mode.

62.20.7 Notes on Program Execution in Boot Mode (USB Interface)

- An oscillator is usable in boot mode (USB interface) when its frequency is 20 or 24 MHz and when the setting value of the main clock oscillator driving ability 2 switching bits (MOFCR.MODRV2[1:0]) are 00b in the result of the matching test conducted by the oscillator manufacturer (the recommended setting value).

- To ensure that the power supply is stable during programming/erasure of flash memory, do not connect a cable via a bus-powered HUB.

62.20.8 Programming/Erasure in Low-Speed Operating Modes 1 and 2

Do not programming/erasure the flash memory when low-speed operating mode 1 or 2 is selected with the operating power control register (OPCCR).

62.20.9 Notes on issuing program commands

When issuing a program command (from writing of E8h to writing of D0h), disable interrupts that may trigger FSCI commands.

If an interrupt occurs during the issuing of an FSCI command and another FSCI command is issued in the interrupt handling routine, the latter FSCI command is either ignored or recognized as an illegal command, and normal operation thus cannot be expected. We recommend disabling such interrupts during the issuing of FSCI commands as well as program commands.